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Dynamic Modeling of Environmental Systems

With 87 Illustrations and a CD-ROM



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*To
Jamie, Deaven, Sam, and Kate Winebrake,
and
to Kim and Brett Deaton,
who have given us countless lessons in sometimes
unpredictable dynamic systems; and to Susan Winebrake
and JoEtta Deaton, who have helped us apply effective
control strategies*

Series Preface

The world consists of many complex systems, ranging from our own bodies to ecosystems to economic systems. Despite their diversity, complex systems have many structural and functional features in common that can be effectively simulated using powerful, user-friendly software. As a result, virtually anyone can explore the nature of complex systems and their dynamical behavior under a range of assumptions and conditions. This ability to model dynamic systems is already having a powerful influence on teaching and studying complexity.

The books in this series will promote this revolution in “systems thinking” by integrating skills of numeracy and techniques of dynamic modeling into a variety of disciplines. The unifying theme across the series will be the power and simplicity of the model-building process, and all books are designed to engage the reader in developing their own models for exploration of the dynamics of systems that are of interest to them.

Modeling Dynamic Systems does not endorse any particular modeling paradigm or software. Rather, the volumes in the series will emphasize simplicity of learning, expressive power, and the speed of execution as priorities that will facilitate deeper system understanding.

Matthias Ruth and Bruce Hannon

Preface

Background

This book evolved from a need to share with undergraduate students and professionals methods and models for understanding dynamic environmental problems. When we first undertook the challenge of educating students in these matters, we were faced with various textbooks—none of which satisfied our objectives. Textbooks in the general environmental science and technology field provided useful background materials for students, but did not allow them to explore environmental problems through microcomputers and modeling applications. Textbooks in the environmental modeling field, however, tended to be designed for graduate student work, often focusing on one category of environmental media (e.g., groundwater modeling). The emphasis of these books was on modeling techniques; hence, the mathematical depth was too great and the environmental subject matter too narrow for our broad educational goals and expectations.

Thus, we set forth to develop an environmental modeling text that had at least two objectives. First, the book needed to discuss some of the broad concepts of “modeling,” particularly dynamic systems modeling. Second, the book needed to apply these concepts to *environmental* systems. We felt that it was important that the main objective of these modeling efforts should be to help the student better understand the dynamics that drive any particular environmental system. This is a different approach than using models to get the “right answer” to an environmental problem. For example, when studying pollution transport in a surface water environment one can create a model that will predict contaminant concentrations at a certain point in time, which is all well and good. Beyond these predictions, however, it is the *process* of developing, applying, and manipulating a model by which one begins to truly understand the physical and environmental system that the model mimics. Hence, models become tools for solving problems as well as tools for gaining a better understanding and appreciation for these problems.

Another issue we faced in writing this text was the distinction between two classes of modelers: model-users and model-builders. In fact, the book

can be used for both audiences. We like to think of model-users as those readers who may find themselves exposed to model output or who may be operating existing environmental models to address environmental problems. These readers need to be aware of how model assumptions, variables, and sensitivities can impact model results. They need to know what questions to ask and how to ask them. Model-builders, on the other hand, must relate generic modeling structures with environmental phenomena and link these structures to develop more complex and powerful models. The chapters in this book are written at a level useful for both types of readers. The distinction is most clear in the exercises found throughout the text. For the model-user, predeveloped models are given on the accompanying CD-ROM, and the user is asked to manipulate those models to explore relationships and assumptions. For the model-builder, exercises are given that require the manipulation and expansion of certain models, as well as the outright development of models to explore specific environmental problems. We encourage readers to use both model-user and model-builder approaches as they read this text.

How to Use the Book

The book is unique in both its approach and content. The approach employs a combination of general modeling concepts applied to specific examples in the environmental sciences. Applications are developed in a “workbook” fashion: An environmental problem is introduced, some technical and scientific background is provided, and the modeling approach needed for exploring the problem is discussed. Throughout each application, generic systems constructs and diagrams are presented, with some attention paid to the mathematical relationships behind these constructs. We have attempted to keep the mathematical content of the book at a level consistent with a single-semester course in calculus. Some of the mathematics may be skipped without compromising our objective to develop dynamic modeling skills. For readers with some background in calculus, however, the mathematics is worth studying and sheds light on the basic building blocks of dynamic systems models.

The book is arranged in two parts. The first part of the book (Chapters 1–3) provides the reader with an overview of dynamic systems modeling, including sections on systems terminology, the uses of systems models, modeling constructs, positive and negative feedback, and sensitivity analysis. The book focuses on dynamic modeling (as opposed to static modeling, in which system changes through time are not a concern) because most environmental problems have dynamic components, and this characteristic is often ignored in traditional environmental modeling texts.

The second part of the book (Chapters 4–9) offers applications that illustrate the use of dynamic models for exploring a variety of environmental

problems. Each chapter focuses on a major environmental problem (e.g., surface water pollution, matter cycling disruptions, global warming). The reader is first provided an illustrative scenario that presents a problem the reader is asked to explore. Each scenario is designed to provide a context for the larger problem at hand. By using this “problem scenario” approach, the reader is forced to apply the skills of a problem-solver in the context of a real problem.

The first step in the problem-solving process usually involves defining the problem. Thus, the reader is provided scientific or technical information that offers background for the topic. This information may be a review for those readers who are versed in environmental science. It is intended simply to provide enough information to establish a foundation on which the systems model can be formulated. Readers who have little background in the problems presented in these chapters are encouraged to refer to additional readings on these topics.

Following each chapter’s background section are sections that help the reader consider the following questions:

1. What are the important components of this system?
2. How are these components related conceptually and mathematically?
3. How can a systems diagram be constructed that illustrates these relationships?
4. What generic systems constructs can be identified, and what do these constructs imply about system behavior?
5. How does the system react to various “perturbations” or changes?
6. How sensitive is the system to changes and what does this imply for system stability?
7. Where might the human-natural system interface occur, and what impacts are expected from human-caused perturbations in the system?

Finally, each chapter includes run-time models that will help readers explore the preceding questions. End-of-chapter exercises are also provided that ask readers to think about the preceding questions in more depth, and that encourage the development of new models or extensions of those given in this book.

Using the Book in a Classroom Setting

We have used earlier versions of this book to teach an environmental modeling class for undergraduate students. We have found that the book can more than fill a full semester of a three-credit-hour course. We usually spend about 4–6 weeks on the first three chapters, ensuring that the students are comfortable with the new terminology and methods of “systems modeling.” We then begin to address individual chapters, taking about 2 weeks per chapter. Because Chapters 4–9 are each self-contained, they can be used in any order,

depending on class interest. In the 2 weeks or so we spend on each chapter, we usually divide the sessions into some short lecture periods and some “hands-on” working periods. In the lecture sessions we present background material. In the working periods, usually held in computer labs, we facilitate the students’ study of models provided in the text (students can also do this independently). For certain chapters, we have also found it rewarding to allow the students several class periods to develop their own original models related to the current chapter. These models are usually presented to the entire class for review and discussion.

Finally, we decided to use the STELLA[®] software package for all of our modeling examples and applications. STELLA[®] is a graphically interfaced, dynamic modeling software system. It allows the user to “program” a model graphically (i.e., without the burden of a cumbersome programming language—although that feature is available). We found this function to be extremely useful because it forces the student thoroughly to think through the problem and the interconnections of all aspects of the system being explored.

Conclusion

We have attempted to provide a wide variety of environmental examples (both in complexity and diversity) throughout the book. We hope that this makes the book more enjoyable to read and use. We also believe this helps in the reader’s fundamental understanding that modeling is not relegated to a priesthood of mathematicians and computer scientists, nor to a particular field such as “groundwater analysis.” The act of “modeling,” we believe, should be a large part of our everyday educational experience (in academe and beyond). It should not be feared, but embraced.

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Michael L. Deaton and James J. Winebrake

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1 Overview of Environmental Systems

Chapter Objectives—

After you finish this chapter, you should be able to:

1. *Recognize many environmental phenomena as coming from dynamic systems.*
2. *Name the **four components of a system** and use those components to construct a simple model of a system.*
3. *Describe how **difference equations** are used to calculate the behavior of a dynamic system over time.*
4. *Distinguish between **systems thinking** and other kinds of thinking.*
5. *Explain how dynamic systems models can be used to understand environmental problems.*
6. *Define **feedback** and **steady-state behavior** and explain why these features are important to environmental systems.*

1.1 Introduction

This book is about change. In particular, it is about how our environment changes. The purpose of this text is to teach you how to model, understand, and analyze the dynamic nature of many real-life environmental phenomena. In doing so, it is our hope that you will develop an intuitive feel for the extraordinary collection of systems that govern the behavior of the environment. It is also our hope that you will learn to use some important tools for evaluating how human beings can potentially upset those systems or significantly alter their behavior.

This book is also about modeling. Virtually all environmental problems are inherently dynamic systems problems. They all deal with environmental phenomena that change over time (i.e., they are **dynamic**) and involve numerous interrelated components (i.e., they are **systems**).

Scientists who study environmental issues now commonly employ computer-based models of environmental systems to help them understand how the environment changes and to make predictions on how it will evolve in the future. These models are not academic curiosities. Their predictions help shape public policy, which in turn has significant impacts on the environment and the economy.

This is where you and this text come in. This book will help you more effectively participate in the scientific and political discussions of the environment by equipping you to describe and study environmental problems

within a **systems analysis** framework. You will learn how to evaluate computer-based models of environmental phenomena, and then how to use those models to better understand the underlying systems and to predict future outcomes. You will also learn some of the underlying principles and thinking skills that are used to build these models.

In order to accomplish these goals, you must first learn to use some tools associated with systems thinking. This chapter introduces some of the basic systems modeling tools and concepts that we will use throughout the text.

1.2 An Example of a Simple System

We all use the word *system* in a variety of contexts in everyday conversation. In dynamic systems modeling, however, the concept of a system has a very specific meaning. We will use the term **system** in this text to refer to any collection of entities that includes the four components discussed in the following.

1.2.1 Reservoirs

A *reservoir* can be thought of as a repository where something is accumulated, stored, and potentially passed to other elements in the system. For example, suppose we wish to model the growth of a population of deer in a particular ecosystem. This model would possibly include a *vegetation reservoir* that represents the food supply of the deer. The system would also have one or more *predator reservoirs* representing the populations of predators that kill the deer. We would also include a *deer reservoir* to represent the population of deer. It is important to note that a reservoir does *not* represent a geographical location. Our deer reservoir should not be thought of as a location in which all the deer reside; rather, it is an accounting mechanism that enables us to keep track of how many deer live in the system at any point in time.

1.2.2 Processes

A *process* is an ongoing activity in the system that determines the contents of the reservoirs over time. Examples of processes in our deer population model might be:

- Birth process (process by which the deer reservoir increases in size)
- Death process (process by which the deer reservoir decreases in size)
- Predation process (process by which the predator stalks and kills the deer)

1.2.3 Converters

Converters are system variables that can play several different roles within a system. Their most important role is to dictate the rates at which the processes operate and therefore the rates at which reservoir contents change. An example of a converter is the *birth rate* of the deer population. This constant will clearly dictate the rate at which the *birth process* generates new deer. It will also affect the size of the deer reservoir over time.

1.2.4 Interrelationships

Interrelationships represent the intricate connections among all components of the system. These relationships are usually expressed in terms of mathematical relationships. For example, we can define a simple mathematical expression that describes the interrelationship among the birth process (i.e., the number of new deer born in a year), the birth rate, and the size of the deer reservoir. Suppose that the birth rate is equal to 0.2 deer born per capita per year. If we let $D(t)$ stand for the size of the deer reservoir in year t , then we can calculate the number of births in year t as follows:

$$\# \text{ births} = 0.2 \cdot D(t)$$

The specific manifestation of the four system components listed previously depends on the context. Different combinations of these components will be used to model different systems. In addition, any given problem can involve one or more systems, each of which is interrelated with the others.

We will now further illustrate these concepts by constructing a simple model involving an imaginary group of 20 tourists (10 males, 10 females, and no children) who have been shipwrecked on an uninhabited and uncharted, but lush, tropical island. This group is hopelessly lost with no chance of rescue in the foreseeable future; hence, they will have to make the best of it. Let us suppose that they build a small village of huts and settle in for a new life of tropical living. Further suppose that one of these villagers is a systems modeler who has a laptop computer (solar powered, of course) along with the latest version of a systems modeling software package. This villager has decided to model the growth of this population of shipwrecked tourists to better understand how its future might unfold on this isolated island. In particular, the modeler wishes to determine:

- The conditions under which the population will survive and flourish, and the conditions under which it will die off
- The time frame over which the population would likely die off, if it should not survive
- The number of people that can be realistically sustained on the island.

Based on this description, the system with which the modeler is interested is the island ecosystem as it relates to the survival (or demise) of the popu-

lation of shipwrecked tourists. Note that there are many “systems” that the modeler could study. For example, the modeler could model the ecosystem of the barrier reef around the island, or the weather system in the regions around the island. The list could go on and on. Whenever we focus our attention on the modeler’s three goals as stated earlier, however, these other systems end up playing at most a secondary role. This assumption admittedly limits the scope of the model (after all, perhaps the ecosystem of the barrier reef will have some impact on the human population). We will err on the side of simplicity (an important principle of systems modeling), however, and then add more detail as needed. We will now discuss examples of the four components for this imaginary system.

1. The reservoirs. In order to identify the reservoirs in this system, we should always answer the following simple question: *Are there important objects or entities in the system that will accumulate and (possibly) diminish over time?*

The modeler is clearly most interested in tracking the number of people that live on the island. In addition, the growth (or death) of the population is dependent on the long-term viability of the island’s resource base. Both of these collections of entities can be expected to accumulate or deplete over time; hence, two important reservoirs for this example are:

- *First reservoir:* The human population on the island (measured as the number of individuals)
- *Second reservoir:* The island resources available for sustaining the human population (measured in generic resource units)

2. The ongoing processes. The processes are those activities (either natural or otherwise) that determine the size of the reservoir contents over time. In our island community, there are two basic processes that will dictate the size of the population of humans. There is a “birth” process, which increases the size of the population, and there is a “death” process, which decreases the population. These processes and the population reservoir can be represented graphically as shown in Figure 1.1.

Figure 1.1 illustrates some modeling conventions that we will use throughout this text. The reservoirs (e.g., *People on the Island*) are represented with rectangles (which we will also call **stocks**). The processes (e.g., *Birth* and *Death*) are represented with directed double-line arrows and attached bubbles (which we will call **flows**) that flow either into or out of the reser-

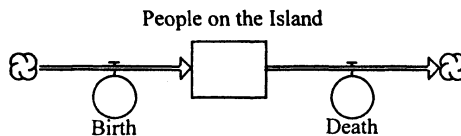


FIGURE 1.1. Population reservoir, with Birth and Death process.

voir. Flows into a reservoir will increase its contents. Flows out of a reservoir will decrease its contents. The reservoirs (stocks) represent stored quantities. The flows represent the processes by which those stored quantities are accumulated or diminished.

The value of each flow (or, equivalently, each process) is expressed as the amount of change it causes in the reservoir in one full-time unit. If time is measured in years, then the *Birth* flow will be expressed in units that correspond to the number of people born in 1 year. The *Death* flow will likewise, be expressed as the number of people that die in a single year.

This interpretation of the units of the flow processes and Figure 1.1 imply an important set of equations that dictate how our model will calculate (and hence predict) future contents of the *People on the Island* reservoir. In particular, the simulation model will calculate the contents of the reservoir at each point in time in the following way.

$$\text{Future contents} = \text{previous contents} + \text{all inflows} - \text{all outflows}$$

We can rewrite this using some simple mathematical variables. Let $R(t)$ stand for the contents of a reservoir at time t . Because the flow processes are expressed as the change in the reservoir contents in a full-time unit, we can calculate the future contents of the reservoir one unit ahead in time as follows:

$$R(t+1) = R(t) + \{\text{sum of all inflows} - \text{sum of all outflows}\}$$

If we wanted to predict the contents of R only one-half unit ahead in time, we would use the expression

$$R\left(t + \frac{1}{2}\right) = R(t) + \{\text{sum of all inflows} - \text{sum of all outflows}\} \cdot \frac{1}{2}$$

In general, if we wanted to predict the contents of R at a point in time that is Δt time units in the future, we would use the expression

$$R(t + \Delta t) = R(t) + \{\text{sum of all inflows} - \text{sum of all outflows}\} \Delta t \quad (1.1)$$

Equation (1.1) is called the ***difference equation*** for the reservoir $R(t)$. A difference equation of a reservoir is an equation for calculating future values of the reservoir from past values.

For our island population model, the difference equation for the *People on the Island* is

$$\begin{aligned} &\text{People on the Island } (t + \Delta t) = \\ &\text{People on the Island } (t) + \{\text{Birth flow} - \text{Death flow}\} \Delta t \end{aligned} \quad (1.2)$$

There are two similar processes that will dictate the size of the *Island Resources* reservoir. To aid in clarity, we will give the inflow process the name *Renewal* and give the outflow process the name *Depletion*. Figure 1.2 presents a diagram representing these two processes and their associated reservoir.

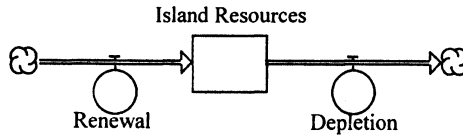


FIGURE 1.2. Island Resource reservoir, with Renewal and Depletion processes.

The choice of units for expressing the elements in Figure 1.2 is not as evident as it is in the case of the *People on the Island* reservoir and the *Birth* and *Death* processes in Figure 1.1. In fact, it is often the case that a natural choice of units is not obvious. In such a case, the model builder can arbitrarily create a new unit and define its meaning. For example, let the *Island Resources* reservoir be measured in a generic unit called a resource unit, where a resource unit stands for the amount of resources needed to sustain one person for a single month. If time is measured in years, then the *Renewal* and *Depletion* processes are expressed, respectively, as the number of resource units created or lost in 1 year. A single person would need 12 of these “resource units” to survive for 1 year.

In accordance with Equation (1.1), the underlying difference equation for calculating the contents of the *Island Resources* reservoir at any point in time is

$$\begin{aligned} \text{Island Resources}(t + \Delta t) = \\ \text{Island Resources}(t) + \{\text{Renewal flow} - \text{Depletion flow}\}\Delta t \end{aligned} \quad (1.3)$$

3. Converters or system constants. The two sets of reservoirs and associated processes in Figures 1.1 and 1.2 comprise the “backbone” of our system. All other elements in our system will regulate the rates at which the processes in this backbone operate. These additional system elements are the converters. The converters will dictate, for example, that rate at which the *Renewal* process adds new resource units to the *Island Resources*, or the rate at which the *Depletion* process removes resource units. In order to identify the converters to include in our system, consider the following question: *What additional quantities or system characteristics regulate the rates at which the processes run (thereby dictating the rates at which the reservoir contents change over time)?*

Consider the *Birth* process in Figure 1.1. This process dictates how many births occur in a given time interval. What determines the number of births? There are clearly complex biological processes involved; however, for our purposes, a way of calculating the average number of births that will occur in a given time interval is all that is needed. Some common sense suggests that the number of births in a given time interval ought to be proportional to the number of people in the population. If the population size were doubled, then we would expect the number of births also to

double (all else being equal). We can write this relationship mathematically as:

$$\# \text{ births} = b \cdot (\# \text{ people in the population}) \quad (1.4)$$

This equation includes a constant, b , which will serve to regulate the number of births that are generated. The constant b represents the number of people born per person in the population each year. Hence, b is a birth rate, expressed in births per capita per year. We will add a converter to our model to represent the quantity b and give it the name *Birth Rate*.

Using similar reasoning, we can also determine that we need a converter in the system to represent the death rate, which is expressed in deaths per capita per year. Call this converter the *Death Rate*. Another converter (call it the *Renewal Rate*) regulates the rate at which the *Island Resources* are renewed. The second converter (the *Depletion Rate*) regulates the rate at which the *Island Resources* are consumed or lost. We will express the *Depletion Rate* as the number of resource units consumed by a single person per year. We can add these converters to our system by augmenting Figures 1.1 and 1.2, as shown in Figure 1.3. Table 1.1 summarizes the information in Figure 1.3 by listing each entity, along with its units.

4. Interrelationships between the reservoirs, processes, and converters. Now that we have formulated a first-draft schematic of the major components in our system (Figure 1.3), we need to specify how these components are interrelated. We will graphically display these relationships by using single-line arrows to show what we understand to be the cause-effect relationships among the components of the system.

You may have noticed that some relationships are already implied by Figure 1.3. For example, we have already pointed out that there is a relationship between each reservoir (stock) and its associated processes (flows).

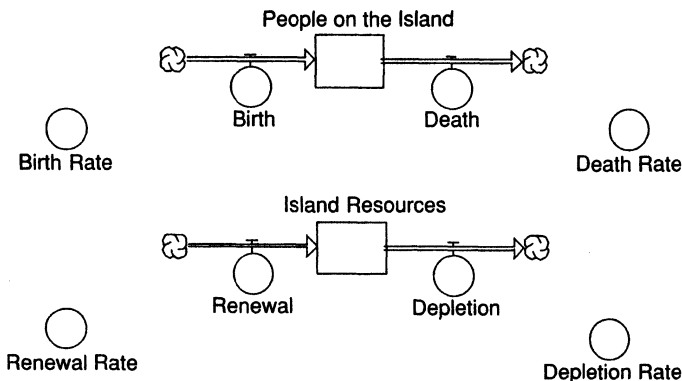


FIGURE 1.3. Island Community reservoirs, processes, and converters.

TABLE 1.1. Reservoirs, processes, and converters in the island system.

System entity	Type of entity	Units
People on the island	Reservoir (stock)	# people
Birth	Process (flow)	# people born per year
Death	Process (flow)	# people dying per year
Island resources	Reservoir (stock)	# resource units (1 unit = amount needed to sustain one person for 1 month)
Renewal	Process (flow)	# resource units added per year
Depletion	Process (flow)	# resource units lost or consumed per year
Birth rate	Converter	# people born per capita per year
Death rate	Converter	# people dying per capita per year
Renewal rate	Converter	# resource units added for each existing resource unit per year
Depletion rate	Converter	# resource units lost or consumed by each person per year

In fact, *in every system model that we develop, we will always assume that the only system entities that can directly affect the values of the reservoirs are the inflows and outflows associated with that reservoir*. It is important to note that other entities that are not flow processes can influence a reservoir's contents. The preceding assumption, however, requires that the only way other nonflow entities can affect a reservoir is by affecting the processes that flow into or out of it.

This assumption closely matches what you would expect in real life. Consider the *Birth* and *Death* processes that affect the population reservoir (see Figure 1.1). One could argue that something like the food supply (i.e., the *Island Resources* reservoir) will also affect the population of humans. The only way this will happen, however, is by affecting the *Birth* or *Death* processes into and out of the *People on the Island* reservoir (i.e., the only way to impact the size of a population is by affecting the number of births and deaths in that population). We have assumed, of course, that there are no emigration or immigration processes in this model: No one can leave the island, and no one can migrate onto the island.

Let us now see if we can specify which system entities are related to which. Remember that we will use a single-line arrow to show the direction of the relationship. The arrow will run from the entity that is the "cause" toward the entity that is "affected." We will refer to these single-line arrows as **connectors**. The **connectors** are used to display the cause-effect relationships between the various entities in the system. Figure 1.4 gives a first cut at specifying these relationships. The numbers on the arrows are provided so that we can briefly discuss the rationale for each. In general, our system diagram would not include these numeric identifiers.

Explanation of the Connectors in Figure 1.4

- *Connectors 1 and 2* indicate that the number of births in the island community is a function of the *Birth Rate* and the number of people in

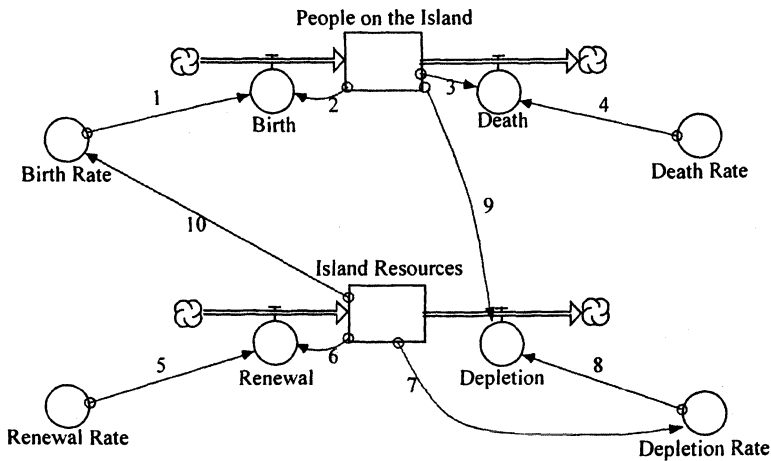


FIGURE 1.4. Island Community system diagram with connectors to show relationships between reservoirs, flows, and converters.

the population. This follows from the discussion leading up to Equation (1.4).

- *Connectors 3 and 4* are justified with the same rationale as connectors 1 and 2 [see the discussion following Equation (1.4)].
- *Connectors 5 and 6* imply that the number of resources that are added to the *Island Resources* reservoir is a function of only the *Renewal Rate* and the number of existing resources. This association makes sense if we consider the *Island Resources* essentially to be a renewable food supply. The growth of that supply will occur through natural processes that are analogous to a birth process in the human population (i.e., the more edible plants—or animals—on the island, the more “offspring” they will have over the year).
- *Connector 7* runs from the *Island Resource* reservoir to the *Depletion Rate* converter to indicate that the rate at which resources are consumed or lost depends on the size of the existing resource base. For example, it is likely that individuals in the island community will consume more resources per person whenever there is an abundant resource base than whenever the resource base is more scarce.
- *Connectors 8 and 9* signify that the number of resource units that are consumed or lost over any time interval is a function of the *Depletion Rate* and the number of *People on the Island*.
- *Connector 10* indicates that the *Birth Rate* in the human population is affected by the size of the existing resource base. This effect could come about because of conscious decisions by the island community to reduce the number of births in the face of limited resources. It could also come about because of the fact that limited resources may impact the overall

health of individuals in the population, thereby reducing their ability to bear children.

Once the connectors are drawn, our modeling friend must formulate mathematical expressions to explain how each quantity in the model is to be calculated at each point in time. The equations that dictate the size of each reservoir will have the general form that was given in Equation (1.1). If any other entity does not have any connectors or flows entering into it, then its value will typically be *exogenous* (i.e., not determined within the model, but defined by the model builder at the outset). If an entity has connectors running into it, then the modeler must specify how the inputs are to be used to calculate that entity's value.

For example, consider the *Birth* process in Figure 1.4. The diagram indicates that the number of births is a function of the number of *People on the Island* and the *Birth Rate*. That is,

$$\text{Birth flow} = f(\text{People on the Island}, \text{Birth Rate}) \quad (1.5)$$

We can determine the exact form of the function $f(\)$ in Equation (1.5) by considering the discussion leading up to Equation (1.4). Hence, the expression for the right-hand side of Equation (1.5) is

$$\text{Birth flow} = \text{Birth Rate} \cdot \text{People on the Island} \quad (1.6)$$

Using similar reasoning, we can derive many of the mathematical expressions for the other system entities.

It is important to use mathematical expressions that are very simple and which match our common-sense understanding of how things work. It will often be the case that simple addition, multiplication, or division operations will do the job.

Figure 1.5 provides another version of Figure 1.4 with several of the mathematical relationships superimposed on the diagram. Some of the chapter exercises will require that you carefully examine Figure 1.5 and understand the mathematical expressions given. In addition, you will be asked to develop mathematical expressions for those entities for which equations are not given.

You may have noticed that the mathematical expression for two of the converters in our system in Figure 1.5 (the *Birth Rate* and the *Depletion Rate*) are not defined with mathematical expressions. Their numeric values are instead “provided by a graph.” In some cases, the exact form of the mathematical relationship that defines a system entity may not be obvious; however, we can often describe the *shape* of the relationship between an entity and the quantities that determine it. For example, our system diagram indicates that the *Birth Rate* is determined by the value of the *Island Resources* reservoir. We cannot (at this point) credibly define a mathematical relationship in which the *Island Resources* value is used to calculate the *Birth Rate* value; however, the *Birth Rate* should decrease as the size of the *Island Resources* reservoir decreases. We could therefore construct a graph

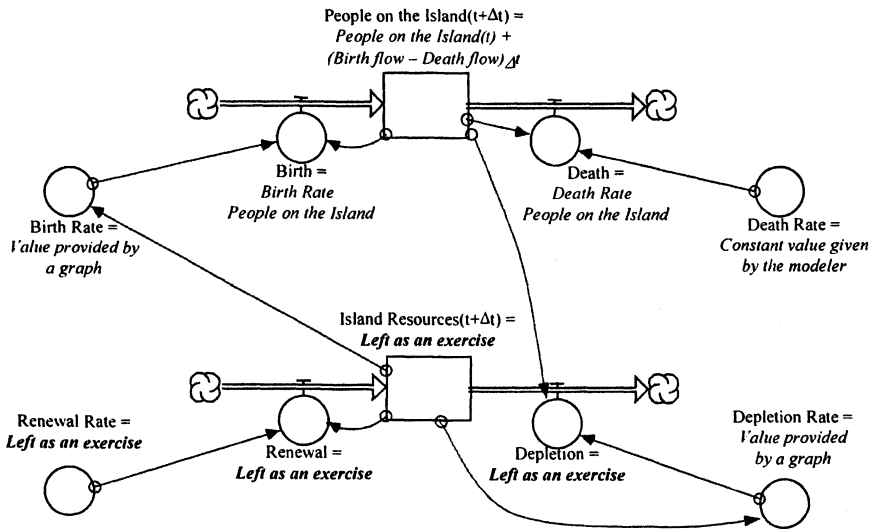


FIGURE 1.5. Island Community system diagram with some mathematical relationships defined.

in which the size of the *Island Resources* is on the *X*-axis and the *Birth Rate* is on the *Y*-axis. This graph should show an overall upward trend as the *Island Resources* reservoir increases. We would also expect that the *Birth Rate* would never drop below zero (a physical impossibility!) and that it would never increase beyond some theoretically maximum value (can you explain why?). Hence, the relationship between the *Birth Rate* and the *Island Resources* reservoir would probably look something like the graph in Figure 1.6. The scales on the *X* and *Y* axes in this graph must be specified

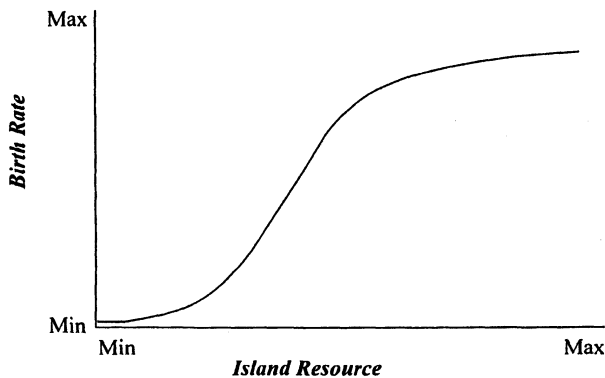


FIGURE 1.6. Suggested graphical relationship between Birth Rate and Island Resources.

by the modeler, based on an understanding of the *Island Resources* as well as an understanding of the reproductive capabilities and tendencies of the shipwrecked tourists.

1.3 Uses of Systems Models

The example model described in the previous section is a rather simple one. We hope, however, that you can see that the shipwrecked tourists on our imaginary island could find many practical uses for a valid and accurate model of their community. In fact, it is often the case that even simple models like the one discussed earlier can yield many useful insights that can in turn guide decision making and policy. To be sure, there are many simple and also some very complex models that are used by researchers and policymakers. How are these models used? Two important uses of these types of models are:

1. To understand the underlying mechanisms dictating how a system works
 - Describe the underlying processes and converters
 - Identify possible mechanisms behind observed cycles and long-term trends
 - Determine how the system maintains stability or identify mechanisms by which its stability is jeopardized
2. To predict future performance of an existing system
 - Project cycles and trends
 - Evaluate the impact of policy options
 - Identify scenarios by which system stability will be jeopardized or restored

1.4 A Systems Approach to Environmental Problems

1.4.1 A Definition of Systems Thinking

We now turn our attention to describing what we mean by *systems thinking*. Our intent is to show how systems thinking differs from other approaches to studying environmental problems. We will then describe a conceptual framework for applying systems thinking to environmental problems in such a way as to integrate scientific principles with the impacts of technology and policy.

A simple example will serve to illustrate some of the distinguishing characteristics of systems thinking. One word of caution: By defining systems thinking we are not implying that you are either a “systems thinker” or that you are an imbecile. The truth is that most of us have already used some sort of systems thinking. For example, if you have ever had to coordinate a large project involving several components and possibly several different people

(like building a house), then you have had to employ some systems thinking.

Suppose that you decided to build a house and that you hired a general contractor named Roger OneStep. You contracted Roger because he was a great finishing carpenter. You had seen some of his work and were particularly impressed with his kitchen cabinets; however, problems soon appeared. Roger did not seem to know where to begin. He was great with cabinets, but he did not understand how all the elements of a new home (e.g., floor plan, materials, heating/cooling systems, etc.) were supposed to fit together. You quickly negotiated a new contract, in which Roger would build and install only your kitchen cabinets. You also identified a new general contractor, Wally WholePlan. Wally made it clear to you that he was not an accomplished finishing carpenter (like Roger), however, he did understand all the components of a successful home construction project. He knew how all the parts of a house fit together to make a dwelling with which you would be pleased.

Wally WholePlan represents the systems thinker in this story. Roger OneStep represents the individual who does not use systems thinking, but who has a thorough understanding of one component of the system of house construction. It is clear from the story that both types of thinking are necessary. It is also clear that these two types of thinking are indeed different from one another. Our emphasis in this text is on developing your systems thinking skills, particularly in the context of environmental modeling and policy analysis. You should keep in mind, however that our focus on systems thinking should not be taken as a de-emphasis on the more specialized type of thinking embodied by Roger OneStep.

We will now discuss six viewpoints and assumptions that characterize systems thinking. Many of these characteristics are not unique to systems thinking; however, all six taken together comprise a powerful approach to analyzing and understanding environmental issues. The six characteristics are:

1. Systems thinking begins with a global description and moves toward the specific. For instance, consider the depleting ozone layer in the upper stratosphere. This ozone layer protects us from ultraviolet radiation, yet some chemicals produced by humans have been causing this protective layer to decrease for decades. The systems thinker might first characterize changes in stratospheric ozone in terms of general processes like “atmospheric convection,” “ozone formation,” and “ozone depletion,” and then move toward a more specific description of each process, as needed. A chemist, on the other hand, might begin by describing in detail the photochemistry behind ozone formation. A meteorologist might begin by describing the atmospheric flows that affect ozone levels. To be a systems thinker, you must first grasp the BIG picture.

2. Systems thinking focuses on dynamic processes. The systems thinker interprets system behavior as the product of possibly numerous underlying processes that are always changing and moving. The systems

thinker recognizes the dynamic processes of the system. For example, in our ozone depletion example, a systems thinker would consider both the present level of ozone concentrations as well as the factors affecting these concentrations and how these factors might change or have changed over time.

3. Systems thinking seeks a closed-loop explanation for how things work. The systems thinker attempts to define the system so that its behavior is dependent on only the elements within the system (i.e., system behavior is not dependent on things outside the system). The systems thinker tries to capture all the important factors in his/her systems model while avoiding unnecessary complexity. Factors that are truly outside the system, or which cause little if any effect on the system, are ignored and not considered.

4. Systems thinking identifies feedback loops. The systems thinker assumes that the flow from cause to effect is not in one direction. According to this thinking, changes at point A in the system will cause changes at point B (and possibly elsewhere), which then cause changes that eventually come back to influence point A again.

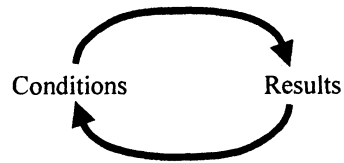
5. Systems thinking looks for checks, balances, and potential for runaway processes. Many systems involve some competing processes or feedback loops that tend to “compete” (e.g., *Birth* processes and *Death* processes). In such cases the system may eventually stabilize around a constant set of conditions. Other systems involve processes that can run “out of control.” The systems thinker seeks to identify those competing or runaway processes, and to understand how they work to affect the overall system.

6. Systems thinking focuses on causal relationships. The systems thinker defines relationships among the elements of the system to reflect true cause-effect relationships. For example, a model that predicts the number of drowning deaths on a given day from the revenues of ice cream sales might give reasonably accurate predictions. This model, however, does not represent a causal relationship (buying ice cream does not cause one to drown!). Hence, the systems thinker would not incorporate such a relationship in his/her model.

An individual who studies environmental problems from a systems perspective is someone who describes what is observed in nature in terms of ever-changing, interdependent processes and conditions. This individual understands the behavior of the environment as coming from the ongoing, dynamic give-and-take between those underlying components. In addition, the systems thinker pays attention to identifying sources of feedback in the system and the conditions under the system will reach a steady state or run out of control.

This approach to understanding environmental problems is facilitated by using the simple modeling constructs of reservoirs, processes, converters, and connectors that were described earlier in this chapter. In addition, it is also clear that using a systems approach requires that we understand the concepts of feedback and steady-state behavior.

FIGURE 1.7. Feedback: A closed-loop circle of cause and effect.



1.4.2 Definition of Feedback

A **feedback loop** in a dynamic system can be defined as a closed-loop circle of cause and effect in which “conditions” in one part of the system cause “results” elsewhere in the system, which in turn act on the original “conditions” to change them. This is represented schematically in Figure 1.7.

Feedback is very common in dynamic systems. For example, consider the island population model introduced in Section 1.2 and reproduced in Figure 1.8. This system includes several feedback loops. One such loop is highlighted in the figure.

The size of the *Island Resources* (a “condition”) affects the *Birth Rate*, which thereby affects the number of births in the *People on the Island* (a “result”). If this causes an increase in the size of the *People on the Island*, then more *Island Resources* will be consumed by the increased number of people on the island. This is shown in the model by the connector running from the *People on the Island* to the *Depletion* process flowing out of the *Island Resources* stock.

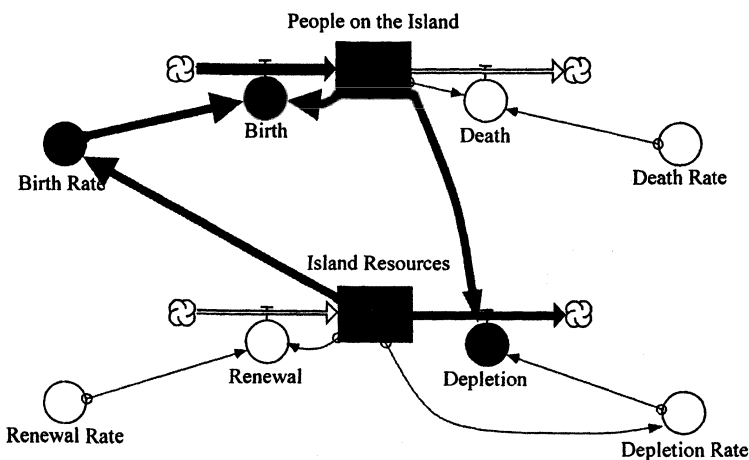


FIGURE 1.8. Island Community system diagram example of feedback highlighted in bold.

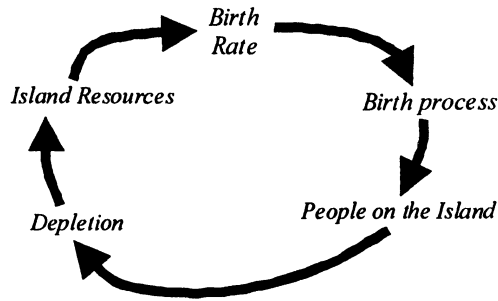


FIGURE 1.9. Example feedback loop in the Island model.

This feedback loop is also shown in a slightly different way in Figure 1.9. The designation of the “conditions” and the “results” in this loop is arbitrary. The important thing is that any node on this loop can be seen to “cause” results at the next node, which in turn eventually come back to affect the original node.

There are two types of feedback loops that can occur. These are (1) *positive feedback* (also called *reinforcing feedback*) and (2) *negative feedback* (also called *counteracting feedback*). Both types are common in the environment. In fact, being able to recognize and distinguish between these two types in a real-life environmental system can lead to significant understandings of how the system works.

1.4.3 Positive Feedback

Positive feedback (also called *reinforcing feedback*) exists whenever changes at one point on a feedback loop eventually work their way back to reinforce or amplify the original change. Such systems tend to eventually run out of control. Many environmental problems are closely associated with naturally occurring positive feedback loops whose influence on the overall system has been accentuated by changes due to human activity.

One example of a positive feedback loop can be found in models of global climate change. It is hypothesized that increases in carbon dioxide (CO_2) emissions into our atmosphere will cause the earth’s global temperature to rise (a phenomenon we will discuss in detail in a later chapter). This in turn will reduce the ability of the earth’s oceans to hold gaseous CO_2 , thereby causing the oceans to release additional CO_2 into the atmosphere. This additional increase in atmospheric CO_2 will lead to further warming, which will then lead to even more CO_2 released from the oceans, and so on. According to this theory, increased CO_2 levels (e.g., from the use of fossil fuels) could lead to a “runaway” accumulation of CO_2 in the atmosphere, thereby leading to increased global temperatures and eventual breakdown of the world’s ecosystems. The diagram for this feedback loop is shown in Figure 1.10.

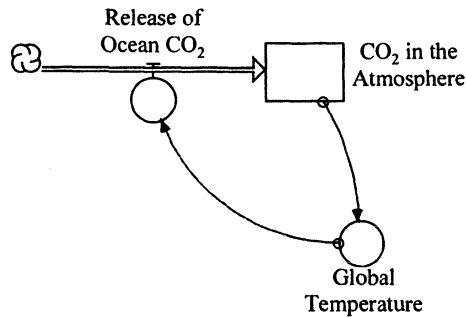


FIGURE 1.10. Global warming positive feedback loop.

1.4.4 Negative Feedback

Negative feedback (also called **counteracting feedback**) exists whenever changes at one point on a feedback loop eventually work their way back through the system to counteract or “damp out” the original change.

Such systems tend to be self-regulating and are not as prone to run “out of control.” Naturally occurring predator-prey ecosystems typically include negative feedback loops. Negative feedback loops help many environmental systems remain stable. In fact, some environmental problems can be attributed to the breakdown of naturally occurring negative feedback loops. Whenever these loops do break down, then the system can lose its stability and can begin to behave in ways that lead to an eventual collapse of the system. For example, take the feedback loop depicted in Figures 1.8 and 1.9. If the size of the *People on the Island* reservoir increases, then the *Island Resources* will be more rapidly consumed. This will in turn reduce the Birth Rate, slowing or even reversing the overall growth of the *People on the Island*. Hence, an initial change at one point the loop (i.e., an increase in the *People on the Island*) eventually works to counteract or “damp out” the original change (i.e., the Birth Rate is reduced and the *People on the Island* either increase more slowly or even decrease).

1.4.5 Steady-State Behavior

Another important type of behavior that occurs in many systems is referred to as **steady-state behavior**. Systems that exhibit steady-state behavior eventually “level off” so that the system reservoirs either change very little or remain constant. When a system finally “levels off” in this fashion, it is said to have **reached steady state**. A system has reached steady state whenever the rates at which its reservoirs change approach zero.

Most environmental systems operate at or near a steady state (i.e., the environment is relatively stable). The underlying systems in the environment

often exhibit just the right mix of positive and negative feedback so that the system never runs “out of control.” It is important to identify those conditions under which a system will exhibit steady-state behavior. In doing so, we can determine those conditions that must be maintained in order for the environment to maintain its remarkable resiliency. On the other hand, we also need to know conditions under which a system will not exhibit steady-state behavior or when it will “run out of control.” In doing so, we can determine what impacts we can have through technology or policies to either upset or help maintain stability in the environment.¹

A reservoir exhibits steady-state behavior whenever a graph of that reservoir's value versus time is a flat (horizontal) line. In other words, whenever a reservoir exhibits steady-state behavior, its rate of change with respect to time is equal to zero. We can use this fact to develop a simple strategy for analyzing the conditions under which a reservoir achieves steady-state behavior. This strategy depends on the use of elementary calculus.

Recall that if $R(t)$ is the value of a quantity at time t , then $\frac{dR(t)}{dt}$ stands for the instantaneous rate at which the quantity $R(t)$ is changing with respect to t . We refer to $\frac{dR(t)}{dt}$ as the **derivative** of $R(t)$ with respect to t . The derivative provides a powerful tool for analyzing the behavior of a reservoir over time. The sign of the derivative indicates whether $R(t)$ is increasing or decreasing over time. Moreover, the larger the magnitude of the derivative, the faster $R(t)$ is changing. For example, if $\frac{dR(t)}{dt} > 0$ at a particular time t , then we know that $R(t)$ is increasing at that point in time. If $\frac{dR(t)}{dt} < 0$, then $R(t)$ similarly is decreasing at that point in time. If $\frac{dR(t)}{dt} = 0$, then $R(t)$ is holding at constant value (at least for an instant). Hence, if $R(t)$ has achieved a steady state after some point in time, then we know that $\frac{dR(t)}{dt} = 0$ during that steady-state period.

This interpretation of the derivative of a reservoir leads to a simple strategy for identifying those conditions under which the reservoir achieves steady-state behavior. This strategy will briefly be described. We will illustrate its use in Chapter 2.

¹ Note that we are using the terms *stability* and *steady state* somewhat interchangeably here. These concepts, however, are not equivalent. For example, many predator-prey populations exhibit oscillatory behavior through time. This behavior is stable (i.e., it does not “run out of control”), but it is not the same as steady-state behavior (i.e., the populations do not hold at constant, unchanging levels). For the purposes of this present discussion, however, this distinction is not important.

1. Develop the systems diagram
2. Use the systems diagram to develop the difference equation for the reservoir of interest
3. Use the difference equation to develop an expression for the derivative of the reservoir with respect to time.
4. Determine conditions under which the derivative is equal to zero.

These four steps require that we develop a mathematical expression for the derivative of each reservoir in the system in order to determine the conditions under which that system will reach steady state. This mathematical expression will be an equation that we will refer to in this text as the **rate equation** for the reservoir. The **rate equation** of reservoir $R(t)$ is a mathematical equation for determining the derivative of $R(t)$. That is, the rate equation will have the general form

$$\frac{dR(t)}{dt} = \dots$$

where the right hand side is some sort of mathematical expression. We find the steady state conditions by finding the expression for the right hand side of the above equation. Then we use basic algebra and common sense to find conditions under which that expression evaluates to zero. This is the process referred to in the four steps given above.

How do we find the rate equation for a reservoir? This is done rather simply by beginning with the difference equation for the reservoir. Recall that the difference equation for a reservoir $R(t)$ is given by Equation (1.1). This equation is reproduced here for clarity.

$$R(t + \Delta t) = R(t) + \{\text{sum of all inflows} - \text{sum of all outflows}\} \Delta t$$

Recall from elementary calculus that the derivative of $R(t)$ is defined by

$$\frac{dR(t)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{R(t + \Delta t) - R(t)}{\Delta t}$$

By subtracting $R(t)$ from both sides of the difference equation given earlier and then dividing by Δt , the difference equation is transformed into the equation

$$\frac{R(t + \Delta t) - R(t)}{\Delta t} = \{\text{sum of all inflows} - \text{sum of all outflows}\}$$

By taking the limit of both sides as Δt approaches zero, we find the following expression for the derivative of $R(t)$.

$$\begin{aligned} \frac{dR(t)}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{R(t + \Delta t) - R(t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \{\text{sum of all inflows} - \text{sum of all outflows}\} \\ &= \{\text{sum of all inflows} - \text{sum of all outflows}\} \end{aligned}$$

Hence, the rate equation for any reservoir $R(t)$ is given by

$$\frac{dR(t)}{dt} = \{\text{sum of all inflows} - \text{sum of all outflows}\}$$

It is often the case that the right hand side of this equation involves fairly complicated expressions. Nonetheless, regardless of the type of system studied, the derivation of the rate equation for any reservoir in the system will have this general form. We will use this approach throughout the text to derive the rate equations and steady state conditions for the systems covered in this text.

1.5 Applying Systems Thinking to Environmental Problems

Given this introduction to systems thinking, this text will now help you apply these skills to a variety of environmental problems. Chapters 4–9 are each devoted to an important environmental issue. The discussion of each issue will always follow along the lines now outlined.

Introduction to a problem context. We will describe a hypothetical but realistic problem scenario that involves a dynamic environmental system to be understood, modeled, and potentially affected by human action or technology. This scenario will provide a motivational context within which we will build and analyze a model.

Introduction to the science underlying the naturally occurring system. Underlying each problem scenario is a naturally occurring system. By *naturally occurring*, we mean that the underlying system would be in operation even if humans were not on the scene. We will provide a brief introduction to the scientific principles behind this natural system.

Development of the systems model. In each case, you will be given a systems diagram similar to the diagram in Figure 1.5 and representing a model found on your CD-ROM. You may also be asked to design your own systems diagram. You will learn how to implement, enhance, and run the model using the STELLA® system modeling software. Your goal at this stage should be to apply each of the six systems thinking skills described in Section 1.4.1 in order to learn how the basic system works. In so doing, you will then be ready to explore the role that technology and policy can potentially play to influence the behavior of the system.

Exploring the model. You will run “experiments” on the systems model using STELLA®. These experiments will be aimed at deepening your understanding of the dynamic behavior of the system, when and why it reaches steady-state behavior, and the impact of the feedback loops that the system possesses.

Exploring the impact of technology and policy on the performance of the system. You will explore several technology options and policy

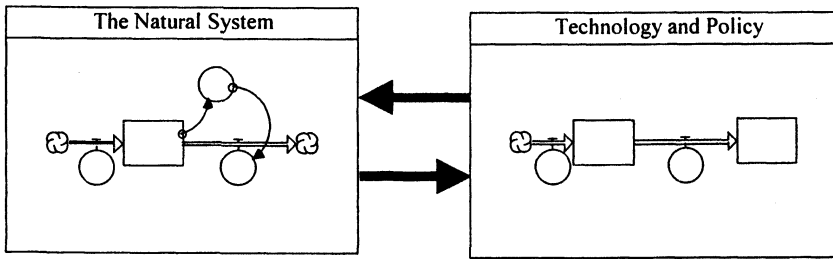


FIGURE 1.11. Environmental systems overview.

options for affecting the underlying natural system. You will develop approaches for incorporating those options into the systems model. It should be noted that, even though policy and technology options are designed to improve system performance, some might be ineffective or even detrimental. In addition, there are often feedback mechanisms whereby the underlying natural system will influence and change the technologies or policies over time. Hence, the entire environmental system may be seen to have at least two major components, as depicted in Figure 1.11.

1.6 Exercises

Section 1.2

1. Modify Figure 1.1 to include flow processes for *immigration* (people moving onto the island) and *emigration* (people leaving the island). Write down the difference equation for the *People on the Island* reservoir to incorporate these new flows.

2. Suppose you wish to build a model for a lake that is fed by one river and three smaller streams that drain a 200 square mile watershed. Your goal is to determine how the turbidity of the lake water will change during a 100-year rainstorm (i.e., a rainstorm so severe that it is expected to occur only once every 100 years). Define two different reservoirs that you would use in a systems model. In addition, identify the flow processes that affect each reservoir. List the reservoirs and their flows and specify the units of each. Do not try to identify any converters or mathematical relationships. Draw a system diagram for each reservoir, similar to Figures 1.1 and 1.2.

3. Write down expressions or equations for the following quantities in Figure 1.5 and briefly justify your answers: (a) *Renewal Rate* (i.e., you will need to specify a constant value), (b) *Renewal*, (c) *Island Resources* (you should specify the difference equation), and (d) *Depletion*.

4. Specify maximum and minimum values for the scales in Figure 1.6. Briefly explain the rationale for your choices. Note that there are no strictly

right or wrong answers to this problem. There are, however, some reasonable and unreasonable answers.

5. Sketch a possible graph for defining how the *Depletion Rate* depends on the *Island Resources* in Figure 1.5. Make sure you specify the maximum and minimum values for the axes in the graph. Write a brief paragraph explaining the rationale for the shape of the graph and the scales of the axes.

6. Open the STELLA® model **CHAP1a.STM** on your CD-ROM. This model corresponds to the diagram in Figure 1.5. Assume that the *Death Rate* = 0.07. Complete the model by filling in the constants, equations, and graphs that you specified in Questions 3–5 (refer to the Appendix at the end of this chapter for an introduction to using STELLA®). In addition, document within STELLA® the units for each model entity. Make a graph showing the *People on the Island* and the *Island Resources* reservoirs over time. Run the model for 50 years, then write a brief paragraph explaining why the system behaviors in the way shown in the graph.

Section 1.4

7. Consider the Global Warming phenomenon in Section 1.4.3. You will find here several descriptions of other system elements that affect the earth's temperature. For each description, create a diagram similar to Figure 1.9 showing the feedback loop that is indicated, then specify if the feedback is negative or positive.

- a. Plants consume CO₂ through photosynthesis. High CO₂ levels have been shown to increase plant growth. This will in turn lead to higher consumption of CO₂.
- b. Increases in global temperatures, will lead to more evaporation of the ocean waters, thereby leading to an increased cloud cover over the earth's surface. This increased cloud cover will increase the earth's reflectivity (called the earth's *albedo*). This increase in albedo will reflect more sunlight away from the earth and allow the earth to "cool off."
- c. As global temperatures rise, the polar ice caps may begin to melt. This will increase the surface area of water on the earth and decrease the surface area of the ice caps. Because water is less reflective than ice, the earth's albedo will decrease, and more sunlight will be absorbed by the earth's surface. Temperatures will climb as this happens.
- d. Increased temperatures from higher atmospheric CO₂ concentrations will cause more people to run their air conditioning units for longer periods of time. This increased demand for energy will necessitate the generation of greater quantities of energy from fossil fuels. Burning fossil fuels pump CO₂ into the atmosphere.

8. Consider the simple population model given in Figure 1.12. Let $P(t)$ stand for the number of people in the *People* reservoir; let $Birth(t)$ stand for

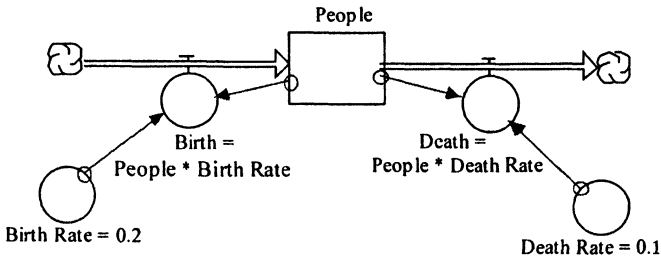


FIGURE 1.12. Model diagram for exercise 8.

the value of the *Birth* process during year t (i.e., $Birth(t)$ = number of births in year t), and let $Death(t)$ stand for the value of the *Death* process during year t . Assume that the *People* reservoir begins at time 0 with 20 people [$P(0) = 20$ people]. Assume also that the *Birth Rate* is equal to 0.2 births/capita/year and the *Death Rate* is equal to 0.1 deaths/capita/year. Calculate the value of $P(t)$ for $t = 0, 0.5, 1, 1.5, 2.5, 3, 3.5$, and 4 years. The first two rows of Table 1.2 are filled in to show how the calculations are done. (*Hint:* Write down the difference equation for calculating $P(t + \Delta t)$ from $P(t)$ and use this to fill in the values in the table. Note that year 0 refers to the first year of the simulation. Hence, the value of $P(0)$ is equal to the initial value defined earlier. The values of $Birth(0)$ and $Death(0)$ are equal to the number of births and deaths, respectively, during the first year).

1.7 Appendix: Getting Around in STELLA®

This section provides a very brief introduction to STELLA®. It is assumed that the reader is familiar with basic interface conventions associated with the MacIntosh® and Windows® platforms. For a more detailed introduction to STELLA®, the reader is referred to the text, *Getting Started with STELLA®*.

TABLE 1.2. Calculations for exercise 8.

t (years)	$p(t)$ (people)	$Birth(t)$ (people/year)	$Death(t)$ (people/year)	$P(t + \Delta t)$ (people)
0	20	4	2	21
0.5	21	4.2	2.1	22.05
1.0				
1.5				
2.0				
2.5				
3.0				
3.5				
4.0				

Software: *A Hands-On Experience*, available from High Performance Systems, Inc.

1.7.1 Some General Background on the STELLA® Environment

STELLA® is a software tool for building dynamic systems models. The user can work on three different levels within STELLA®, depending on the complexity of the problem and the level of detail and organization that is desired in the model. These three levels are:

1. The ***high-level mapping layer***, where the user can design high-level system maps and forms of user interaction, and explore model dynamics. We will use this level only in a limited sense in this text.

2. The ***model construction layer***, where the user sketches a system diagram similar to those introduced in this chapter. A systems diagram displays the flows, reservoirs, converters, and interrelationships (connectors) described in Section 1.2. In addition, the model construction layer is used to specify the mathematical relationships that are used to run the model. An example of a system diagram and the underlying mathematical relationships is shown in Figure 1.5. Most of the exercises and discussions of environmental systems in this text can be handled in the model construction layer. The rest of this appendix will provide an introduction to using the features in this layer.

3. The ***equations layer***, where the user can view the underlying equations in the model and (if desired) make modifications to those equations. This layer is used only by the more advanced STELLA® user and will not be explicitly addressed in this text.

Whenever STELLA® is launched, it automatically starts up in the construction layer; however, you can readily move up to the high-level mapping layer or down to the equations layer simply by clicking on the up or down arrow buttons in the upper left corner of the model building window.

1.7.2 How to Build a Systems Diagram in STELLA®

When STELLA® is launched, a blank model building window is shown. This is the “canvas” on which you will draw a systems diagram. A “palette” of tools (the icon toolbar) that will allow you to graphically construct systems models is above this canvas. This feature makes STELLA® a very user-friendly modeling platform. Users can create a systems diagram on the canvas and STELLA® will transform that diagram into mathematical relationships. In order to build the diagram, you simply “click and drag” various elements onto the “canvas” (i.e., reservoirs, flows, converters, and connectors) and arrange them as you see fit.

The following STELLA® conventions should be kept in mind when creating models on the model-building canvas:

- Reservoirs are placed on the canvas by clicking on the reservoir icon on the toolbar and then clicking on the canvas where you would like the reservoir placed. If you are unhappy with your placement, you can always click and drag the reservoir to another location.
- Flows are created by clicking on the flow icon button on the toolbar and then clicking on the canvas where you want the flow to begin. Drag and release the mouse button at the spot on the canvas where you want the flow to end.
- If you wish to create an *inflow* to a reservoir, you should first place the reservoir on the canvas, then create a flow whose ending point is inside the reservoir. The reservoir will become highlighted when it is ready to receive this flow. When you release the mouse button, the flow will be connected to the reservoir as an inflow.
- If you wish to create an *outflow* from a reservoir, you should first place the reservoir on the canvas, then create a flow that begins inside the reservoir (the reservoir will become highlighted), and drag outside the reservoir to the location where you want the flow to end. When you release the mouse button, the flow will be depicted as an outflow of that reservoir.
- Converters can be placed on the canvas by simply clicking on the converter icon, moving your mouse to the location where you want the converter, and clicking the mouse button.
- Connectors are created in a similar fashion as are flows. Select the connector icon button on the toolbar, click on the system element that will be the “cause,” and drag to the system element that will be the “effect.” When the “effect” is highlighted, you can release the mouse button and the connector will be placed. You can arrange the arc on the connector by clicking on the circle at the originating end of the connector and dragging.
- Once objects are placed on the canvas, you can click and drag them to new locations. All connectors and flows are preserved.
- You can also click once on any object on the canvas (except connectors) and then type in a descriptive name.
- In order to input a text description of an object in the diagram (e.g., to describe its units or physical meaning), you should make sure that the button just below the arrows on the left bar of the model construction window shows an icon of a globe. When you double click on any object on the canvas (except for a connector), you will be presented with a dialogue box that allows you to type in any text description that you wish.
- Mathematical equations (e.g., those for the Birth flow in Figure 1.5) that define how a given flow or converter is calculated cannot be specified until all the connectors into that flow or converter are drawn. In order to

input the mathematical equation for calculating a flow or converter, you should make sure that the button just below the arrows on the left bar of the model construction window shows the icon " χ^2 " (by clicking on that button you can alternate between the globe and the χ^2 icon). When you double click on the flow or converter of interest, you are presented with a "calculator" dialogue that allows you to specify the mathematical expression.

- Objects on the canvas can be deleted by first clicking on the dynamite icon button on the toolbar, then clicking on the object you wish to delete, and releasing the mouse button. That object and all connectors running into or out of it will be deleted. Note that when you place the dynamite over an object and then click and hold, the object to be deleted will be highlighted. If you do not wish to delete what is indicated, just drag the pointer off of the object before you release.

1.7.3 How to Create a Graph or Table in STELLA®

Once the systems diagram and underlying equations are specified in STELLA's® model construction window, you can create a graph or table in which to display output from simulation runs. Click on either the graph or table icon button on the toolbar and then place the graph or table anywhere on the canvas. Double click on the resulting graph or table and fill in the dialogue to specify which variables you want displayed. After you are finished, you can close the graph or table window. An icon will remain on the canvas to indicate what you have created. In order to view the output, just double click on the table or graph icon. You can run models with graph or table windows open, although this may slow the model's execution speed.

1.7.4. Getting Ready to Run the Model

In order to prepare to run the model, you must make some important decisions.

- *For how many time periods should I run the model?* In general, the length of time you run the model depends on the questions that you wish to answer. For example, if you just want to know how large a particular reservoir will be after 20 years, you need to run the model for 20 simulated years. If you wish to know, on the other hand, how long it takes the system to reach steady state, you may need to run the model for a much longer time. In the absence of any other criteria, you will generally want to run the model long enough for the system to pass through the transient startup behavior that may reflect more the starting values you chose for the reservoirs rather than the long-term system behavior. Some trial and error may be necessary to find a reasonable run length. Once you choose a run length, you can specify this in Stella® by select-

ing *Time Specs* under the *Run* menu. Then type in the run length under the *TO* value in the resulting dialogue.

- *How often should the system update the values of the reservoirs, processes, and other system elements?* This is tantamount to deciding how fine a grid of time values you wish to use to update the system values and generate a time plot of system behavior. In Stella® terminology, this decision is embodied in the quantity *DT* in the *Time Specs* dialogue under the *Run* menu. If you choose a *DT* value of 1, then Stella® will use a *DT* value of 1.0 in the difference equation (1.1) and will update the system values once every time unit. If you select a *DT* of 0.25, then Stella® will update the system once every 0.25 time unit (four times per time unit). The more complicated the behavior of your system, the smaller your *DT* value should be. For example, a system whose behavior follows a straight line will not need a small *DT* value, whereas a system whose behavior oscillates up and down will need a much smaller value. Unless otherwise indicated, *DT* values of 0.25 or less are best. In order to see if your *DT* value is small enough, you should make successive runs, using a smaller *DT* value each time. At some point, smaller *DT* values will not make any appreciable difference in the model output. This will help you decide how small *DT* must be in order to get numerically accurate results.
- *What integration method should I use?* The integration method is found under the *Time Specs* dialogue under the *Run* menu. This refers to the specific form of difference equations used to update the model at each *DT* time step. In general, use Euler's method unless you expect your system to oscillate. However, if your system uses conveyors (chapter 3), you should use Euler's method even if it oscillates.

1.7.5 How to Run a STELLA® Model

Once the model is built, you can run the model by clicking on the *Run* menu and selecting *Run*, or by pressing *CTRL-R* on the keyboard. You can also modify the time span over which the model runs and the time step (*DT*) over which the model quantities are evaluated by selecting *Time Specs* under the *Run* menu.

2

Basic Modeling Concepts in Environmental Systems Models

Chapter Objectives—

After you finish this chapter, you should be able to:

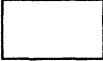
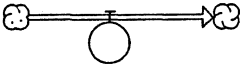
- 1. Describe five **behavior patterns** that are present in many dynamic systems. These five patterns serve as the building blocks for many complicated systems models. The five **behavior patterns** are:*
 - Linear growth or decay*
 - Exponential growth or decay*
 - Logistic growth*
 - Overshoot and collapse*
 - Oscillation*
- 2. Construct and recognize systems diagrams for systems exhibiting each type of behavior.*
- 3. Describe the underlying mathematical relationships behind each type.*
- 4. Identify the types of feedback required to create each type of behavior and determine conditions under which each pattern achieves steady-state behavior.*
- 5. Identify environmental systems where each type of behavior exists.*

2.1 Introduction: Building Blocks for Environmental Systems Models

In Chapter 1 we discussed an imaginary construction project in which you were going to build a new home. Let us suppose that your contractor has the building plans and building materials (i.e., lumber, nails, cement, etc.) and is now ready to begin construction. The builder will combine these materials to construct your home.

Building an environmental systems model is much like building a house. You need some “building materials.” The building materials for a systems model, fortunately, are much simpler than those used for building a house. There are only four basic building blocks. These are the four system components introduced in Section 1.2. Those same components are reproduced in Table 2.1 for your reference. In addition, we have displayed the symbol used to represent each component. These symbols are not universal, but they do reflect modeling conventions used in this text. Every environmental system described in this text can be modeled using only these four “build-

TABLE 2.1. Four systems components and their modeling symbols.

Name	Description	Symbol
Reservoir	A component of a system where something is accumulated. The contents of the reservoir may go up or down over time.	The stock 
Processes	Activities that determine the values of reservoirs over time.	The flow 
Converters	System quantities that dictate the rates at which the processes operate and the reservoirs change.	The converter 
Interrelationships	Define the cause-effect relationships between system elements.	The connector 

ing blocks." In fact, models of very complex systems can be built from these four simple components.

If a homebuilder has access to the right building materials, there is still no guarantee that he or she will build a home that will meet your needs or stand the test of time. The contractor also needs to use good construction principles. For example, the foundation must be sufficiently deep and the floor joists need to be large enough to support the weight of the floor and the occupants.

In the same way, using the four building blocks in Table 2.1 does not guarantee that any model built with them will be correct or even useful. Certain rules of construction must also be followed. We will now outline five rules for building a systems model that must be followed if your model is to be useful and reliable. These have been illustrated in Chapter 1. We state them explicitly here for your reference, and will follow these rules throughout the text to guide our modeling activities. Other modeling principles will be given later to assure that your model is *valid* (i.e., that it accurately represents the real-life system) and is useful for addressing the questions you want to answer.

Rules for Building Systems Models

1. Keep the systems diagram (and hence the model) as simple as possible. Add complexity only as it is needed.
2. Use common-sense mathematical expressions to define the relationships between elements in the system.

3. If a credible mathematical expression is not readily available, define relationships by using graphs.
4. Make sure you identify the units in which time is to be measured in the model. Also identify the units of measurement used for each element in the system, and make sure that the units are compatible with the mathematical expressions you defined under Point 2.
5. Be sure that the only system entities that directly affect the values of a particular reservoir are the inflows and outflows associated with that reservoir.

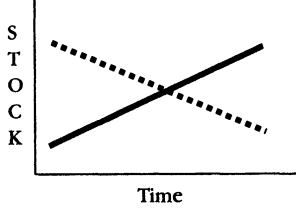
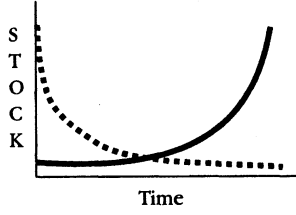
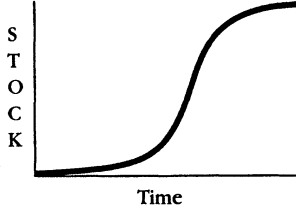
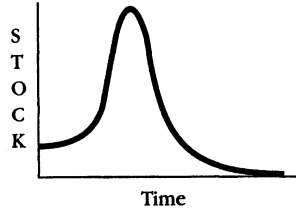
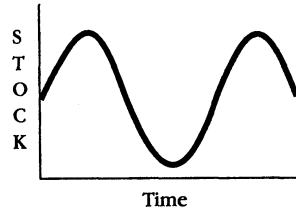
Let us now use our home building example to illustrate how the rest of this chapter is organized. Builders seldom construct a home completely from “scratch.” For example, even though the doors in the home are made from wood, the builder does not actually construct the door on the building site. A builder usually purchases a prefabricated door and frame. In fact, builders have even developed names for the various sizes and styles of door units. By examining the architect’s drawings, the builder identifies how many door units of each type are needed and then orders them from the factory. This is much more efficient (and much less expensive!) than building the doors from scratch.

The same is true about systems models. Many environmental systems have common features. These features tend to lead to certain types of behaviors that can be predicted and easily modeled. Much of the art behind environmental modeling consists of recognizing these features and then using the proper modeling constructs (i.e., the right combinations of stocks, flows, converters, and connectors) to model them. We will refer to these commonly occurring features as *behavior patterns*. By calling something a behavior pattern, we are simply referring to a common pattern of behavior exhibited by one or more of the system reservoirs. The main body of this chapter is devoted to defining and illustrating the five common behavior patterns described in Table 2.2. For each type, the discussion will follow this outline:

1. Illustrative example. We will employ an example of a simple system having one or more reservoirs that exhibit that particular type of behavior. System diagrams and underlying mathematical relationships for the examples will be provided.

2. System features, diagram, and equations. We will identify the underlying features that are necessary for a system to exhibit this type of behavior. A generic system diagram and the underlying mathematical relationships will be given. We will identify the types of feedback present in each case. We will also borrow from the tools of calculus to develop equations that describe the *rate* at which a reservoir changes whenever it exhibits the type of behavior discussed. These equations are called *rate equations*. They will play an important role in describing dynamic systems and in understanding and predicting how such systems behave. The rate equations will

TABLE 2.2. Five common behavior patterns in dynamic systems.

Type of dynamic	Behavior over time
1. Linear growth or decay	
2. Exponential growth or decay	
3. Logistic growth	
4. Overshoot and collapse	
5. Oscillation	

also be used to determine if the system will reach a steady state and under what conditions it will do so.

3. Summary table. We will provide a table to highlight the distinguishing characteristics of each type of behavior and to identify the conditions under which each type of behavior can occur.

By covering all five types of behaviors, you will develop a set of prefabricated “modeling constructs” that you can use (like prefabricated doors) to help you understand and build environmental systems models.

2.2 Behavior Pattern #1: Linear Growth or Decay

2.2.1 Linear Growth or Decay: Illustrative Example

Consider an underground reserve of 10 million barrels of oil that is being consumed at a fixed rate of 10,000 barrels per day. A diagram for this simple system is shown in Figure 2.1. Note that time is measured in days in this model.

It should be obvious that the *Oil Reserves* will begin with a value of 10 million barrels at time zero and will decrease by 10,000 barrels each day until it is finally depleted after 1,000 days. In fact, the equation for calculating the size of the *Oil Reserves* one day ahead is (see Equation 1.1):

$$\text{Oil Reserves Tomorrow} = \text{Oil Reserves Today} - 10,000 \text{ barrels}$$

Note that the preceding equation is applicable only as long as the *Oil Reserves* reservoir has a positive value. In other words, the equation is only applicable for 1,000 days, after which the *Oil Reserves* will be a constant value of zero because it cannot be negative.

If we wanted to know the size of the *Oil Reserves* 12 hours from now, we would subtract only 5,000 barrels (because 12 hours is one half of a day). If we wanted to know the size of the *Oil Reserves* at a point in time n days from now, we would subtract $n \cdot 10,000$ barrels. We can write this in a more general form by letting $R(t)$ represent the size of the *Oil Reserves* at time t (where t is measured in days). The **difference equation** for calculating the size of the *Oil Reserves* at a future time $t + \Delta t$ is given by Equation (2.1). A plot of the behavior of the *Oil Reserves* over time is given in Figure 2.2.

$$R(t + \Delta t) = R(t) - (10,000 \cdot \Delta t) \quad (2.1)$$

In the *Oil Reserves* example, there is a single outflow process that has a constant value of $-10,000$ barrels/day until the reservoir is depleted. The rate of decrease is equal to the value of the outflow process. This rate is depicted graphically as the slope of the line in Figure 2.2.

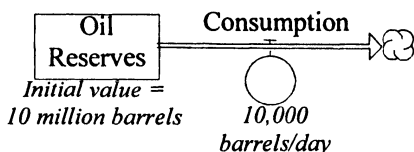


FIGURE 2.1. Oil consumption model.

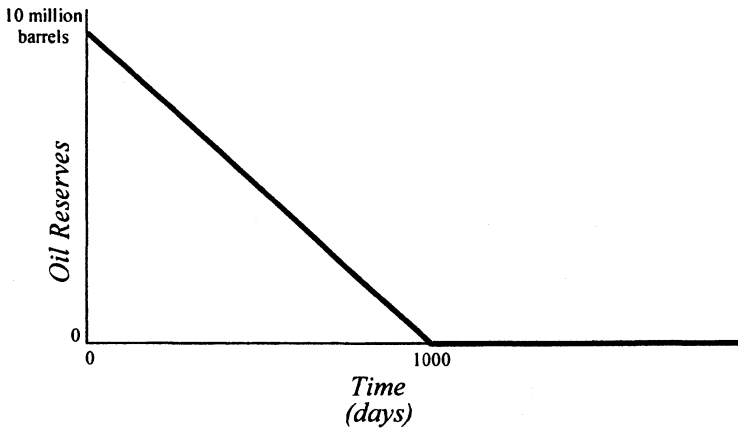


FIGURE 2.2. Oil reserves versus time.

The system in Figures 2.1 and 2.2 (prior to $t = 1,000$ days) illustrates the **linear decay** behavior pattern. A system undergoing **linear growth** would also follow a straight-line plot, but would slope upward instead of downward. In order for a reservoir to exhibit linear growth or decay, *the sum of all the inflows into the reservoir, minus the sum of all its outflows must be constant*. This can happen whenever each of the inflow and outflow processes affecting the reservoir is constant. In addition, one of the end-of-chapter problems demonstrates that linear growth or decay can also occur even if some of the inflows and outflows are not constant.

2.2.2 Linear Growth or Decay: System Features, Diagram, and Equations

A linear behavior pattern is one in which the reservoir of interest changes at a constant rate over time. In order for a reservoir to exhibit linear growth or decay, therefore, the sum of all the inflows into the reservoir, minus the sum of all its outflows must be constant. If the constant is positive, then the system will display linear growth. If the constant is negative, then the system will exhibit linear decay. If the constant is equal to zero, then the reservoir will remain constant through time. A generic linear system with multiple inflows and outflows is shown in Figure 2.3.

Note that such a system can have any number of inflows and outflows. In addition, it is not necessary for each of the flows to be constant through time in order for the system to exhibit linear growth or decay. It is necessary, however, that the sum of the inflows, minus the sum of the outflows be constant. In other words,

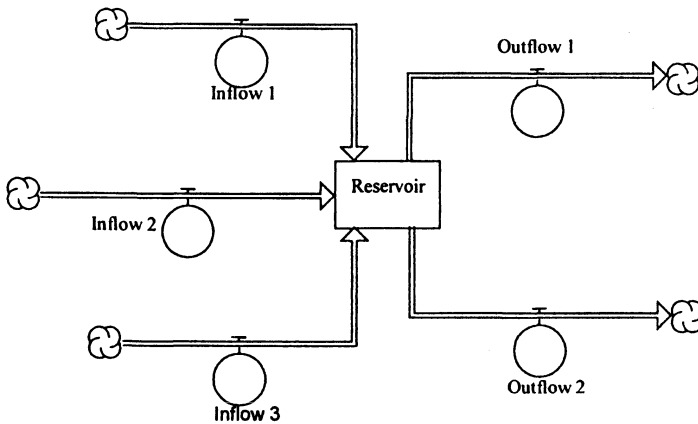


FIGURE 2.3. Generic system diagram for linear growth or decay.

Linear growth or decay occurs if and only if changes in the reservoir over an interval from time t to $t + \Delta t$ are constant for all t . If this change is positive, the reservoir will exhibit **linear growth**. If the change is negative, the reservoir will exhibit **linear decay**.

This property is demonstrated for the *Oil Reserves* model. A simple rearrangement of the terms in Equation (2.1) yields the following expression for the total net change in the *Oil Reserves* in a time interval of Δt days.

$$R(t + \Delta t) - R(t) = -10,000 \cdot \Delta t \quad (2.2)$$

Because this change does not depend on it and is negative the *Oil Reserves* model exhibits linear decay for 1,000 days. After $t = 1,000$ days, $R(t)$ remains at a value of zero.

A careful examination of the system diagram in Figure 2.3 demonstrates that the simple linear system does not contain any kind of feedback. Remember that a feedback loop will always cause an initial change in the system to eventually be either damped out (in the case of **counteracting feedback**) or amplified (in the case of **reinforcing feedback**). Because a linear system changes at a constant rate, no damping or amplification can occur; therefore, feedback is absent.

To generalize the linear model mathematically, let $R(t)$ stand for the value of the reservoir at time t in Figure 2.3. The difference equation for $R(t)$ for this generic system is given by the following:

$$R(t + \Delta t) = R(t) + \{(\text{Inflow}_1 + \text{Inflow}_2 + \text{Inflow}_3) - (\text{Outflow}_1 + \text{Outflow}_2)\} \cdot \Delta t \quad (2.3)$$

Because all of the inflows and outflows in this equation are constant values, the change in the reservoir from time t to time $t + \Delta t$ (given by the bracketed term on the right side of Equation (2.3)) will also be constant. This equation can be generalized for any number of inflow or outflow processes.

We understand from calculus that the derivative of $R(t)$ with respect to time stands for the rate at which the value of $R(t)$ changes over time. Hence, the derivative of any reservoir exhibiting a linear behavior pattern must be a constant; therefore, if we let $\frac{dR}{dt}$ stand for the derivative of the reservoir with respect to time, the linear decay model can be represented as:

$$\frac{dR(t)}{dt} = k \quad (2.4)$$

where $R(t)$ is the value of the reservoir at time t , and where k is a constant. If k is positive, the system will exhibit linear growth. If k is negative, the system will exhibit linear decay. In addition, the value of k is also equal to the slope of the graph of $R(t)$ versus time, as depicted in Figure 2.4.

The rate constant k also has a direct relationship to the inflow and outflow processes in Figure 2.3. From the difference Equation (2.3) we can write

$$R(t + \Delta t) - R(t) = \{(\text{Inflow}_1 + \text{Inflow}_2 + \text{Inflow}_3) - (\text{Outflow}_1 + \text{Outflow}_2)\} \cdot \Delta t$$

Dividing both sides by Δt gives the following.

$$\frac{R(t + \Delta t) - R(t)}{\Delta t} = \{(\text{Inflow}_1 + \text{Inflow}_2 + \text{Inflow}_3) - (\text{Outflow}_1 + \text{Outflow}_2)\}$$

By taking the limit of both sides as Δt goes to zero, we obtain the derivative of $R(t)$ as follows.

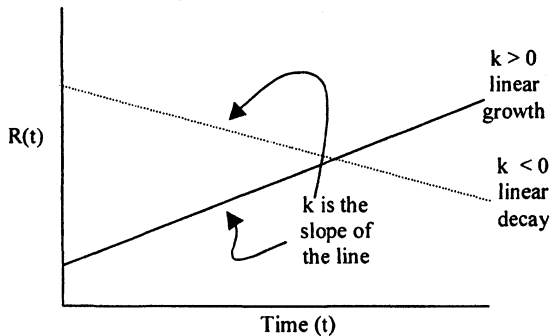


FIGURE 2.4. Linear growth or decay: $\frac{dR(t)}{dt} = k$.

$$\lim_{\Delta t \rightarrow 0} \frac{R(t + \Delta t) - R(t)}{\Delta t} = \frac{dR(t)}{dt} = \{(\text{Inflow}_1 + \text{Inflow}_2 + \text{Inflow}_3) - (\text{Outflow}_1 + \text{Outflow}_2)\} \quad (2.5)$$

A comparison of Equation (2.4) with Equation (2.5) reveals that the rate constant k is equal to $\{(\text{Inflow}_1 + \text{Inflow}_2 + \text{Inflow}_3) - (\text{Outflow}_1 + \text{Outflow}_2)\}$. In general, when there are several inflow and outflow processes for a system exhibiting linear growth or decay, the following relationship holds between the rate constant k in Equation (2.4) and the inflow and outflow processes:

$$k = (\text{sum of all inflows}) - (\text{sum of all outflows}) \quad (2.6)$$

Consider the *Oil Reserves* example in Figure 2.1. Because there is only one outflow and no inflows into the reservoir, the right side of Equation (2.6) simplifies to

$$k = -10,000 \text{ barrels/day.}$$

This implies that the *Oil Reserves* reservoir will follow a linear decay. The graph of the *Oil Reserves* versus time will display a straight line with a downward slope equal to $-10,000$ barrels/day.

In order to analyze the steady-state behavior of a linear system, we will give a more detailed analysis of the derivative of the reservoir in such a system. Recall from Section 1.4.5 that whenever a reservoir $R(t)$ exhibits steady-state behavior, its derivative with respect to time will be zero; that is, $\frac{dR(t)}{dt} = 0$. For the linear growth/decay system, we have already shown in

Equations (2.4) and (2.6) that $\frac{dR(t)}{dt}$ is a constant value k , where k is equal to the difference in all inflows minus all outflows associated with that reservoir. A linear system, therefore, cannot achieve steady state unless $k = 0$. This occurs only if the sum of all inflows minus the sum of all outflows is zero. This happens only if the system is *always* at steady state. The only exception to this is whenever there is some external constraint on the reservoir that does not allow it to either exceed or fall below some predetermined value. In the case of the *Oil Reserves* example, we know ahead of time that the *Oil Reserves* cannot fall below zero. Because all of the original *Oil Reserves* are depleted after day 1,000, the *Oil Reserves* must necessarily remain at a constant value of zero after that time. In this case, the *Oil Reserves* are not at a steady state for the first 1,000 days and are then at a steady state after 1,000 days.

2.2.3 Linear Growth and Decay: Summary

Table 2.3 shows a summary of the linear growth and decay behavior pattern. You may have noticed that this table includes something called the **solution**

TABLE 2.3. Defining characteristics of linear growth or decay.

Description	The reservoir increases at a constant rate (linear growth) or decreases at a constant rate (linear decay).
Rate equation	$\frac{dR(t)}{dt} = k, \text{ where } k \text{ is a constant.}$ • k is the slope of the graph of $R(t)$. • $k > 0$ leads to linear growth; $k < 0$ leads to linear decay • k gives the net change in the value of $R(t)$ per unit of time. • $k = (\text{sum of all the inflows}) - (\text{sum of the outflows})$
Solution to the rate equation	$R(t) = R_0 + kt$, where R_0 is the initial value of $R(t)$ at time $t = 0$.
Graphical behavior	Graph of $R(t)$ versus t is a straight line
Steady-state behavior	None, unless $k = 0$, or (in the case of linear decay) the reservoir reaches zero and cannot go below zero.
Example applications	Resource management with constant inflows and outflows

to the rate equation. This represents the mathematical expression for $R(t)$ that satisfies the requirements implied by the rate equation.

The rate equation for a linear growth/decay system says that the rate at which the reservoir $R(t)$ changes is constant. This means that a graph of $R(t)$ versus time must follow a straight line. Hence, we know that $R(t)$ must have the form

$$R(t) = a + bt \quad (2.7)$$

where b is the rate of change and a is the value of $R(t)$ at time $t = 0$. Substituting k for b and R_0 for a in Equation (2.7) gives the expression in Table 2.3.

2.3 Behavior Pattern #2: Exponential Growth or Decay

2.3.1 Exponential Growth or Decay: Two Illustrative Examples

The linear growth/decay behavior pattern is very simple, but its applicability is limited. Most naturally occurring systems do not change at a constant rate. Two examples are used to illustrate this.

Suppose that a young child in your household brought home a pair of white mice (one male, one female) that her science teacher gave to her. Further imagine that (due to poor game management practices), the mice escaped from their cage. Your family would probably see few (if any) white mice at first. After a period of time, however, these two mice might mate and bear some offspring. Within a short period of time, this second generation

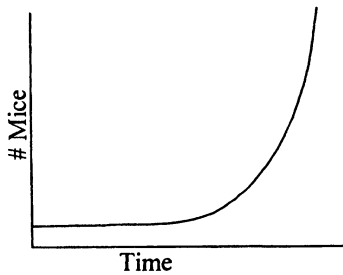


FIGURE 2.5. Population of mice versus time.

of mice would be ready to mate and bear offspring. The number of mice born over time would increase as the number of mice available to bear offspring increased. Hence, the white mouse population would grow slowly at first, and then more rapidly as the number of mice increased. A graph of the number of mice versus time might look something like Figure 2.5.

The graph demonstrates that the rate of growth of the mouse population is not constant. This system does not exhibit linear growth. We will show that this is an example of *exponential growth*.

The system diagram and equations for the mouse population are given in Figure 2.6. Note that we have specified a birth rate of 1.1 mice per capita per month. This means that (on the average) there will be 11 mice born every month for every 10 mice in the population. This is admittedly a very prolific mouse population! In addition, we have specified a death rate of 0.08 mice per capita per month. This means that 8 of every 100 mice in the population will die each month. It would be important to obtain realistic values of the birth and death rates before using the model to make predictions. We will use these rather arbitrary values right now, however, to illustrate how this model works.

Suppose we let $W(t)$ represent the number of white mice in the population. Based on what we learned in Chapter 1, we know that the equation for

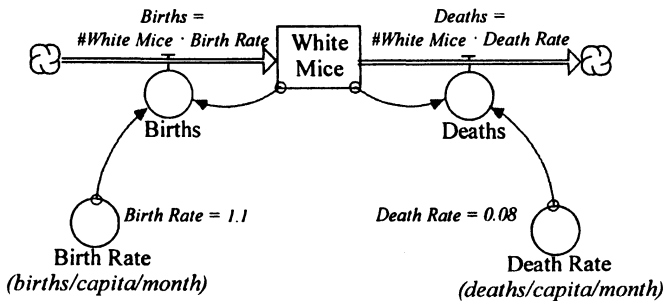


FIGURE 2.6. System diagram for the mouse population.

$W(t + \Delta t)$, the number of white mice at a future time $t + \Delta t$ is determined by the following *difference equation*.

$$\begin{aligned}
 W(t + \Delta t) &= W(t) + \{(\text{sum of all inflows}) - (\text{sum of all outflows})\} \cdot \Delta t \\
 &= W(t) + \{\text{Births} - \text{Deaths}\} \cdot \Delta t \\
 &= W(t) + \{\text{Birth Rate} \cdot W(t) - \text{Death Rate} \cdot W(t)\} \cdot \Delta t \\
 &= W(t) + \{1.1 - 0.08\} \cdot W(t) \cdot \Delta t \\
 &= W(t) + \{1.02\} \cdot W(t) \cdot \Delta t
 \end{aligned} \tag{2.8}$$

For another example, imagine a perfectly cylindrical bucket full of water, with a small outlet valve at the very bottom (see Figure 2.7). If the valve is opened, water will flow from the bucket onto the ground. The rate at which the water flows out will be greater at the beginning because the pressure of the water still in the bucket pushes down, forcing water out. As the water volume in the bucket decreases, however, the downward pressure lessens. Hence, the rate at which the water will flow from the bucket will also decrease. A graph of the water volume (in cubic centimeters) versus time would look like Figure 2.7.

This is an example of *exponential decay*. The rate at which the water volume changes is proportional to the amount of water still in the bucket. For example, the water will flow out twice as fast when there are 1,000 cc of water in the bucket as it will when there are 500 cc of water. Over time, the remaining water volume decreases more slowly and eventually (*asymptotically*) approaches zero. The system diagram for this example is given in Figure 2.8. Note that we have again picked an arbitrary value for the *Flow Rate*. This value can (and should!) be determined from other research or even from experimentation, if necessary. We use this arbitrary value now only to illustrate the behavior of the model. The Flow Rate is expressed as the volume of water exiting the bucket per second (in cubic centimeters per second) for every cubic centimeter of water still in the bucket. The specific value of 0.1 cc/sec/cc indicates that if there are 100 cc of water in the bucket, then 10 cc will flow out over the next second. If there are only 10 cc of water in the bucket, then only 1 cc will flow out over the next second. This

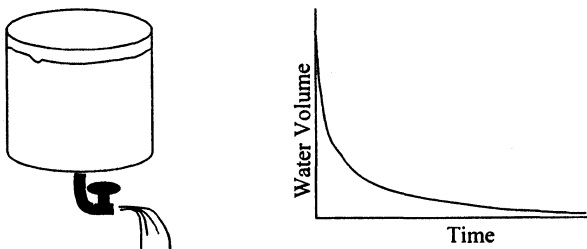


FIGURE 2.7. Exponential decay water example.

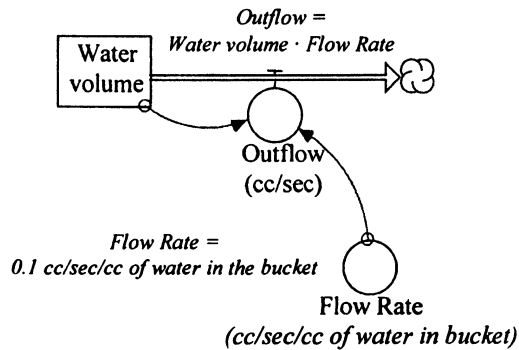


FIGURE 2.8. System diagram for the water bucket.

corresponds to the original description of this system in which it was noted that water flows out at a slower rate whenever there is less water in the bucket to help “push” it out.

These two examples illustrate the fundamental feature of systems exhibiting exponential growth or decay.

Exponential growth (or decay) occurs if and only if the reservoir increases (or decreases) at a rate that is proportional to its size. If the reservoir is increasing in size, then the system exhibits *exponential growth*. If the reservoir is decreasing in size, then the system exhibits *exponential decay*.

2.3.2 Exponential Growth or Decay: System Features, Diagram, and Equations

Figure 2.9 shows a generic example of a simple exponential growth/decay system. Make sure that you can identify the differences between this type of system and the simple linear system displayed in Figure 2.3. In particular, the flow processes in Figure 2.9 all operate at rates that are proportional to the current size of the reservoir. This is different than the linear system, where

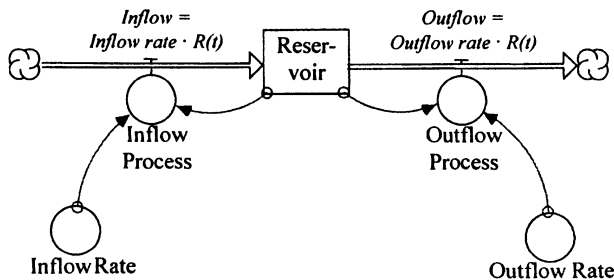


FIGURE 2.9. Generic system diagram for exponential growth or decay.

the net rate of the inflow and outflow processes is constant, independent of the size of the reservoir.

In addition, note that the exponential system has **feedback loops**. Figure 2.9 has two such loops. The first loop includes the inflow process and the reservoir. If the inflow process increases, then the reservoir will also increase, which in turn drives the inflow to a higher level. This is a **reinforcing (positive) feedback** loop. The other loop in Figure 2.9 involves the reservoir and the outflow process. This loop is a **counteracting (negative) feedback** loop.

The **difference equation** for the reservoir $R(t)$ in Figure 2.9 is

$$\begin{aligned} R(t + \Delta t) &= R(t) + \{ \text{Inflow Rate} \cdot R(t) - \text{Outflow Rate} \cdot R(t) \} \cdot \Delta t \\ &= R(t) + \{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot R(t) \cdot \Delta t \end{aligned} \quad (2.9)$$

The term $\{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot R(t) \Delta t$ on the right side of Equation (2.9) demonstrates that the change in $R(t)$ from time t to $t + \Delta t$ is proportional to $R(t)$. The constant of proportionality is $\{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot \Delta t$.

If we subtract $R(t)$ from both sides and then divide by Δt , we get

$$\frac{R(t + \Delta t) - R(t)}{\Delta t} = \{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot R(t)$$

Taking the limit of both sides as Δt goes to zero gives the derivative of $R(t)$:

$$\begin{aligned} \lim_{\Delta t \rightarrow 0} \frac{R(t + \Delta t) - R(t)}{\Delta t} &= \{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot R(t) \\ \frac{dR(t)}{dt} &= \{ \text{Inflow Rate} - \text{Outflow Rate} \} \cdot R(t) \\ &= k \cdot R(t) \end{aligned}$$

where $k = \{ \text{Inflow Rate} - \text{Outflow Rate} \}$. Hence, the rate equation for the reservoir $R(t)$ in the exponential system in Figure 2.9 is

$$\frac{dR(t)}{dt} = kR(t), \text{ where } k = \{ \text{Inflow Rate} - \text{Outflow Rate} \} \quad (2.10)$$

It can be shown that the solution to this rate equation is

$$R(t) = R_0 e^{kt} \quad (2.11)$$

where R_0 = the value of $R(t)$ at time $t = 0$, and where k is given in Equation (2.10). This is left as an exercise for the reader.

The following facts about the rate constant k will aid in its interpretation. Figures 2.10a and 2.10b graphically illustrate these concepts.

Interpreting the Rate Constant k in the Exponential System

1. Recall from Equation (2.10) that $k = \text{Inflow Rate} - \text{Outflow Rate}$. Hence, k is the **net growth (or decay) rate** in the system.

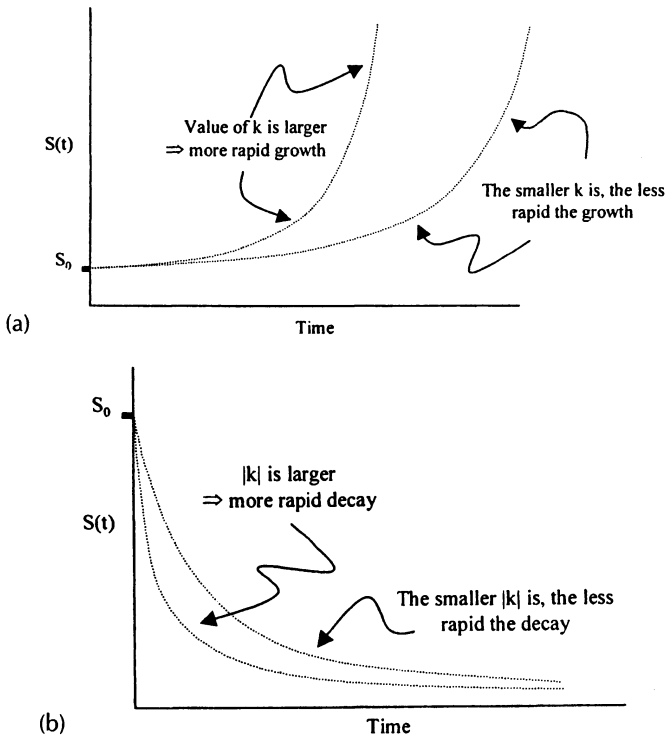


FIGURE 2.10. (a) Graphical interpretation of the rate constant k in an exponential growth system ($k > 0$). (b) Graphical interpretation of the rate constant k in an exponential decay system ($k < 0$).

2. If $k > 0$ then the system will exhibit **exponential growth**.
3. If $k < 0$ then the system will exhibit **exponential decay**.
4. The **ratio of increase (if k is positive) or decrease (if k is negative)** in the reservoir over one unit of time is e^k . In other words, after time advances one unit, the reservoir will change from $R(t)$ to $(e^k) \cdot R(t)$.
5. The **larger $|k|$ is, the more rapid the growth or decay**.

Recall that if any reservoir $R(t)$ exhibits steady-state behavior after a time t_0 , then the derivative of $R(t)$ will also be zero at all times after t_0 . We know from Equation (2.10) that the rate equation for an exponential system is

$\frac{dR(t)}{dt} = k \cdot R(t)$. It is easy to see that this derivative can be zero only if $k = 0$ or if $R(t) = 0$. Both of these situations are of no practical value because they correspond to the situation in which $R(t)$ is a constant value for all t (a very easy system to model!). With an exponential decay model, however, $R(t)$ does get closer and closer to zero the longer the system runs. As this happens, the

TABLE 2.4. Defining characteristics of exponential growth and decay.

Description	The reservoir increases at a rate that is proportional to its current size.
Rate equation	$\frac{dR(t)}{dt} = kR(t)$, where k is a constant. • k is the net growth rate or net decay rate of $R(t)$ • $k = \text{Inflow Rate} - \text{Outflow Rate}$ • If $k > 0$ then the system will exhibit exponential growth • If $k < 0$ then the system will exhibit exponential decay • After time advances one unit, the reservoir will change from $R(t)$ to $e^k \cdot R(t)$. • The larger k is, the more rapid the growth or decay
Solution to the rate equation	$R(t) = R_0 e^{kt}$, where R_0 is the initial value of $R(t)$ at time $t = 0$.
Graphical behavior	Exponential growth: Graph of $R(t)$ increases slowly at first and then more rapidly as time passes Exponential decay: Graph of $R(t)$ decreases rapidly at first, and then more slowly as time passes; Graph of $R(t)$ eventually approaches a horizontal asymptote.
Steady state solution	Steady state of $\bar{R} = 0$ occurs only in the exponential decay system as $t \rightarrow \infty$.
Example applications	Population dynamics; heat transfer; fluid dynamics

derivative of $R(t)$ also gets very close to zero and the graph of $R(t)$ versus t approaches a horizontal asymptote. We will use the notation $\bar{R}$ to stand for the steady-state value that the reservoir $R(t)$ achieves (or approaches asymptotically). Hence, based on the preceding discussion, we can summarize these observations as follows:

An exponential system will never exhibit perfect steady-state behavior; however, if the system involves exponential decay, then $R(t)$ will asymptotically approach a steady-state value of $\bar{R} = 0$ the further out in time we go.

2.3.3 Exponential Growth or Decay: Summary

Table 2.4 summarizes important concepts for the exponential growth and decay system.

2.4 Behavior Pattern #3: Logistic Growth

2.4.1 Logistic Growth: Illustrative Example

An example graph of a logistic behavior pattern is given in Figure 2.11. It is clear from the graph why this behavior pattern is sometimes referred to as

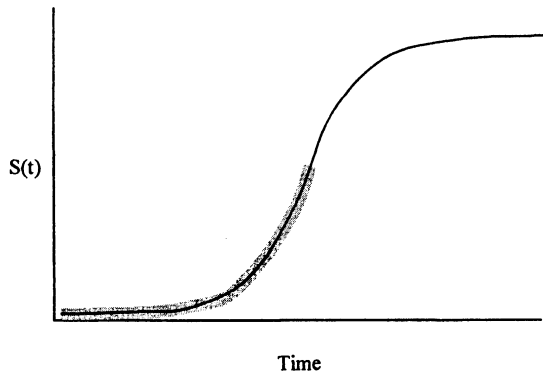


FIGURE 2.11. The logistic S-curve.

the “S-curve.” An examination of Figure 2.11 suggests that a logistic system has some similarities with an exponential system. Notice in particular the section of the figure that is highlighted. This part of the curve looks like an exponential growth system. Unlike the exponential growth system, however, the growth in a logistic system eventually levels out and the system approaches a steady-state value. The underlying mechanisms that force this leveling off are what distinguish a logistic system from an exponential system.

Consider how a human population grows. Early on, whenever the resources necessary for survival are plentiful, and no other constraints on the reproductive choices of the people exist, the population grows exponentially. As time passes and the population gets very large, however, the resources available to support that population begin to be “stretched” to the point that further unchecked growth will lead to starvation, overcrowding, and disease. The death rate will likely begin to climb until it finally reaches a level that is the same as the birth rate. As this happens, the growth of the population will slow and eventually level off.

One way of describing the circumstances under which logistic behavior occurs is as follows. Logistic growth occurs whenever an exponential system is constrained so that the reservoir achieves a maximum level that is sustainable by the system. In such a case, the reservoir increases at a rate that is initially similar to an exponential system. As the reservoir approaches that maximum sustainable level, however, the rate of growth decreases and the system approaches a steady state.

2.4.2 Logistic Growth: System Features, Diagram, and Equations

A generic diagram for a logistic system is given in Figure 2.12. This Figure corresponds to the description given in Section 2.4.1. Notice from Figure

2.12 that the *Inflow* process operates in exactly the same way as in an exponential system: The size of the *Inflow* at any point in time is proportional to the current size of the reservoir. The *Outflow* process, however, does not operate the same way as in an exponential model. Notice that the size of the *Outflow* is determined by the equation

$$\text{Outflow} = R(t) \cdot \text{Unconstrained Growth Rate} \cdot \frac{R(t)}{\text{Carrying Capacity}} \quad (2.12)$$

The *Carrying Capacity* converter in this model stands for the maximum size of the reservoir (i.e., the human population in our example) that can be sustained by the system. What is the proper value for the *Carrying Capacity*? This depends on the nature of the system that is being modeled. In our example involving a human population, the *Carrying Capacity* represents the maximum number of individuals that the system can support over the long term.

Equation (2.12) looks similar to the equation for the outflow in an exponential system. Recall that the outflow to an exponential system is calculated as:

$$\text{Outflow} = R(t) \cdot \text{Outflow Rate} \quad (2.13)$$

A comparison of Equation (2.13) with Equation (2.12) reveals that the logistic system calculates the Outflow by multiplying $R(t)$ by a “rate” that changes over time. That “rate” is given by the expression

$$\text{Outflow Rate} = \text{Unconstrained Growth Rate} \cdot \frac{R(t)}{\text{Carrying Capacity}} \quad (2.14)$$

Consider how this rate will behave in our example involving a human population. When the population is small compared with the *Carrying Capacity*

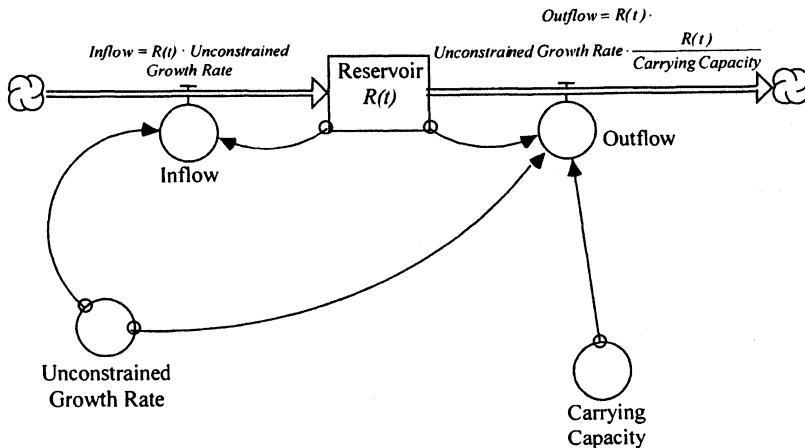


FIGURE 2.12. Generic system diagram for a logistic system.

of the system, the ratio $\frac{R(t)}{\text{Carrying Capacity}}$ will initially be close to zero. Hence, the *Outflow Rate* given in Equation (2.14) will be very small. This means that the inflow (i.e., the “births”) will exceed the outflow, and the system will grow exponentially; however, as the system progresses and the population approaches the *Carrying Capacity*, the ratio $\frac{R(t)}{\text{Carrying Capacity}}$ will get closer to 1.0 and the *Outflow Rate* in Equation (2.14) will increase and approach the *Unconstrained Growth Rate*. Whenever this happens, the number of “deaths” will be very close to the number of “births,” and the population’s growth will slow down.

Notice also that the logistic system in Figure 2.12 includes both reinforcing and counteractive feedback loops. It will be left as an exercise to identify the feedback in this system.

The difference equation for the reservoir in Figure 2.12 is given by

$$\begin{aligned} R(t + \Delta t) &= R(t) + \{\text{Inflows} - \text{Outflows}\} \cdot \Delta t \\ &= R(t) + \{\text{Unconstrained Growth Rate} \cdot R(t) - \text{Outflow Rate} \cdot R(t)\} \cdot \Delta t \end{aligned} \quad (2.15)$$

The rate equation for the reservoir is given by

$$\frac{dR(t)}{dt} = k(t) \cdot R(t),$$

where

$$k(t) = \text{Unconstrained Growth Rate} \cdot \left\{ 1 - \frac{R(t)}{\text{Carrying Capacity}} \right\} \quad (2.16)$$

A comparison of Equation (2.16) with the rate equation for the exponential system Equation (2.10) reveals some important similarities between the exponential and logistic systems. Similar to the exponential system, the rate at which $R(t)$ changes is proportional to the current size of $R(t)$. Unlike the exponential system, however, the proportionality constant in the logistic system [i.e., $k(t)$] changes with time. In fact, if the initial value of $R(t)$ (i.e., R_0) is much smaller than the *Carrying Capacity*, then Equation (2.16) shows that $k(t)$ will be close to the *Unconstrained Growth Rate* and the system will initially behave like an exponential growth system. As time passes and $R(t)$ grows to values that approach the *Carrying Capacity*, $k(t)$ will approach zero, and the rate at which $R(t)$ grows will level off toward zero (no growth).

On the other hand, if R_0 is much larger than the *Carrying Capacity*, then $k(t)$ will start off with large negative values. Hence, $R(t)$ will begin to *decrease* rapidly toward the *Carrying Capacity*. As $R(t)$ shrinks to values near the *Carrying Capacity*, $k(t)$ will again approach zero, and $R(t)$ will shrink much more slowly until it finally levels off at the *Carrying Capacity*.

A common-sense understanding of the logistic system suggests that it will reach a steady state whenever the reservoir $R(t)$ approaches the *Carrying*

Capacity of the system. This is easily confirmed by examining the rate Equation (2.16). Recall that if the system reaches a steady state, then the derivative of $R(t)$ will be zero. That is, $R(t)$ reaches steady state if $\frac{dR(t)}{dt} = 0$, or, equivalently, if

$$\text{Unconstrained Growth Rate} \cdot \left(1 - \frac{R(t)}{\text{Carrying Capacity}}\right) \cdot R(t) = 0$$

This condition is achieved if and only if one of the following is true.

$$\text{Unconstrained Growth Rate} = 0, \text{ or}$$

$$R(t) = 0, \text{ or}$$

$$1 - \frac{R(t)}{\text{Carrying Capacity}} = 0$$

We can assume that the *Unconstrained Growth Rate* > 0 (i.e., the reservoir will grow if there is an unlimited *Carrying Capacity*). We can also assume that $R(t) > 0$ (the reservoir is not empty). Hence, the only possible

way for $\frac{dR(t)}{dt}$ to be zero and remain zero for all later values of t , is if

$$1 - \frac{R(t)}{\text{Carrying Capacity}} = 0 \text{ or, equivalently, } R(t) = \text{Carrying Capacity. Thus,}$$

we can see that the steady-state solution to the logistic system is $\bar{R} = \text{Carrying Capacity}$.

This makes sense whenever we consider that the *Carrying Capacity* represents the maximum reservoir size that can be sustained by the system. Based on the discussion in the previous paragraph, $R(t)$ will either grow or shrink toward this *Carrying Capacity*, depending on the size of the reservoir at the beginning.

The solution to the rate Equation (2.16) is

$$R(t) = \frac{\text{Carrying Capacity}}{1 + Ae^{-\text{Unconstrained Growth Rate} \cdot t}},$$

$$\text{where } A = \frac{\text{Carrying Capacity} - R_0}{R_0} \quad (2.17)$$

An analysis of Equation (2.17) confirms that $R(t)$ will behave so that it is driven to its *Carrying Capacity* value. Figure 2.13 illustrates this behavior for the case when R_0 is less than the *Carrying Capacity*. Make sure that you convince yourself that Figure 2.13 is consistent with the preceding description and that it matches the behavior you would expect from Equation (2.17).

2.4.3 Logistic Growth: Summary

Table 2.5 summarizes the logistic growth behavior pattern.

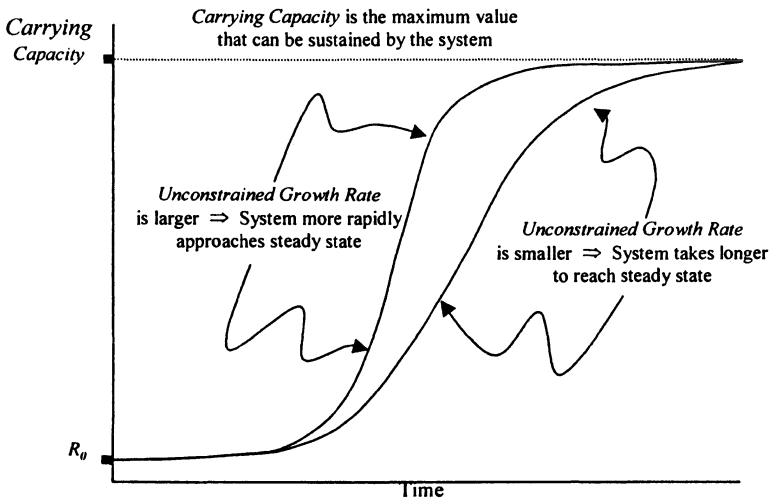


FIGURE 2.13. Graphical interpretation of the logistic system.

2.5 Behavior Pattern #4: Overshoot and Collapse

2.5.1 Overshoot and Collapse: Illustrative Example

Over the past several decades, there has developed an increased awareness of our culture's excessive consumption of petroleum. This nonrenewable resource (i.e., nonrenewable over any realistic time frame) is a source of energy and consumer products. The use of petroleum and petroleum products touches nearly every person's life in some significant way. If the global oil reserves should be completely used up, many life-critical products, services, and resources would be lost or jeopardized.

There are many historical examples of societies or animal species that have disappeared because of the loss of resources that were critical to their survival. Any system in which a population is dependent on a nonrenewable resource for survival is subject to a potential collapse as that resource is depleted. A population dependent on a *renewable* resource may also collapse if that resource is consumed at a rate much greater than the resource's rate of renewal. This latter situation is particularly true if the population's overconsumption of the resource reduces the overall ability of the resource to renew itself. An example might be cattle overgrazing in a field. If too many cattle graze on such a field, all the grass might be eaten, and the soil may be damaged to a point that the growth of new grass is limited. The new grass will only be able to support a smaller population of cattle than it would have initially.

We refer to this kind of system as an **overshoot and collapse system**. Figure 2.14 illustrates the behavior of an overshoot and collapse system given a nonrenewable resource. Notice that as the population increases, the resource is consumed at a faster and faster rate. As the resource reaches dangerously low levels, the well being of the population is jeopardized. When this happens, the population begins to slow its growth, then eventually begins to decline until it collapses to a minimal level or disappears altogether.

TABLE 2.5. Defining characteristics of a logistic behavior pattern.

Description	Whenever the initial value R_0 is much smaller than what can be sustained over the long term, the system initially exhibits exponential growth, which flattens out as the reservoir approaches a maximum sustainable value (called the <i>Carrying Capacity</i> of the system). If the initial value is above the <i>Carrying Capacity</i> , then it will exhibit exponential decay, eventually approaching a steady-state value equal to the <i>Carrying Capacity</i> .
Rate equation	$\frac{dR(t)}{dt} = k(t) \cdot R(t),$ where $k(t) = \text{Unconstrained Growth Rate} \cdot \left\{ 1 - \frac{R(t)}{\text{Carrying Capacity}} \right\}$ • The <i>Unconstrained Growth Rate</i> represents the rate at which $R(t)$ would <i>initially</i> grow (if resources were unlimited) • The larger the <i>Unconstrained Growth Rate</i>, the more rapidly the system will approach steady state. • <i>Carrying Capacity</i> represents the maximum value of $R(t)$ that the system can sustain over the long term.
Solution to the rate equation	$R(t) = \frac{\text{Carrying Capacity}}{1 + Ae^{-\text{Unconstrained Growth Rate} \cdot t}}$ where $A = \frac{\text{Carrying Capacity} - R_0}{R_0}$
Graphical behavior	If $R_0 < \text{Carrying Capacity}$, then $R(t)$ increases, eventually leveling off at the <i>Carrying Capacity</i>. If $R_0 \ll \text{Carrying Capacity}$, the early behavior of $R(t)$ will resemble exponential growth. If $R_0 > \text{Carrying Capacity}$, then $R(t)$ decreases over time, eventually leveling off at the <i>Carrying Capacity</i>. If $R_0 \gg \text{Carrying Capacity}$, the early behavior of $R(t)$ will resemble exponential decay.
Steady state solution	$\bar{R} = \text{Carrying Capacity}$ (achieved only as $t \rightarrow \infty$)
Example applications	Population growth under limited resources; Epidemiology; Information and policy dissemination

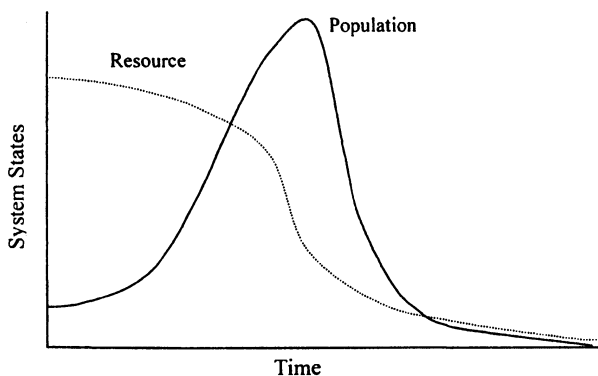


FIGURE 2.14. Example of overshoot and collapse behavior.

2.5.2 Overshoot and Collapse: System Features, Diagram, and Equations

Figure 2.15 displays an example diagram of an overshoot and collapse system. There are many different model constructs that will lead to similar behavior. The system diagram in Figure 2.15 was chosen for its overall simplicity and in order to illustrate the main features of this type of system. The use of the words *Population* and *Resource* in Figure 2.15 are not meant to imply that overshoot and collapse behavior can happen only with populations using resources. They are used here only for convenience. The

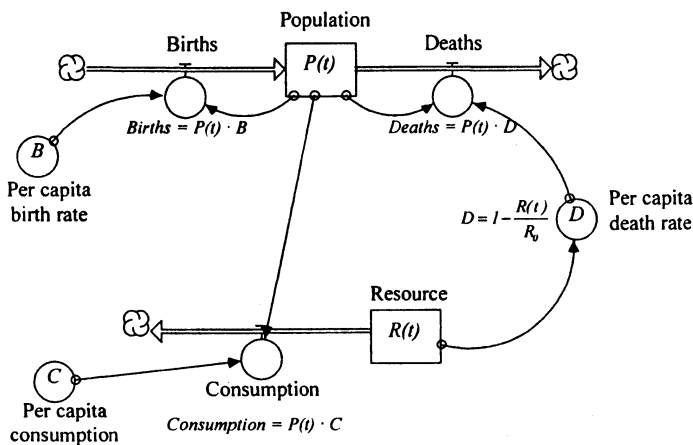


FIGURE 2.15. Generic system diagram for overshoot and collapse behavior.

exercises in this section provide an opportunity for you explore other overshoot and collapse systems.

Study Figure 2.15 very carefully to make sure you see how this system will lead to overshoot and collapse behavior. The following important characteristics are evident from the figure.

1. There are two reservoirs: the *Population* and the *Resource*.
2. Because the *Resource* has only an outflow process attached, it is nonrenewable; the system does not provide any means by which the *Resource* can be replenished.
3. Each individual in the *Population* consumes C *Resource* units in a single unit of time. Hence, as the *Population* size increases, the rate at which the *Resources* are consumed also increases. The value of the constant C depends on the particular system being modeled.
4. The size of the remaining *Resource* base affects the *Death Rate* in the *Population*. As the *Resource* base decreases, the *Death Rate* increases. The expression for relating the *Death Rate* to the *Resource* stock is defined in this example to be $\text{Death Rate} = 1 - \frac{R(t)}{R_0}$ where $R(t)$ is the size of the *Resource* base at time t , and where R_0 is the initial size of the *Resource* base. This particular expression is somewhat arbitrary. Overshoot and collapse behavior will occur as long as the *Resource* is nonrenewable, and the *Death Rate* increases as the *Resource* decreases.
5. The system depicted in Figure 2.15 includes three different feedback loops. These are displayed in Figure 2.16 as well. Note that the loop involving the *Population* and *Births* and the loop involving *Population* and *Deaths* are reinforcing and counteractive feedback loops, respectively. The large loop involving *Population*, *Consumption*, *Resources*, and *Death Rate* is a counteracting feedback loop. This last loop is the feature in this system that leads to the ultimate collapse of the *Population*.

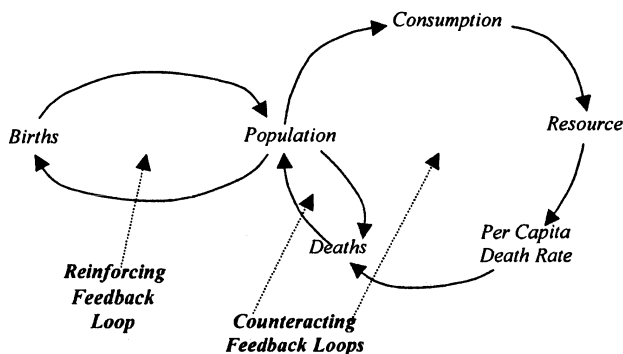


FIGURE 2.16. Feedback loops in the overshoot and collapse model.

We can describe the circumstances under which the overshoot and collapse behavior pattern will occur as follows:

A system will exhibit overshoot and collapse behavior whenever one reservoir (i.e., a “population”) depends on another nonrenewable reservoir (i.e. a “resource”) for survival. As the population increases in size and overconsumes the resource, the resource becomes depleted to a point where the population cannot survive, leading to eventual collapse.

Using the same approach used for the linear, exponential, and logistic systems, the difference equations corresponding to Figure 2.15 can be derived. It is important to note that overshoot and collapse systems different from Figure 2.15 can be constructed, and those constructs would lead to a different set of difference equations. If you understand how this particular system and the associated equations work, however, you will be able to apply these same concepts to other overshoot and collapse constructs.

There are two reservoirs in Figure 2.15. Hence, we provide here two difference equations, one for each reservoir. By referring to Figure 2.15, the difference equation for the *Population*, is seen to be

$$P(t + \Delta t) = P(t) + \left\{ B - \left(1 - \frac{R(t)}{R_0} \right) \right\} \cdot P(t) \cdot \Delta t \quad (2.18)$$

Equation (2.18) shows that future values of the *Population* depend both on past values of $P(t)$ as well as on values of the *Resource* $[R(t)]$. We will elaborate on this relationship shortly. Equation (2.19) gives the difference equation for the *Resource* reservoir. Make sure that you see how this expression [and Equation (2.18)] are obtained from Figure 2.15.

$$R(t + \Delta t) = R(t) - P(t) \cdot C \cdot \Delta t \quad (2.19)$$

This expression likewise indicates that future values of the *Resource* $[R(t)]$ depend on past values of both the *Resource* and the *Population*.

Using Equations (2.18) and (2.19), we can develop the rate equation for each reservoir in this system. By rearranging terms in Equations (2.18) and (2.19) and taking the limit as Δt approaches zero, we get

$$\frac{dP(t)}{dt} = \left\{ B - \left(1 - \frac{R(t)}{R_0} \right) \right\} \cdot P(t), \text{ and} \quad (2.20)$$

$$\frac{dR(t)}{dt} = -C \cdot P(t) \quad (2.21)$$

where B is the *per capita birth rate (per unit of time)*, and C is the *per capita consumption rate of the Resource base (per unit of time)*. The quantity R_0 is also fixed and stands for the value of $R(t)$ at time $t = 0$ (the initial size of the *Resource*).

Equations (2.20) and (2.21) are referred to as a ***coupled set of rate equations***. They are coupled because the rate at which the *Population* $[P(t)]$

changes [in Equation (2.20)] is a function of the *Resource* $[R(t)]$. Likewise, the rate at which the *Resource* changes [Equation (2.21)] is a function of the *Population*. That is, neither the *Population* nor the *Resource* can vary in size without affecting the rate at which the other changes.

A study of Equations (2.20) and (2.21) provides some important insights into the behavior of this system. From Equation (2.20) we see that the rate at which the *Population* changes is proportional to the current size of the *Population*; however, just as in the case with the logistic behavior pattern, the proportionality constant [i.e., the quantity $\left\{B - \left(1 - \frac{R(t)}{R_0}\right)\right\}$] varies over time. Early on, $R(t)$ will be close to its initial value R_0 . Hence, $\left(1 - \frac{R(t)}{R_0}\right)$ will be close to 0, and the proportionality constant $\left\{B - \left(1 - \frac{R(t)}{R_0}\right)\right\}$ will be close to the per capita birth rate (B). Thus, early on, the *Population* will increase in a roughly exponential fashion with a rate constant close to the birth rate B . As time passes, more resources will be consumed and the value of $R(t)$ will eventually be much smaller than the initial value R_0 . As this happens, the expression $\left(1 - \frac{R(t)}{R_0}\right)$ will approach 1.0 and the proportionality constant will approach the value $B - 1$. Assuming that the per capita birth rate (B) is much less than 1.0 (a reasonable assumption), the quantity $\left\{B - \left(1 - \frac{R(t)}{R_0}\right)\right\}$ in Equation (2.20) will be negative. This means that the *Population* will then decrease in size.

The negative sign on the right side of the rate equation for $R(t)$ [Equation (2.21)] indicates that $R(t)$ will always be decreasing at a rate that is proportional to the size of the *Population*. As the *Population* increases in size, $R(t)$ will decrease more rapidly. This matches our intuition: the larger the *Population*, the more rapidly the *Resource* base will be consumed.

Because the system involves two reservoirs, it is necessary that both reservoirs reach a steady state in order for the overall system to reach a steady state. Hence, the system reaches a steady state whenever the following conditions are met.

$$\frac{dP(t)}{dt} = 0 \text{ and } \frac{dR(t)}{dt} = 0 \quad (2.22)$$

Setting the right-hand sides of Equations (2.20) and (2.21) equal to zero indicates that the system reaches steady state whenever

$$\begin{aligned} 0 &= \left\{B - \left(1 - \frac{R(t)}{R_0}\right)\right\} \cdot P(t), \text{ and} \\ 0 &= -C \cdot P(t) \end{aligned} \quad (2.23)$$

It is clear that these conditions are satisfied under each of the following two cases:

$$\left\{ B - \left(1 - \frac{R(t)}{R_0} \right) \right\} = 0 \text{ and } C = 0, \text{ or} \\ P(t) = 0$$

We will assume that the consumption rate C is greater than zero. In other words, we will assume that the *Population* does consume some of the *Resource* in each time unit. Hence, the first case is impossible. The second case corresponds to the situation in which the *Population* has totally collapsed and no longer exists. This happens only if the *Resource* is completely consumed so that the *per capita death rate* (i.e., $1 - \frac{R(t)}{R_0}$) reaches 100%, thereby “killing off” the entire *Population*. This occurs asymptotically as we go further out in time. The system reaches steady state only in the limit as $t \rightarrow \infty$. The final steady-state values for the *Population* [$P(t)$] and the *Resource* [$R(t)$] are:

$$\bar{P} = 0 \text{ and } \bar{R} = 0 \quad (2.24)$$

By considering the rate Equations (2.20), (2.21), and (2.24) together, we can now predict how this system will behave. This description corresponds to Figure 2.14. At the beginning, the derivative for the *Population* [Equation (2.20)] will be positive; therefore, $P(t)$ will increase in size. As the *Population* increases, the *Resource* base $R(t)$ will decrease more and more rapidly. At some point, $R(t)$ will be so much less than its initial value R_0 that the derivative of the *Population* will become negative, and the *Population* will begin to drop. Both the *Population* and the *Resource* will asymptotically approach steady-state values of zero as we go further out in time.

2.5.3 Overshoot and Collapse: Summary

Table 2.6 summarizes the overshoot and collapse model characteristics.

2.6 Behavior Pattern #5: Oscillation

2.6.1 Oscillation: Illustrative Example

Many systems in the environment exhibit oscillatory behavior. For example, consider the cyclic behavior of the weather, the ocean tides, and the sun's energy output. The classic predator-prey relationship is another such example. Imagine a population of predators that live off of a renewable population of prey. As the number of prey increases, the number of predators also increases. As the number of predators increases, however, more prey

TABLE 2.6. Defining characteristics of an overshoot and collapse system.

Description	Two reservoirs. One is a nonrenewable "resource" and the other is a "population" that continually consumes the resource and also depends on it for survival.
Rate equations	For the Population: $\frac{dP(t)}{dt} = \left\{ B - \left[1 - \frac{R(t)}{R_0} \right] \right\} \cdot P(t)$ For the Resource: $\frac{dR(t)}{dt} = -C \cdot P(t)$ • $P(t)$ = Population reservoir at time t • $R(t)$ = Resource reservoir at time t • B = per capita "birth" rate in the Population (per unit of time) • C = per capita consumption rate of the Resource (per unit of time) • R_0 = initial value of the Resource reservoir
Solution to the rate equation	Not provided
Graphical behavior	Initial exponential growth of the Population, followed by a peak and then collapse; Resource base continually decreases; Both approach a steady state after the system collapses.
Steady state solutions	$\bar{P} = \bar{R} = 0$ (achieved only as $t \rightarrow \infty$)
Example applications	Population growth with nonrenewable resources; epidemiology

are killed and consumed, thereby leading to a reduction in the prey population. As the prey population decreases, the predator population also decreases (because the predators cannot find as much to eat). This reduction in predators leads to a "rebound" in the number of prey (because there are not as many predators around to kill them). On and on this cycle goes, creating a sinusoidal pattern similar to the one in Figure 2.17.

2.6.2 Oscillation: System Features, Diagram, and Equations

The main feature of an oscillating system is the presence of a strong counteracting feedback loop that forces the system to oscillate around an equilibrium set of conditions. Consider the predator-prey example described earlier. Figure 2.18 illustrates this loop.

Figures 2.17 and 2.18 highlight several features that will cause a system to exhibit oscillatory behavior. These features will now be described.

1. The system contains at least two interdependent reservoirs. One reservoir can be thought of as the *Consumer* in the system, and the other can be thought of as the *Resource*. In some cases, the context will clearly identify

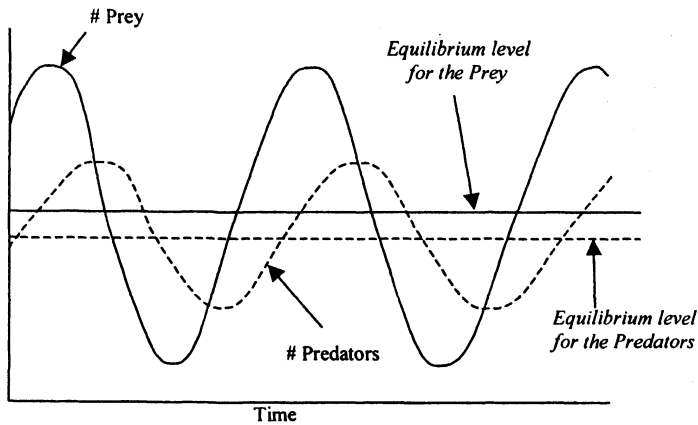


FIGURE 2.17. Typical predator-prey oscillatory behavior.

which role each reservoir plays. In many other cases, the designation of *Consumer* and *Resource* may be arbitrary.

2. The *Consumer* and *Resource* reservoirs have **equilibrium values** around which they oscillate.

3. The further one reservoir is from its equilibrium value, the more influence the other reservoir exerts to “pull it back” toward equilibrium. For example, whenever our example *Prey* reservoir is significantly above its equilibrium value, the *Predators* will grow rapidly and enthusiastically hunt down and kill the *Prey*, thereby driving them back toward equilibrium. If the *Predator* reservoir is significantly above its equilibrium level, then the *Prey* reservoir will rapidly shrink, thereby forcing the *Predators* back toward equilibrium.

Figure 2.19 gives a simplified example of an oscillatory system. Study this diagram to confirm that it includes the features described in the previous

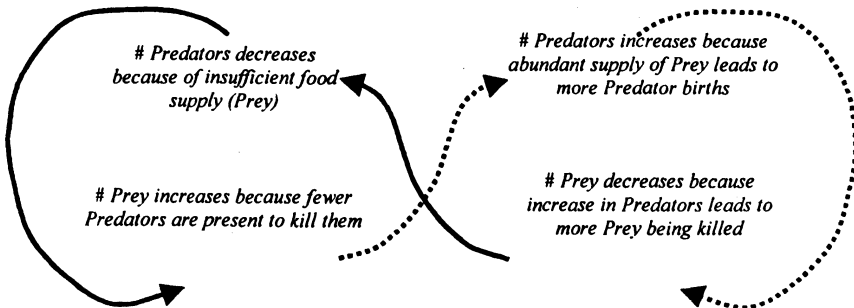


FIGURE 2.18. Counteracting feedback loop in the predator-prey example.

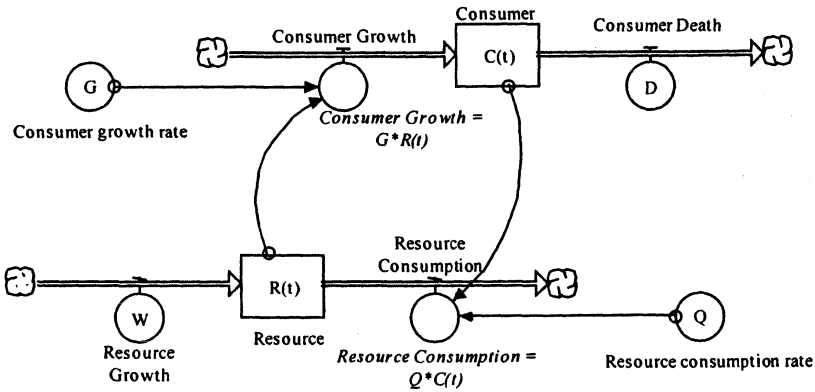


FIGURE 2.19. System diagram for a simple oscillating system.

paragraphs. In particular, note how the *Consumer* reservoir directly affects the rate at which the *Resource* reservoir is depleted (through the *Resource Consumption* outflow). Note also that the *Resource* reservoir affects the rate at which the *Consumer* reservoir grows (through the *Consumer Growth* inflow).

The difference equations for the two reservoirs in this system are given by

$$C(t + \Delta t) = C(t) + \{G \cdot R(t) - D\} \cdot \Delta t \quad (2.25)$$

$$R(t + \Delta t) = R(t) + \{W - Q \cdot C(t)\} \cdot \Delta t \quad (2.26)$$

We can rearrange terms in Equations (2.25) and (2.26) in the usual way and then take the limit as $\Delta t \rightarrow 0$ to get the following rate equations.

$$\frac{dC(t)}{dt} = G \cdot R(t) - D \quad (2.27)$$

$$\frac{dR(t)}{dt} = W - Q \cdot C(t) \quad (2.28)$$

A careful examination of Equations (2.27) and (2.28) will show that the two reservoirs will behave in some sort of cyclic pattern. For example, if the *Resource* reservoir is large, then Equation (2.27) indicates that the derivative of the *Consumer* reservoir will be a large positive value. Hence, the *Consumer* reservoir will rapidly grow in size. If the *Consumer* reservoir grows too large, however, then Equation (2.28) shows that the derivative of the *Resource* reservoir will turn negative, thereby indicating that the *Resource* will shrink in size. As $R(t)$ shrinks, the derivative Equation (2.27) will also get smaller and (eventually) turn negative, thereby indicating that the *Consumer* reservoir will no longer grow, but shrink instead. This will produce

the kind of cyclic behavior described earlier. It can be shown that any system with the rate equations given in Equations (2.27) and (2.28) will oscillate around equilibrium values given by

$$\text{Equilibrium level for the Resource, } R(t), \text{ is } \frac{D}{G} \quad (2.29)$$

$$\text{Equilibrium level for the Consumer, } C(t), \text{ is } \frac{W}{Q} \quad (2.30)$$

The oscillatory system given here will reach a steady state whenever $\frac{dR(t)}{dt} = \frac{dC(t)}{dt} = 0$. This occurs whenever both reservoirs are equal to their equilibrium values given in Equations (2.29) and (2.30). Hence, the steady-state solution to the simple oscillatory system in Figure 2.19 is given by

$$\bar{R} = \frac{D}{G} \text{ and } \bar{C} = \frac{W}{Q} \quad (2.31)$$

Notice that the system reaches steady state only if the reservoirs are equal to their equilibrium values at the same time. For example, if only $R(t)$ is at the equilibrium value, then it will not be “allowed” to stay at equilibrium. If the *Consumers* are above their equilibrium, then they will consume too many of the *Resources* and drive the $R(t)$ reservoir below equilibrium. If the *Consumers* are below their equilibrium value, then the *Resources* will begin to proliferate (due to their being too few *Consumers* to keep the population in check).

2.6.3 Oscillation: Summary

Table 2.7 summarizes the important characteristics of oscillating systems.

2.7 Exercises

Section 2.2

1. Consider a new landfill in which solid waste is deposited at a rate of 25 metric tons/day. Draw a system diagram depicting this system. Fill in constant values for each of the quantities in the system. Make a sketch showing the volume of waste in the landfill as a function of time. Write down the rate equation for the volume of waste in the landfill and specify the value of the rate constant.

2. Define three systems in real life that will exhibit linear growth or decay. For each example, identify the following: (a) The reservoir that will exhibit this linear dynamic (specify its units); (b) All of the inflows and outflows

TABLE 2.7. Defining characteristics of a simple oscillating system.

Description	Two interdependent reservoirs (i.e., a Consumer reservoir and a Resource reservoir). The Consumer "consumes" Resource units. The Resource enables the Consumer to grow. Both oscillate around equilibrium values. The further one reservoir is from its equilibrium value, the more the system "works" to drive it back toward equilibrium.
Rate equations (refer to Figure 2.17)	For the Consumer $\frac{dC(t)}{dt} = G \cdot R(t) - D$ For the Resource: $\frac{dC(t)}{dt} = W - Q \cdot (t)$ • G = Consumer Growth Rate (reflect how the growth of the Consumer reservoir depends on the size of the Resource) • D = Consumer Deaths per time unit • W = Resource Growth per time unit • Q = Resource Consumption Rate (reflects how the depletion of the Resource depends on the size of the Consumer reservoir)
Solution to the rate equation	Not provided
Graphical behavior	Graph of each reservoir resembles a sinusoidal wave. Each wave oscillates about an equilibrium value. The equilibrium value for the Consumer is $\frac{W}{Q}$. The equilibrium value for the Resource is $\frac{D}{G}$. The period and amplitude of each sine wave is determined by D, G, W, Q, and the initial values of the reservoirs R_0, and C_0.
Steady state solutions	The system will run at steady state only if both reservoirs begin at their equilibrium values. That is, the steady state is reached at $\bar{C} = \frac{W}{Q} \text{ and } \bar{R} = \frac{D}{G}.$
Example applications	Predatory-Prey systems; Systems with consumers and renewable resources

(specify units); (c) Whether the system will typically exhibit linear **growth** or linear **decay**.

3. Assume that the reservoir in Figure 2.3 represents the mass of a particular pollutant in a lake at time t , where t is measured in years. Assume the initial pollutant mass in the lake is 1,000 kg. Specify the units for each of the flow processes in this model.

4. You will find nine different sets of values for the flows in the Figure 2.3 listed in Table 2.8. For each case (a) Specify whether the resulting system will exhibit linear growth, linear decay, or neither (assume that time is measured in minutes); (b) If the system exhibits linear growth or decay, determine the value of the rate constant k and write down the equation [corresponding to equation (2.7)] for the reservoir value $R(t)$ that is the solu-

TABLE 2.8. List of data for exercise 4.

Case	Inflow 1	Inflow 2	Inflow 3	Outflow 1	Outflow 2
A	10	20	35	35	20
B	10	20	35	35	35
C	10	20	$\sin(t)$	30	$\sin(t)$
D	10	20	$\sin(t)$	30	0
E	$5t$	0	0	$5t$	0
F	10	0	0	0	0
G	0	0	0	0	10
H	10	20	$\sin(t)$	$0.75\sin(t)$	$0.25\sin(t)$
I	10	20	$\sin(t)$	$0.75\sin(t)$	0

tion to the rate equation in Table 2.3. Make sure you give numeric values for all quantities.

Section 2.3

5. Use STELLA® to construct the mouse population example system in Figure 2.6. Assume that the system starts with two mice. Try different values for the *Birth Rate* and *Death Rate* and run the model to see what effect they have on the system. Under what conditions does the mouse population increase over time? When does it decrease over time? When is the population at a steady state?

6. Define three systems in real life that will exhibit exponential growth or decay. For each example, identify the following: (a) The reservoir that will exhibit this type of behavior (specify its units); (b) All of the inflows and outflows (specify units); (c) Whether the system will typically exhibit exponential **growth** or exponential **decay**.

7. Briefly describe what factor(s) you think would determine the value of the “flow rate” converter in Figure 2.8.

8. Explain why the feedback loop involving the reservoir and outflow process in Figure 2.9 is a counteracting feedback loop.

9. Show mathematically that the solution to the exponential rate Equation (2.10) is the expression given in Equation (2.11). [Hint: Take the derivative of the expression in Equation (2.11) and show that it is equal to the right-hand side of Equation (2.10)].

10. In STELLA®, modify the model in Figure 2.8 to include a constant water inflow of 10 cc/sec and do the following: (a) Write down the difference equation for the water volume in the bucket; (b) Derive the rate equation for the water volume in the bucket in this new system; (c) Will this new system behave in an exponential fashion? Under what conditions?

11. Is it reasonable to assume that populations of people or other living organisms will exhibit exponential behavior indefinitely? Why or why not?

12. Consider what happens whenever a 90°F can of soft drink is placed in a 38°F refrigerator. The can’s temperature will immediately begin to drop

and will rapidly approach the ambient temperature in the refrigerator. As the can's temperature gets closer to 38°F, however, the rate at which its temperature drops will decrease until (after a long time) the temperature of the can is virtually identical to the temperature inside the refrigerator. Do the following: (a) Sketch a graph of the can's temperature versus time; (b) Draw a system diagram for modeling the temperature of the can. Use a reservoir to represent the can's temperature. Note that the rate at which the can's temperature changes is proportional to the difference between the can's temperature and the ambient temperature in the refrigerator.

Section 2.4

13. Assume that the Reservoir in Figure 2.12 represents a population of people and that time is measured in years. Specify the units for all the system elements in Figure 2.12.

14. Define three systems in real-life (other than those described in this chapter) that will exhibit logistic growth. For each example, identify the following:

- The reservoir that will exhibit this type of behavior (specify its units)
- All of the inflows and outflows (specify units)

15. Use STELLA®, to construct the logistic model Figure 2.12. Assume that the reservoir represents a population of people and that time is measured in years. Run the model for 100 years with each of the following sets of values for R_0 , the *Unconstrained Growth Rate*, and the *Carrying Capacity*. In each case, make a graph of the reservoir versus time and explain why the system behaves as it does. In addition, for each case, write down Equation (2.17), giving numerical values for all the parameters in that expression.

- $R_0 = 10$, Unconstrained Growth Rate = 0.1, Carrying Capacity = 1,000
- $R_0 = 10$, Unconstrained Growth Rate = 0.5, Carrying Capacity = 500
- $R_0 = 2,000$, Unconstrained Growth Rate = 0.1, Carrying Capacity = 500
- $R_0 = 2,000$, Unconstrained Growth Rate = 0.5, Carrying Capacity = 1,000

16. Sketch a graph like Figure 2.13 showing what the system will behave like if the initial value of the reservoir is greater than the *Carrying Capacity* of the system. Make sure you show how the value of the *Unconstrained Growth Rate* will affect the shape of the graph.

17. Identify and describe the feedback loops in Figure 2.12. Specify whether each is positive or negative feedback.

18. Show that Equation (2.16) is the rate equation for the logistic system in Figure 2.12. [Hint: Begin with the difference equation in Equation (2.15). Then use an approach similar to what was used to derive the rate equation for the exponential system, Equation (2.10)].

19. In the logistics model described in Section 2.4, we assumed that the “death” rate would increase and the birth rate would remain constant as the population approached the Carrying Capacity of the system. In more advanced cultures, however, people may adopt a more proactive approach before resources are seriously depleted. In this scenario, individuals in the population may choose to have fewer children, thereby reducing the overall birth rate. In this case, the birth rate will decrease over time and the death rate will remain constant. The growth should again level off as the population approaches the Carrying Capacity. Create a systems model in STELLA®, to match this scenario. Make sure you specify all the equations and units for calculating the system quantities specify number using for the initial size of the population, the carrying capacity and any other values necessary to run the model. Include a graph that shows the Population and Resource reservoirs over time. Derive the rate equation for the system. This equation should be similar to (but not exactly the same) as Equation (2.15). Run the model for 50 years.

Section 2.5

20. Assume that the *Resource* reservoir in Figure 2.15 is measured in generic “resource units.” In addition, assume that time is measured in years. What are the units for the *Consumption* flow process?

21. Assume that in Figure 2.15 the initial value of Resource is 10,000 units, the initial value of Population is 10 people, the birth rate is 0.50 births per capita per year, and each person consumes two resource units per year. Use these values and Figure 2.15 to construct the overshoot and collapse model in STELLA®. Make a graph showing the *Population* and *Resource* reservoirs over time. Run the model for 50 years.

- Decrease the *Birth Rate* to 0.1 and rerun the model. Make sure you run long enough for the system to stabilize. What differences do you see from the case in which the *Birth Rate* is 0.5? Make sure you pay attention to the scale on the vertical axis of the graph! Can you explain why the system’s behavior changes in this way?
- Change the *Birth Rate* back to 0.50 and change the *Consumption Rate* to 20 units/person/year. Rerun the model and compare the results with the original case. Can you explain why the system’s behavior changes in the ways that it does?
- Summarize what you have learned from these experiments with the model. What conditions lead to more rapid collapse of the system?
- See if you can identify values for the *Birth Rate* and *Consumption Rate* that will not lead to a total collapse of the *Population* for the preceding exercise. Can this be done? If so, how? Why is this the case?

22. The expression used for the *Per Capita Death Rate* in Figure 2.15 is $D = 1 - \frac{R(t)}{R_0}$. Under this formulation, the *Death Rate* will eventually reach

levels that are very close to 100%. It may be the case, however, that the *Population* is not completely dependent on the *Resource* for survival. Whenever this is so, then the *Death Rate* may not approach 100% as the *Resource* is depleted. Suppose that the *Death Rate* D will reach only 60% as $R(t)$ approaches zero. How could you modify the expression for D to accommodate this fact? Incorporate this modification into the model corresponding to Figure 2.15 and then run the model with the original values for the *Birth Rate*, *Population*, and *Resource*, given in Exercise 21. How does the system behave? Can you explain why?

23. Define three systems in real life (other than the oil consumption example described in this section) that can exhibit overshoot and collapse behavior. Make sure that you sketch a system diagram for each case. Clearly identify all reservoirs, flows, and converters by giving them descriptive names. Clearly define the units for each quantity in the system.

24. Some systems involve a *Resource* that can be replenished. For example, the *Resource* might represent a food supply that is renewable through agricultural methods or natural growth. Modify your STELLA® model of the original system described in Exercise 21 so that the *Resource* can be renewed via a *Renewal* flow process. Assume that the *Renewal* inflow is proportional to the size of the *Resource* (as in an exponential model). Also assume that a converter called the *Renewal Rate* represents the proportionality constant. Set the *Renewal Rate constant* to be equal to 0.05. How does the behavior of this new system compare with the behavior of the original system? Can you explain why?

Section 2.6

25. Assume that time is measured in hours in Figure 2.19, that the *Consumer* reservoir is expressed as a number of organisms, and that the *Resource* reservoir is expressed as the number of generic resource units. Specify the units for all the other quantities in the model.

26. Use STELLA® to build the model in Figure 2.19. Populate the model with the values specified later. Make a graph showing both the *Consumer* and the *Resource* reservoirs over time. Run the model for 25 years, and answer the following questions. The graph of the *Consumer* and *Resource* reservoirs should exhibit simple oscillatory behavior. [Note: The oscillatory model has a complicated enough behavior that we must take special care to avoid serious round-off errors in STELLA®. This is done by using a very small time step and by using a more sophisticated numerical algorithm. You can make these changes by selecting **Time Specs** under the **Run** menu. The value of **DT** specifies the time step. Change **DT** to **0.0625** hours. In addition, instruct STELLA® to use the **Runge-Kutta 2** method of integration in the **Time Specs** window of STELLA®.]

$$G = 1 \quad D = 20 \quad Q = 1 \quad W = 15 \quad C_0 = 10 \quad R_0 = 15$$

- a. What are the **equilibrium values** for the *Consumer* and *Resource* reservoirs? Note that the equilibrium values are those midpoint values around which each reservoir oscillates.
- b. What is the **period** of this system? (A *period* is the length of time it takes for the system to complete one full cycle).
- c. When is the first time that the *Consumer* reservoir reaches its equilibrium value in the simulation run? Where is the *Resource* reservoir in relationship to its equilibrium value during this same time? How does the level of the *Resource* affect the behavior of the *Consumer* from this point in time until the *Consumer* next reaches its equilibrium?
- d. Do the *Consumer* and the *Resource* ever achieve equilibrium at the same time? What do you think would happen if they did?
- e. Change the value of W to 20 and re-run the model. What has changed? What explanation can you give?
- f. Change W back to 15 and then change the value of G to 0.5 and re-run the model. What has changed? What explanation can you give?
- g. Now change G back to 1. Set the starting value of $C(t)$ and $R(t)$ to be $C_0 = 15$ and $R_0 = 20$. Run the model. Can you explain why the system behaves this way?

27. Imagine that you are a wildlife manager, responsible for keeping the population of a certain deer species as close to equilibrium as possible. Suppose that the only predators in your system are a population of wolves. How might knowledge of the steady-state behavior of this system aid you in your job? What would you do to try and maintain the deer population to a level as close to steady state as possible?

Model Building Exercise

28. Construct a system model in STELLA® that describes the relationships between the oil supply and the population of fossil fuel-burning vehicles. Your model should include at least three reservoirs: (1) the fossil fuel vehicles, (2) the supply of processed oil that is ready for consumption, and (3) the underground reserves of crude oil. Your model should include at least one feedback loop. Include flow processes for the production and obsolescence of vehicles and for the production and consumption of processed oil. Specify all interrelationships and equations or graphical relationships that are necessary in order for the model to run (apart from the values of constants). Provide descriptions within STELLA® of each of the system components. Specify their units. Prepare a brief (one- or two-page) write-up describing your system and giving the rationale for the system diagram. Include a copy of the system diagram and a graph of the model output as part of the write-up. Answer the following questions:

- a. Describe the feedback loops in your system. Are they examples of positive or negative feedback?
- b. Which types of behavior patterns will your model exhibit (linear, exponential, logistic, overshoot and collapse, or a combination)? Under what conditions will the type of behavior you identified occur? Briefly justify your answer.

2.8 Suggested Readings

Haberman, R. 1977. *Mathematical Models: Mechanical Vibrations, Population Dynamics, and Traffic Flow*. New Jersey: Prentice-Hall, Inc.

High Performance Systems. 1996. *An Introduction to Systems Thinking*. Hanover, NH: High Performance Work Systems, Inc.

3

Strategies for Analyzing and Using Environmental Systems Models

Chapter Objectives—

After you finish this chapter, you should be able to:

1. *Outline a strategy for analyzing and using an environmental systems model.*
2. *Given a modeling problem, develop a purpose statement for using the model.*
3. *Check the **structural validity** and **predictive validity** of the model by using:*
 - *The systems diagram and a description of the system.*
 - *Baseline behavior patterns*
 - *Steady State behavior patterns*
 - *Runaway behavior patterns*
4. *Conduct an **exploratory analysis** to develop an understanding of the system's dynamic behavior. This analysis will include:*
 - *Experiments to observe the system's response to **pulse, ramp, or step** inputs.*
 - ***Sensitivity analysis** to identify the "high-leverage" variables in the system.*
5. *Perform a **case analysis** with the model in order to evaluate the impact of potential changes to the system.*

3.1 Analyzing a Systems Model: Overview

The analysis of an environmental systems model must be done within the context of the overall "problem" that the model was designed to address. For example, consider the island community model described in Chapter 1 and illustrated in Figure 1.5. That model is useful for describing the projected growth (or death!) of the shipwrecked human population on the new island home. The model is not useful, however, for describing the impact of the human population on, say, the ocean ecosystem just off the island's coast. An entirely different model would be needed—one that would include system elements to describe such things as the amount and type of fishing used by the island inhabitants, the waste that they might dump into the ocean, and so forth. This illustrates that a model may be perfectly suitable for addressing one problem (i.e., describing the projected growth of the human popu-

lation on the island) and completely unsuitable for addressing another problem (i.e., modeling the impact of the humans on the surrounding ocean ecosystem).

In this chapter you will learn a general strategy for analyzing and using an environmental systems model within the context of a clearly defined modeling problem. In applying this strategy, you will learn to use *systems thinking*, as described in Section 1.4, to help you understand environmental systems.

The strategy involves the following major steps.

1. **Problem Definition**, in which we construct a *purpose statement* describing the intended use of the model and the questions we wish to answer. This purpose statement defines the context within which the model is to be applied.
2. **Model Validation**, in which we evaluate whether the model as designed can give reasonable predictions and explanations of the system with which we are concerned.
3. **Exploratory Analysis**, in which the model is used to explore the system and gain insight into how the system works. This step involves two types of analyses:
 - Experiments in which the system is deliberately “upset” by simulating *PULSE*, *RAMP*, or *STEP* changes in one or more system flows or converters.
 - *Sensitivity analysis*, which helps to identify those elements of the system that exercise a high degree of leverage on system behavior.
4. **Case Analysis**, in which we modify the model or change model parameters to mimic how the system will behave under new conditions. These new conditions may arise because of such things as policy changes, increased pollution levels, remediation activities, and the like.

The rest of this chapter is devoted to describing and illustrating these four steps. We will illustrate the strategy by using a model for the spread of an infectious disease through an aquatic ecosystem. The model and the problem will be introduced in the next section.

3.2 An Illustrative Model: Infectious Disease Dynamics

3.2.1 General Description of the Problem

Imagine a fish population in a large aquatic ecosystem. Under normal conditions, the number of fish will remain relatively stable. It is possible, however, for a toxic substance or infectious microorganism to be introduced into the system and upset this balance. This will in turn jeopardize the health of the fish population and, ultimately, the health of the ecosystem.

A real-life example of such a phenomenon is the outbreak of *Pfiesteria piscicida* in the sea bass population of the Chesapeake Bay on the eastern

coast of the United States. Over the last few years watermen and biologists have observed large numbers of dead sea bass in the bay. It is believed that many of these fish were killed after being infested with *P. piscicida*. These fish exhibited numerous lesions, sometimes more than 1 in. in diameter, and running deep enough to expose internal organs. These lesions are highly infectious to other fish. In addition, exposure to this disease is thought to cause skin lesions, memory loss, and respiratory problems in humans.

Some researchers have hypothesized that excessive agricultural runoff is the source of the apparent *Pfiesteria* outbreak. This theory has not been verified; however, the state of Virginia allocated more than \$7 million between the years 1998 and 2000 to study the problem.

Regardless of the underlying cause of the problem, a systems model can help scientists and policy makers understand the dynamics of such an aquatic-based epidemic. This understanding, coupled with the biological understandings developed through laboratory and fieldwork, would allow scientists to predict how the epidemic might progress over an extended time frame. In addition, if a valid systems model were available, scientists could also use the model to explore various options for responding to the outbreak, such as: (1) imposing tighter controls on agricultural runoff, (2) instituting a campaign to catch and eliminate infected fish from the system, or (3) introduce more resistant strands of fish into the ecosystem.

Many models for infectious disease epidemics have been developed. We provide here a brief description of a generic model for an infectious disease in a fish population whose size is relatively stable. We will use this model as a platform for illustrating the concepts covered in this chapter.

Our goal is to use the four-step strategy described in Section 3.1 to build and use a model for the spread of an infectious disease through an imaginary fish population. We will use the model to evaluate the impact of different options for responding to the epidemic. We ultimately want to minimize the long-term impact on the fish population.

3.2.2 A Description of the Aquatic Infectious Disease System and Model

For the sake of illustration, let us call the disease we are modeling *disease X*. Suppose that the fish infected with disease X exhibit symptoms that are similar to those caused by *P. piscicida*. Victims develop severe, open lesions on the skin. These lesions grow and deepen until the fish's internal organs are exposed. In some fraction of cases, the infected fish is eventually weakened to the point of death. In other cases, the fish gradually recovers. Those fish that recover are resistant to the disease. After some time, the resistance wears off and the now healthy fish are once again susceptible to contracting the disease.

Disease X is transmitted from one fish to another whenever a healthy fish comes into physical contact with the lesions on an infected fish. “At-risk contact” with disease X occurs whenever a healthy fish physically touches the lesions on an infected fish by eating it or simply by brushing up against it.

It is important to note, however, that at-risk contact with an infected fish does not necessarily lead to the disease being passed along. Epidemiologists use the term *infectiousness* to refer to the propensity for a disease to be passed from one individual to another. The infectiousness of a disease refers more specifically to the probability that a susceptible individual who comes into sufficient contact with an infected individual will contract the disease. The infectiousness is a numeric value between zero and 1. For example, if the infectiousness of a disease is 0.30, then (on average) 30 out of every 100 susceptible individuals who experience sufficient contact with an infected individual will contract the disease. Values near zero characterize diseases that are not highly contagious. For example, the common cold has an infectiousness value that is much higher than, say, tuberculosis.

The rate at which a disease spreads through a population depends on the infectiousness of the disease and the number of at-risk contacts that susceptible individuals have with infected individuals. In order to model the spread of the disease, we must know

- How many infected fish are present in the population at each point in time
- How many times per day that a susceptible fish has contact with other fish in such a way as potentially to contract the disease (if the other fish is infected)

The calculations for determining the total number of exposures between susceptible and infected fish depend on elementary probability theory. The following discussion outlines the mathematical justification for those calculations. The details are provided here for completeness; however, the less-mathematically inclined reader can proceed on to Section 3.2.3, keeping in mind that the upshot of this discussion is expressed in Equations (3.2) and (3.3). These equations are used in the infectious disease model described in this chapter.

Suppose we let ρ stand for the total number of fish in the ecosystem (healthy fish + sick fish + resistant fish). Let s stand for the number of sick fish. Then s/ρ is the fraction of fish in the ecosystem that are infected. Now suppose that a randomly chosen fish in the population has come into contact with another fish. If we assume that the infected fish are distributed uniformly throughout the ecosystem, the probability that our randomly selected fish has in fact come into contact with an infected fish is s/ρ . Likewise, the probability that the other fish is *not* an infected fish is therefore $1 - \frac{s}{\rho}$. For

example, suppose that we have 100 fish in the ecosystem ($\rho = 100$). Further, assume that 10 of these fish are infected ($s = 10$). If a randomly selected

healthy fish comes into contact with another fish, there is a 10% chance (probability = 0.1) that the other fish is infected. There is a 90% chance (probability = 0.9) that the other fish is not infected.

Now suppose that, on average, each fish in the population will have a total of c at-risk contacts with other fish each day. That is, each fish will come into contact with c other fish during the day in such a way that the disease could be passed from one fish to another. If we assume that these c contacts are statistically independent of each other, then the probability that none of the c contacts a fish has in a given day are with an infected fish is given by

$$P(\text{all contacts are NOT with infected fish}) = \left(1 - \frac{s}{\rho}\right)^c \quad (3.1)$$

By subtracting the quantity in Equation (3.1) from 1.0, we can calculate the probability that a fish has an at-risk contact with at least one infected fish during a 1-day period as follows:

$$P(\text{at least one contact with an infected fish}) = 1 - \left(1 - \frac{s}{\rho}\right)^c \quad (3.2)$$

Equation (3.2) gives the probability that any fish in the population has at least one contact with an infected fish in a given day. Equation (3.2) can also be interpreted as giving the fraction of contacts in a given day that involve an infected fish. This quantity is expressed as *the fraction of at-risk exposures to an infected fish/capita/day*. Hence, if there are H susceptible fish in the ecosystem, we would expect (on average) that the number of contacts between susceptible and infected fish in a given day is given by

$$H \left[1 - \left(1 - \frac{s}{\rho}\right)^c \right] \quad (3.3)$$

3.2.3 System Diagram for the Infectious Disease Problem

Figure 3.1 displays a systems diagram for the type of epidemic described in Section 3.2.2. The contents of all three reservoirs in Figure 3.1 are expressed in numbers of fish. Time is measured in days. The three reservoirs are described as:

1. The *Susceptible Fish*. These individuals do not have disease X , but are susceptible to contracting the disease after exposure to other infected fish (which are represented in the *Sick Fish* reservoir).
2. The *Sick Fish*. This reservoir contains those fish that are infected with the disease. The initial value of this reservoir represents the number of infected fish at the outset of our simulation. The source of this initial infection is not specified; however, it is assumed that the only way other fish

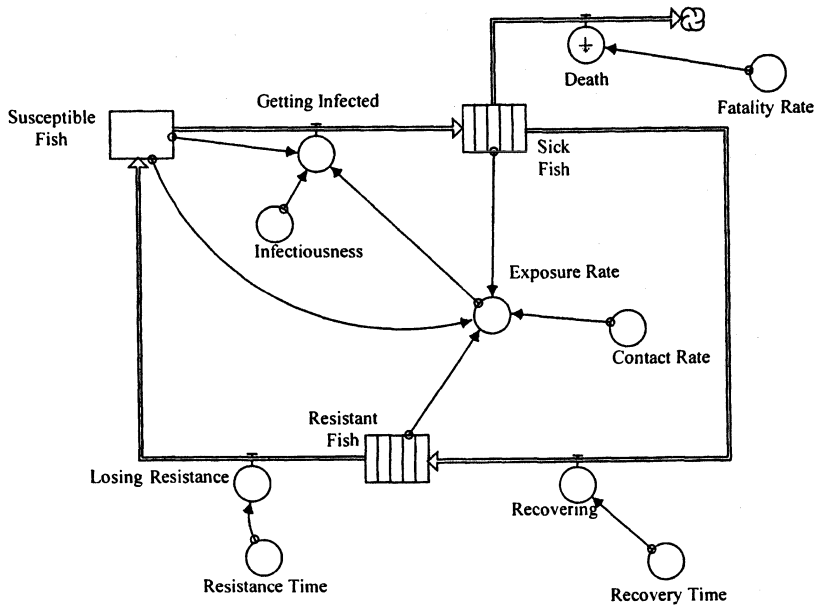


FIGURE 3.1. System diagram for the fish disease example.

in the system can contract the disease is through contact with the fish in this reservoir. Fish in this reservoir will either die from the disease (at a rate specified by the *Fatality Rate*), or will recover and become resistant to the disease.

3. The *Resistant Fish*. Some of the sick fish recover from the disease. These individuals no longer exhibit the symptoms of the disease, nor are they carriers. In addition, for a limited period of time (defined by the *Resistance Time*), these individuals are resistant to the disease. This immune response eventually diminishes so that the fish are once again *Susceptible*.

Figure 3.1 uses a new type of reservoir to represent the *Sick Fish* and the *Resistant Fish*. This representation is used to indicate that fish can stay in these reservoirs for only a fixed period of time. For example, the outflow from the *Resistant Fish* reservoir is controlled by a quantity called the *Resistance Time* (see Figure 3.1). This quantity specifies the length of time that a fish will be resistant to the disease before its resistance breaks down, thereby making the fish once again *Susceptible*. Let us suppose that the *Resistance Time* is 30 days. Hence, the number of fish exiting the *Resistant Fish* reservoir at any point in time will be equal to the number of fish that entered that reservoir 30 days earlier. We use a special modeling construct called a *conveyor* to model this type of reservoir. The important characteristics of a *conveyor* are:

1. The **transit time**. The amount of time that individuals or material entering the conveyor will remain before flowing on to the next step. In Figure 3.1, the *Recovery Time* and the *Resistance Time* specify the transit times for the *Sick Fish* and *Resistant Fish* reservoirs respectively.

2. The **flow through**. The outflow through which individuals exit after residing in the conveyor for the specified transit time. The flow through from the *Sick Fish* reservoir is the outflow running from the *Sick Fish* to the *Resistant Fish*.

3. The **leakage**. An optional outflow from which individuals can “leak” or exit from the conveyor before the transit time is complete. In our model, the *Death* outflow from the *Sick Fish* conveyor is a leakage outflow. *Sick Fish* will die from the disease prior to the *Recovery Time*.

4. The **leakage fraction**. The fraction of individuals that “leak out” of the conveyor over the transit time. In Figure 3.1, the leakage fraction is determined by the quantity called the *Fatality Rate*. This rate specifies the fraction of *Sick Fish* that die from the disease over the time period specified by the transit time.

The interpretation of each of the flow processes in Figure 3.1 should be self-explanatory. Table 3.1 outlines these features. Note that some of the information in Table 3.1 is missing and is left as an exercise for the reader.

3.3 Applying the Strategy: Problem Definition

Before an informed analysis of an environmental systems model can be performed, it is necessary that we have a clear understanding of the overall purpose of our modeling effort. Hence, the first step in our strategy for analyzing and applying a systems model is to develop a clear **purpose statement**. Such a statement will provide the context within which our analysis will be conducted. A good purpose statement is typically no more than three to five sentences long. It focuses our thinking by identifying the following

- The system we wish to study
- The behaviors we are interested in understanding
- The core questions that we wish to address.

Consider the following purpose statement for the infectious disease problem described earlier:

We wish to model the spread of *disease X* through our fish population over a 2-year period. Under normal conditions (i.e., no infected fish are present), the fish population exhibits a stable size over time. Past history has shown that this disease tends to reappear on a regular basis. We wish to predict how the makeup of the population of fish will change over time as a result of recurrent epidemics of disease *X*. We will use the model to evaluate two options for responding to an epidemic of this type: (1) repeated capture and removal of infected fish, and (2) introduction of a new

TABLE 3.1. Explanation of flows and converters in Figure 3.1.

Name	Type of element	Description (units)	Value of equation
Getting infected	Flow	Process of susceptible fish becoming infected (fish/day)	Susceptible fish* exposure rate* infectiousness
Infectiousness	Converter	Probability that an at-risk exposure to an infected fish will lead to contracting the disease (infected fish/exposure)	Constant value; always between 0 and 1.
Contact rate	Converter	Number of at-risk exposures a fish will experience in a day (units: left as an exercise)	Constant value; Corresponds to the quantity c in Equations (3.1) through (3.3)
Exposure rate	Converter	Fraction of at-risk exposures with an infected fish in a given day. (units: left as an exercise)	See Equation (3.2)
Death	Flow (leakage for sick fish)	Death flow of sick fish (units: left as an exercise)	Sick fish* fatality rate Equal to the number of sick fish dying while sick (specified by the fatality rate)
Fatality rate	Converter	Death rate of sick fish (units: left as an exercise)	Constant value; always between 0 and 1
Recovering	Flow through for sick fish	Sick fish that survive the disease recover and become resistant after R days; R = recovery time (units: left as an exercise)	Equal to the inflow into sick fish from R days earlier, minus the fish that died
Recovery time	Converter (transit time for sick fish conveyor)	Time from infection until a sick fish will recover and become resistant (units: left as an exercise)	Constant value
Losing resistance	Flow through for resistant fish	Resistant fish lose resistance after K days and become susceptible; K = resistance time (units: left as an exercise)	Equal to the inflow into resistant fish K days earlier
Resistance time	Converter (transit time for resistant fish conveyor)	Time from recovery until a fish loses its resistance to the disease. (units: left as an exercise)	Constant value

resistant strain of fish for which the infectiousness of disease X will be reduced by 50%.

Notice that this statement identifies the three elements listed earlier:

1. *The system we wish to study.* This system involves the population of fish and the spread of disease X through that population. The purpose does *not* require us to model the complex biological mechanisms by which the disease operates.

2. *The behaviors we are interested in understanding.* We wish to understand how the population size and makeup (susceptible, infected, or resistant fish) will evolve over a 2-year period. The statement also indicates that the disease seems to reappear on some sort of repeating cycle. We wish to understand the mechanism behind this behavior.

3. *The core questions we wish to address.* These are

- How does the makeup of the population of fish change over time as a result of the disease?
- Why does the disease reappear on a regular basis?
- What impact will be realized if we capture and remove infected fish?
- What impact will be realized if we introduce a more resistant strain of fish into the ecosystem?

3.4 Applying the Strategy: Model Validation

3.4.1 Two Aspects of Model Validity

A model is said to be **valid** if its predictions of system performance adequately mimic how the real system behaves. Before a model can be used to understand or predict system behavior it is necessary to check the model and determine if it accurately represents the system we are studying.

It is important to recognize that our ability to validate a model directly depends on how much we already know about the system. The more we know about the real-life system, the more precisely we can evaluate the validity of the model. Hence, any credible check on model validity requires a great deal of up-front work to learn as much about the system as we can.

The degree of accuracy required in order for the model to be valid depends on the overall purpose of the modeling effort. For example, suppose that we wish to use a predator-prey model to evaluate the potential of allowing recreational hunting of the predators. We may be interested in determining if this would lead to eventual collapse of the prey population. In such a case, it is not as important for the model to predict the exact number of predators and prey in the future as it is for the model to predict whether the prey population will collapse at some point in the future.

The process of model validation can be a very complex and time-consuming process. For our purposes, we will concentrate on the following two aspects of model validity:

1. Structural validity. A model is said to be **structurally valid** if its system infrastructure (as represented in the system diagram, units of measurement, and underlying equations) accurately represents the best understanding of the cause-effect relationships in the real system. The system diagram should provide an accurate schematic representation of how the system works. For example, the disease *X* system schematic in Figure 3.1 would not be structurally valid if it did not take into account the infectiousness of the disease or the rate at which *Susceptible Fish* come into contact with *Sick Fish*. The check for structural validity requires that we have a reasonably detailed understanding of how the system works and how the various system elements (i.e., reservoirs, flows, and converters) are interrelated. An expert who is familiar with the system we are modeling should provide this understanding. This person's knowledge serves as the benchmark with which the model's infrastructure is compared.

2. Predictive validity. The model exhibits **predictive validity** if its predictions of system behavior adequately mimic the real system. Running the model for several "benchmark" cases in which the behavior of the real-life system can be either predicted from theory or can be known through direct observation best assesses the predictive validity. Some possible "benchmark" cases that can be used to check predictive validity are:

- **Baseline behavior patterns.** We may know something about how the system behaves under "normal" circumstances even before we build a model. This knowledge may be fairly detailed (e.g., "the system cycles every 25 hours with an equilibrium of 60 units") or it may be general (e.g., "the system cycles about every 20–30 days around an equilibrium of 50–80 units"). The model should always be run under conditions that correspond to this baseline behavior. If the model's output matches what we know about the system, the predictive validity is supported.
- **Steady-state behavior patterns.** If theoretical conditions can be identified under which the real system is known to exhibit steady-state behavior, these conditions can be applied to the systems model and the model can be run. If the model exhibits steady-state behavior, then the predictive validity is supported.
- **Runaway behavior patterns.** It is sometimes possible to identify conditions under which the system will exhibit "runaway" or "out-of-control" behavior. These conditions can be duplicated in the model and checked.

The process of model validation should be viewed as an iterative process that may require several cycles of activity. Each time the model is checked, we may identify changes that need to be made. The values of particular constants may need to be changed, units may be incorrectly specified, equations may be incorrect, or the systems diagram may need to be changed. After these modifications are made, the validity is rechecked until we are convinced that the model will meet our needs.

3.4.2 Checking Structural Validity of the Fish Disease Model

Open up the STELLA® model *CHAP3a.STM* on your CD-ROM. This model corresponds to Figure 3.1. Notice that the units of all the system elements are documented and the equations for calculating system quantities are provided.

A check of this model's structural validity requires us to compare the system diagram, units, and equations with the description of the system that is provided in Sections 3.2.1 and 3.2.2. It is left to the reader to confirm that the units are all correctly specified. The equations for calculating the *Exposure Rate* and the *Getting Infected* flow process are also consistent with the theoretical discussions provided earlier.

In order to evaluate how well the system diagram matches reality, it is good practice to first develop an overall "feel" for how the system diagram is laid out. This is done by carefully studying the system diagram and trying to discern how the diagram does or does not correspond to the "expert's" description of the system. Areas that are not consistent with what we know about the system should be highlighted and reconciled. The key questions to keep in mind are:

1. Does the collection of relationships and activities, as represented in this diagram, make logical sense?
2. Do these relationships match the description provided by the expert?

One word of caution: A careful study of the model's infrastructure will almost always identify ways in which the model could be enhanced or modified to better match the real system. This is to be expected because the model is at best a crude representation of the real system. Priority should be given, however, to correcting those features that are clearly illogical or that are believed to be significant factors in affecting the model's predictive utility. Judgments between those items that are critical and those that can be ignored can be made by using some common sense and after gaining experience with systems thinking.

A comparison of the model in the file *CHAP3a.STM* with the system description in Sections 3.2.1 and 3.2.2 suggests that the model's infrastructure closely matches the real-life system. One might argue, however, that the model is unrealistic in that it assumes that no fish are born into the system during the entire simulation. To counter this argument, one could note that because the fish population is known to be stable when the disease is absent, then we know that the net growth of the population (births minus deaths) must be zero. Hence, it is not necessary to include birth and death processes.

3.4.3 Checking the Predictive Validity of the Fish Disease Model

The purpose statement in Section 3.3 describes some baseline behavior patterns against which we can check our model. These are:

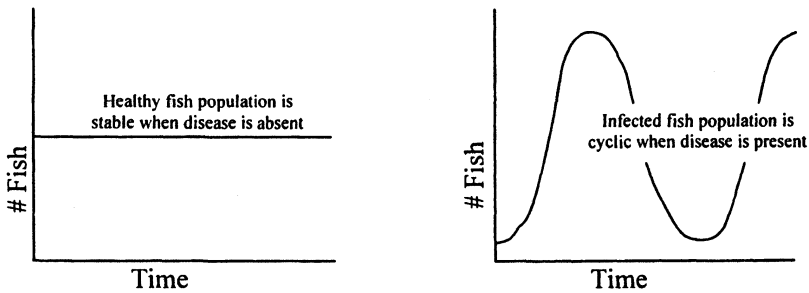


FIGURE 3.2. Baseline behavior patterns for the fish disease model.

1. The population of fish is relatively stable over time in the absence of the disease.
2. The disease seems to reappear on a regular cycle. This implies that the number of infected fish in the system follows some sort of oscillatory behavior.

These baseline behavior patterns are displayed graphically in Figure 3.2. The first graph describes the behavior of the system whenever there are no infected fish in the ecosystem. The second graph describes the behavior of the system whenever infected fish are present. One of the exercises at the end of the chapter will ask you to check the predictive validity of the model against these two baseline behavior patterns.

3.5 Applying the Strategy: Exploratory Analysis

3.5.1 The Goal of Exploratory Analysis

The main purpose of **exploratory analysis** is to use the model to develop an understanding of how the system responds or “adapts” to changing stimuli. Through such an understanding, scientists and problem solvers can appreciate how policy or technology might impact system performance. Exploratory analysis can be thought of as the process of conducting a series of “experiments” with the model to evaluate how the system responds to different conditions. Many of the experiments may involve conditions that we would not seriously consider applying to the real-life system. Our purpose is more to “play” with the system and see how it responds than to try and find optimal conditions for the system to run. We will look for practical ways to change and improve the system later when we discuss the process of **case analysis**.

The mode by which a system responds to changes is determined by something that modelers refer to as the **dynamics** of the system. The dynamics are created by the interrelationships among the system components (i.e., the

reservoirs, flows, and converters). These interrelationships are represented schematically in the system diagram; however, the behavior of a complicated system is difficult (and may be impossible!) to predict from the system diagram alone. In fact, the same system diagram can lead to dramatically different behaviors, depending on the relative sizes of ever-changing flow rates or other quantities in the system. It is therefore useful to run “experiments” on the model to improve our understanding of how and why the system behaves the way it does.

As a result of a thorough exploratory analysis, the modeler should be able to answer questions like the following:

1. What role does each system element play in the overall behavior of the system?
2. Which individual elements have the biggest affect on the overall system? Which seem to have little overall influence?
3. Are there any combinations of system elements that, when taken together, exert an unusually large influence on the system?
4. How does the system behave whenever it is perturbed or upset?
5. Are there conditions under which the system collapses or runs out of control?

Being able to answer the preceding questions can lead to new, unanticipated options for affecting the behavior of the system through policy or technology. In fact, the exploratory analysis of an environmental model can lead to some exciting insights. For example, suppose that we had access to a valid systems model for the formation and depletion of stratospheric ozone. An exploratory analysis of the model would identify those factors that most dramatically affect the formation of ozone. Policy makers and scientists could then direct their attention to developing policies and technologies that would impact those “high-leverage” factors.

We will outline two important types of “experiments” that are part of an exploratory analysis. These two types of experiments are:

1. ***PULSE, RAMP, and STEP experiments*** to deliberately “shock” the system and observe its behavior.
2. ***Sensitivity experiments*** to look for those variables that exert a high degree of influence on the response of the system and those variables that exert a minimal influence.

3.5.2 Using *PULSE, RAMP, and STEP Experiments* to Study System Dynamics

The dynamics of a system are often difficult to observe or understand whenever the system is stable. For example, consider a simple aquatic ecosystem. Under normal conditions, there are many processes at work to keep this system “in balance.” When this balance exists, a casual observer might

conclude that nothing is really happening. We know, however, that there are many complex and active processes working within the ecosystem. Processes such as birth, death, disease, decay, and weather are all operating to create the stable behavior that we observe; unfortunately, the workings and roles that each of these processes play may not be evident when everything is in balance.

Now suppose that the population of a particular organism in the ecosystem was suddenly increased by 100%. Such a “shock” would undoubtedly cause the entire system to adjust in possibly unanticipated ways. We might see, for example, that another species of animal in the system suddenly begins to flourish, whereas another eventually disappears. We might even find that the species whose population was originally doubled eventually dies off. Over time, the entire ecosystem might change to have a dramatically different look. During this period of adjustment, the underlying processes and interrelationships that make up the original system can be observed and better understood.

Even if it were possible to perform such an experiment, we would be highly reluctant to deliberately manipulate an ecosystem in this way. The consequences would be nearly impossible to predict, and could even have a devastating effect; however, we can perform a similar experiment on a validated simulation model with no risk of endangering an entire ecosystem.

Hence, one useful strategy for developing an understanding of how a system works is to deliberately upset or *perturb* the system and then watch how the system reacts. By observing and explaining the behaviors we see during these perturbations, we can gain a richer understanding of the overall system.

For example, recall that the *Susceptible Fish* population in the fish disease model is stable if there are no sick fish in the system. How would the system respond if 10 sick fish were added to the system after 100 days of steady-state behavior? Would the *Susceptible Fish* population eventually collapse? Would there be an overall decline in the number of healthy fish?

We could attempt to answer these questions by setting the initial value of the Sick Fish conveyor to 10 fish. This unfortunately would not allow us to observe how the system would transition from its steady state to a “diseased” state. This is because the 10 sick fish would be in the system from the beginning, rather than suddenly appearing after 100 days of steady-state behavior. Thus, we need some method of *perturbing* the system so that we, in effect, add 10 sick fish on the one-hundredth day of the simulation.

There are many common ways to perturb a system. These can be applied using predefined functions in the system modeling software, or they can be introduced in “real time” as the simulation is running. We will describe here three commonly used perturbations. All three of these can be used to

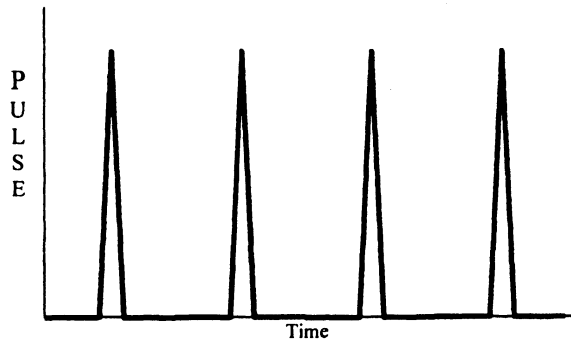


FIGURE 3.3. Graph of a PULSE perturbation.

introduce changes in the sizes of flows or converters at prespecified time points during the simulation.

1. The **PULSE**. This type of perturbation causes the value of a flow or converter to “spike” at a specified point in time. The pulse can be introduced at a single point in time or at multiple points in time. Figure 3.3 illustrates this type of perturbation.

In order to illustrate the use of a **PULSE** to explore the system dynamics, suppose that the initial value of the *Sick Fish* reservoir is set to zero. We could add an inflow of *Infected Fish* into that reservoir and use STELLA’s® **PULSE** function to force the inflow to be zero fish/day until day 100, when the flow would “spike” to 10 fish/day for 1 day. (See the appendix to this chapter for more information about STELLA’s® **PULSE** function). Figure 3.4 shows how the original system diagram in Figure 3.1 can be modified in this way. Figure 3.4 also displays the initial values and other constants in the model. Figure 3.5 displays a graph of the behavior of the system over a 730-day run. A brief study of this graph reveals the following dynamics:

- The system runs in a steady state until day 100 when the 10 infected fish are “spiked” into the system.
- Once a few infected fish are introduced, the disease quickly spreads through the population. Within about 30 days the epidemic is in full swing and more than 200 fish (20% of the population!) have the disease.
- The system oscillates on about a 60-day cycle. These oscillations gradually damp out.
- Each of the three reservoirs are “out of phase” with each other. That is, each reaches the peak in its cycle at a different time than the others.
- All of the reservoirs have a gradual downward trend.

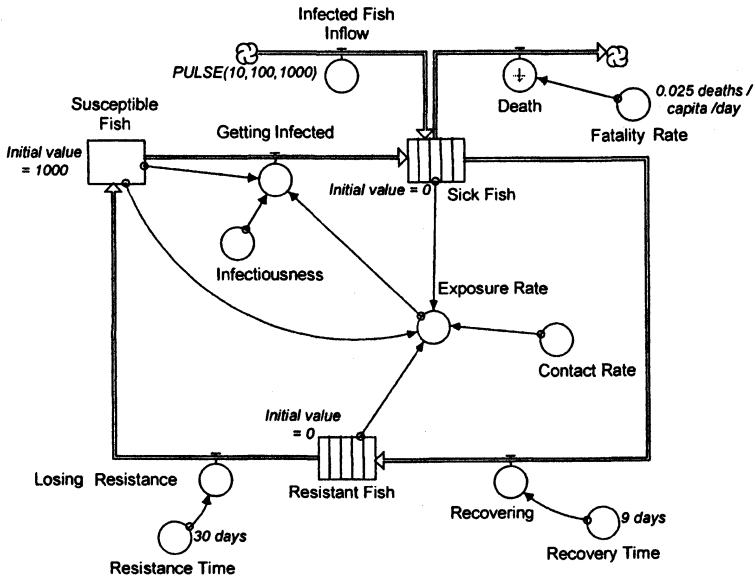


FIGURE 3.4. Modified fish disease model to introduce 10 infected fish on day 100.

It is left as an exercise for you to provide explanations for each of these behaviors.

2. The **STEP** (see Figure 3.6). This STEP perturbation causes the value of a flow or converter to make a one-time step change at a specified point

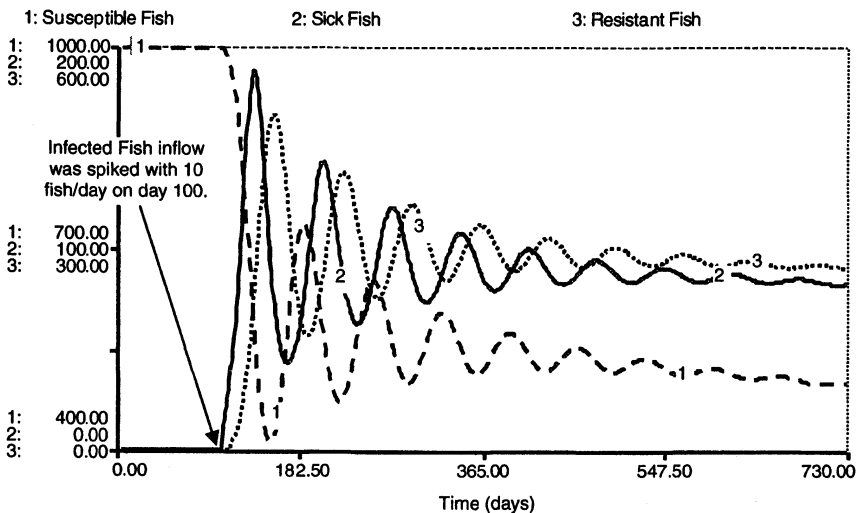


FIGURE 3.5. System behavior when 10 infected fish are added on day 100.

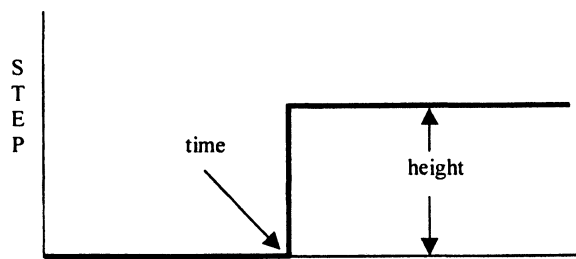


FIGURE 3.6. Graph of the STEP perturbation.

in time. The STEP is different from the PULSE in some important ways. The PULSE introduces a one-time event that occurs at a single, isolated point in time and then ceases. The STEP function, however, will sustain the change for the rest of the simulation. A graph of the STEP perturbation is displayed in Figure 3.6. Figure 3.7 shows the behavior of the system whenever the PULSE inflow in Figure 3.4 is replaced with a STEP inflow of 10 fish/day, beginning on day 100. (See the appendix to this chapter for more information on STELLA's® STEP function). Notice that the PULSE introduces 10 sick fish into the system only on day 100. The STEP perturbation introduces 10 fish on day 100 and on each day thereafter. What behaviors are evident from Figure 3.7? Why do they occur?

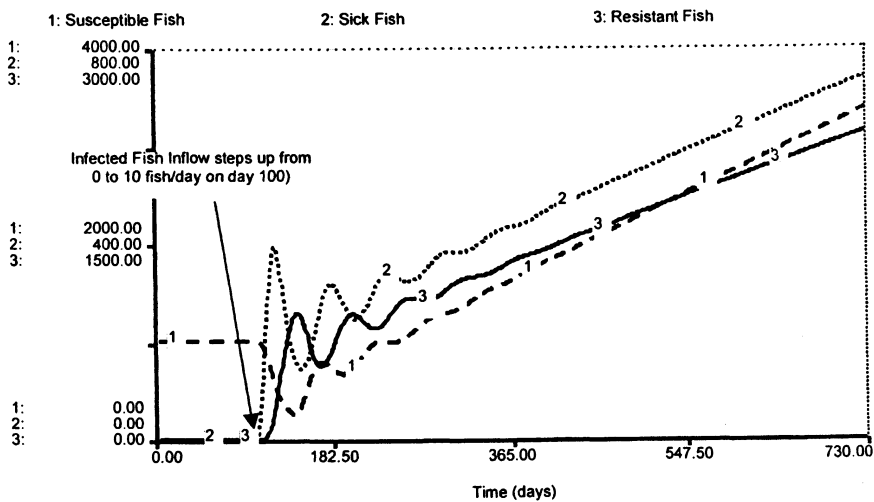


FIGURE 3.7. System behavior whenever a STEP perturbation of 10 fish/day is used for the infected fish inflow of Figure 3.4.

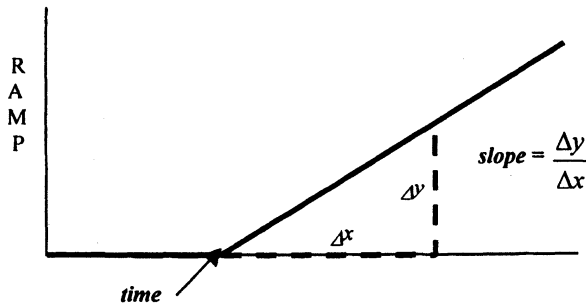


FIGURE 3.8. Graph of the RAMP perturbation.

3. The **RAMP** (see Figure 3.8). The RAMP perturbation increases or decreases at a constant rate, beginning at a specified time point. A graph of the RAMP is displayed in Figure 3.8. Figure 3.9 shows the system behavior when the PULSE inflow of sick fish in Figure 3.4 is replaced with the RAMP inflow with a slope of 10 fish/day that begins on day 100. In other words, there are no infected fish in the system for the first 100 days. Ten sick fish are introduced on day 100. Twenty sick fish are introduced on day 101, and so on.

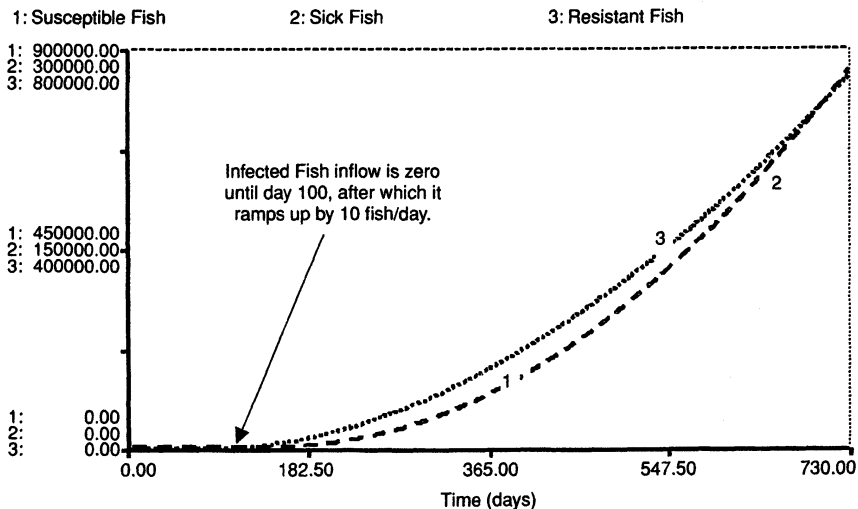


FIGURE 3.9. System behavior whenever RAMP (10,100) is used for the infected fish inflow in Figure 3.4.

3.5.3 Using Sensitivity Experiments to Identify Variable Influence

The goal of *sensitivity experiments* is to identify those variables in the system that fall into the two following categories.¹ These categories are not exhaustive. Most systems will contain some variables that do not fit into either category.

1. High-leverage variables are those variables whose values have a significant impact on the system's behavior. When the values of these variables are changed even slightly, the system behavior can change dramatically. High-leverage variables are those quantities whose values must be carefully validated with the real system. Because they exert such an influence, even a slight error in specifying their value can have a significant impact on the predictions from our model. The high-leverage variables also provide the best opportunities for policy makers to impact an environmental system. If policies or technologies can be instituted that exert even a slight impact on a high-leverage variable, the change to the overall system could be significant.

2. Low-leverage variables are those variables whose values have a minimal impact on the system. The value of a low-leverage variable can be changed (within a reasonable range) without significantly affecting the overall system. The values of these variables need not be as carefully validated against the real system. In addition, the low-leverage variables are important because they provide options for policy makers to change the system in ways that may have important economic or other benefits without adversely affecting the system.

A sensitivity analysis (in its most basic form) involves the following steps:

1. Identify the *exogenous variables* in the system. These are the variables whose values do not depend on other quantities in the system, but are instead set by the user or the model builder. Exogenous variables will correspond to those converters in the system diagram that do not have any connectors pointing into them. Note also that starting values for reservoirs can be considered to be exogenous variables. The influence of the exogenous variables will be evaluated through the sensitivity analysis.

¹ Sensitivity analysis typically involves much more than classifying individual variables as "high-leverage" or "low-leverage." For example, it is often the case that *combinations* of variables work together to exert a high degree of leverage over the system. However, the strategies for identifying synergistic groups of variables are beyond the scope of this text. We will restrict our attention to a single-variable sensitivity analysis.

2. For each exogenous variable, make a series of model runs, changing the value of the variable slightly from run to run. The variable should be varied over a range spanning the reasonable set of values that are possible or likely in the real system. In the absence of a reliable range of “reasonable values,” the variable can be varied over a fixed range of plus or minus 50%.

3. Observe and compare the system behavior for each run. Determine the extent to which the system behavior changes whenever each exogenous variable is changed. It is important to note that significant changes in system behavior can manifest themselves as changes in the overall *shape* of the system response over time, or in the *level* of the response. For example, one high-leverage variable may have values that cause an oscillating reservoir to cycle more and more wildly over time until it runs out of control. Another high-leverage variable may not have settings that lead to ever more extreme oscillations; rather, it may cause that same reservoir to oscillate around a much higher average value. It is advisable to develop some sort of quantitative measure of the change in system behavior that each exogenous variable causes. The particular choice of which measure to use depends on which types of changes in system response are of most interest. Examples would be

- The percentage change of a stock at a particular point in time, when comparing the runs from the lowest setting of the exogenous variable and the highest setting (when interested in changes the *level* of the response)
- The percentage change in the average value of a stock over a particular span of time values (when interested in changes in the *level* of the response)
- The percentage change in the variance of the values of a stock over a particular span of time values (when you are interested in changes in system *variability* through time)

4. Identify those variables that have the most impact and those that appear to have little impact. If possible, try and give a rationale for the way each variable is classified. That is, try to explain why this particular variable exerts so much (or so little) influence on the system.

We will now illustrate the steps in a sensitivity analysis using the fish disease model. Open up the file *CHAP3a.STM* and follow along as we work through the main steps.

Step a: Identify the exogenous variables. It is clear from the system diagram (see Figure 3.1) that there are eight exogenous variables in this model. They are:

- *Infectiousness*
- *Contact Rate*

- *Fatality Rate*
- *Recovery Time*
- *Resistance Time*
- *Sick Fish* Population at time = 0
- *Susceptible Fish* Population at time = 0
- *Recovering Fish* Population at time = 0

Step b: Make a series of runs for each exogenous variable, changing the variable slightly from run to run. We will illustrate this step using the *Recovery Time* variable. It is left as an exercise to repeat this process for each of the other seven exogenous variables in the model. You should make a minimum of three sensitivity runs on each exogenous variable. One run should be made with the variable set at its lowest reasonable value, another with the variable set to its highest reasonable value, and the third with the variable set at a value midway between the two extremes. The choice of what constitutes the highest and lowest “reasonable values” depends on our knowledge of the system, some common sense, and intuition. In the case of our imaginary disease *X*, a reasonable range of values for the *Recovery Time* might be available from experts who are familiar with the disease. It is possible that field or experimental data exist that give information on the duration of disease *X* or of some other disease that is similar to disease *X*. In the absence of any such information, we recommend picking a range that is 50% above and below the nominal level given in the model, unless this violates some physical constraints inherent to the system. We will assume that we do not have any data on the *Recovery Time* of disease *X* other than the nominal value of 9 days that is provided as the default value in the model. Hence, we will make sensitivity runs with the *Recovery Time* set to 4.5 days, 9 days, and 13.5 days.

(*Note:* STELLA®, provides a convenient and powerful facility for making sensitivity runs. By selecting *Sensi Specs* under the *Run* menu, we are presented with a dialogue that allows us to select an exogenous variable, specify the different variable values that we wish to use in the sensitivity runs, and then construct a graph that will display the results of all three runs. See the STELLA® Help facility for a description of how to set up a series of sensitivity runs in this way).

Step c: Observe and compare the system behavior for each run. Figure 3.10 displays the behavior of the *Sick Fish* reservoir for the three *Recovery Time* sensitivity runs. Note that curve #1 shows the behavior when the *Recovery Time* is 4.5 days, Curve #2 shows the behavior when the *Recovery Time* is 9 days, and Curve #3 show the behavior corresponding to 13.5 days. Notice from Figure 3.10 that the *Recovery Time* variable seems to have a dramatic effect on the system (at least as measured by the *Sick Fish* reservoir). Whenever the *Recovery Time* is 4.5 days, the epidemic is short-lived and the recurrent epidemic cycles disappear. When-

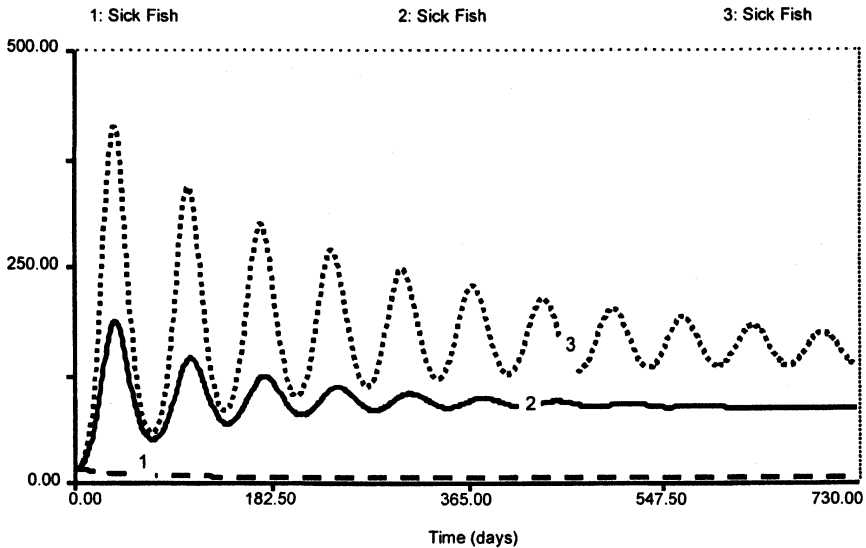


FIGURE 3.10. Results of the sensitivity runs for recovery time where #1 = 4.5-day recovery time, #2 = 9-day recovery time, #3 = 13.5-day recovery time.

ever the *Recovery Time* is 9 or 13.5 days, the cyclic behavior is present. Furthermore, the higher the *Recovery Time*, the more fish there are in the *Sick Fish* reservoir.

Step d: Identify those variables exhibiting a high degree of leverage over the system and those exhibiting little leverage. If possible, give a rationale for why the high-leverage variables are so important and the low-leverage variables are not. *Recovery Time* does appear to exert a significant influence over the system (see Figure 3.10). The differing responses to the *Recovery Time* variable displayed in Figure 3.10 can be explained as follows. Whenever the *Recovery Time* is short, infected fish will not stay sick for long. This means that *Sick Fish* that survive the disease soon become resistant (for 30 days). Furthermore, because the simulation begins with an initial value of 10 fish in the *Sick Fish* reservoir, the spread of the disease is slower than the rate at which fish are recovering. Hence, the *Sick Fish* reservoir decreases in size from the beginning. Once the number of *Sick Fish* reaches zero, the epidemic is over and the system achieves steady state. The fact that the system seems to be so sensitive to *Recovery Time* suggests that we should do some research to determine more closely the actual *Recovery Time* for disease X. In addition, the sensitivity to *Recovery Time* indicates that any corrective measures that could shorten the actual *Recovery Time* for this disease could significantly reduce the long-term impact of this disease on the ecosystem.

3.6 Applying the Strategy: Case Analysis

3.6.1 Overview of Case Analysis

Once we have validated the model and performed an exploratory analysis, we should have a much deeper understanding of the dynamics that dictate how the system works. In addition, the exploratory analysis may have revealed some unanticipated opportunities for impacting the system in either a positive or negative way. These understandings can now be applied to formulate and model changes to the system that are aimed at changing or improving system performance. For example, we are interested in evaluating options for addressing the outbreak of disease *X* in our imaginary aquatic ecosystem. Our overall goal is to minimize the impact of the disease so that we avert a dramatic drop in the fish population.

Recall that the modeling purpose statement described in Section 3.3 includes a description of the types of interventions or changes to the system that we wish to evaluate with the model. It is often the case that individuals who are charged with addressing environmental problems may have a priori ideas of how to address an environmental problem. A reliable and valid systems simulation model provides a powerful tool for evaluating those ideas. The purpose statement for the Fish Disease Model (provided in Section 3.3) described two particular interventions that would be evaluated. Those were:

1. Repeated capture and removal of infected fish.
2. Introduction of a new resistant strain of fish for which the infectiousness of disease *X* will be reduced by 50%.

We must define how the system model in Figure 3.1 would be modified to simulate each of these options. This process of incorporating a policy or corrective action into an existing systems model requires the same modeling and analysis skills covered in the first three chapters of this text. In some cases, simply changing the values of some constants in the original model can simulate the system intervention. In other cases, the intervention can be modeled by using a *PULSE*, *STEP*, or *RAMP* function as discussed earlier. It is also possible that some interventions must be modeled by changing the system infrastructure (as represented in the system diagram). Some elements may need to be deleted and others added or modified.

Once the intervention has been built into the model, it should be validated (if possible) by running the model and checking the model output with some simple, known cases. The newly modified model can then be analyzed using the same techniques used in the exploratory analysis described earlier. The results of the “new” model should be compared with the results of the “old” model to evaluate the impact of the proposed intervention.

3.6.2 Case Analysis for the Fish Disease Model

Case 1: Modeling the first intervention (repeated capture and removal of infected fish). Assume that this intervention involves the use of commercial and private fishing fleets to remove any diseased fish that are caught during routine fishing operations. This intervention can easily be modeled by increasing the *Fatality Rate* that governs the leakage outflow from the *Sick Fish* reservoir. This outflow was originally intended to represent fish that died from the disease; however, we can broaden the meaning of this outflow to include fish that either die from the disease or are caught and removed by local fishing fleets. One approach to modeling the intervention would be to introduce a *STEP* function into the *Fatality Rate* converter to make a step change at some point in time after the beginning of the simulation. The results could be modeled and compared with the baseline case using the original *Fatality Rate*. In order to implement this in the model, we will need to determine the level of increase in the *Fatality Rate*.

Case 2: Modeling the second intervention (introduction of a new resistant strain of fish that are more resistant to the disease, thereby decreasing the infectiousness by 50%). A coarse evaluation of this intervention can be accomplished by decreasing the Infectiousness by 50%. This approach is tantamount to assuming that there was a 100% changeover from the original strain of fish to the new, more resistant strain (can you explain why this is so?). This approach, however, also represents a “best case” analysis of this intervention. By assuming that we have completely replaced the old fish population with a new one, we can see what the impact would be on the spread of the disease. If this “best case” simulation shows little or no reduction in the epidemic, then the intervention is not worthy of further investigation. If the “best case” simulation indicates the potential for significant improvements, however, then we can then consider how to model the case more realistically in which the new fish population gradually displaces or mixes in with the original population. Both of these approaches are left as an exercise.

3.7 Exercises

Section 3.2

1. Fill in the unspecified units for each of the quantities in Table 3.1.
2. The system diagram in Figure 3.1 could be modified so that a conveyor is not used for the *Resistant Fish* reservoir. Using this approach, the *Resistance Time* could still be used to dictate how long a fish would typically remain resistant. Assume that the *Resistance Time* is 30 days and give an equation for calculating number of fish flowing through the *Losing*

Resistance flow under this new setup. (Hint: If the *Resistance Time* is 30 days, what fraction of the *Resistant Fish* would typically lose resistance each day?)

3. What is the difference between the approach used in Question 3 and the approach based on using conveyors?

Section 3.3

4. You will find here two example purpose statements for other modeling problems. For each example, identify (1) the system to be studied, (2) the behaviors to be understood, and (3) the core questions to be addressed.

- a. The purpose of this modeling effort is to understand the underlying mechanisms by which global CFC production is depleting stratospheric ozone levels, and to evaluate the long-term impact of the Montreal Protocol on those levels.
- b. The purpose of this modeling effort is to understand why the effluent from the local waste treatment facility in a university community exhibits a spike in biochemical oxygen demand for the 2 weeks after the end of a university holiday. We also wish to determine if this spike can be eliminated by seeding the bacteria population in the settling pond.

5. You have been hired by a solid waste management company to analyze the potential for solid waste collection in a small college community whose population is increasing at a rate of 3% per year. In particular, the company is interested in evaluating the growth of waste in this community over a 20-year period. They are also interested in determining how this growth may be affected by a recycling and reuse campaign expected to begin within the community in the fifth year. Write a purpose statement for this modeling problem.

Section 3.4

6. Identify three possible enhancements to **CHAP3a.STM** to improve its structural validity. Modify the model to incorporate the enhancement that you think is most important among the three.

7. Check the predictive validity of **CHAP3a.STM** by running it under the two sets of conditions specified in baseline behavior patterns in Figure 3.2.

- a. Identify modeling conditions to use to check against the first graph Figure 3.2. Run the model for 2 years (730 days) and check the system behavior. What do you see? Are there any changes to the model that are suggested from this check?
- b. Select conditions to use to in order evaluate how well the model matches the second graph in Figure 3.2 (note that several different

- conditions are possible). Run the model for 730 days and check the system behavior. What do you see?
- Does the model duplicate the baseline behavior patterns? If it differs from Figure 3.2, in what ways does it differ? Does this make physical sense, given the context of the problem? Are any modifications to the model suggested by this check?

Section 3.5

- Briefly explain the behaviors you see in Figures 3.5, 3.7, and 3.9.
- Open the STELLA® model **CHAP3a.STM**. Modify this model to incorporate each of the following perturbations by using the PULSE, STEP, and RAMP functions. Make sure you run only one case at a time (i.e., do not combine the cases described later). For each case, run the model for 2 years. Observe the behavior of the system and write a brief paragraph explaining why the system exhibits the particular behavior that you see. You can refer to the appendix at the end of this chapter to see how to use combinations of the PULSE, STEP, and RAMP functions to model each case.
 - The *Contact Rate* stays at a level of 2 contacts/fish/day until day 100, at which time the *Contact Rate* increases to 4 contacts/fish/day. (*Hint*: Add a constant value to a STEP function).
 - The *Contact Rate* stays level at 2 contacts/fish/day until day 100, at which time the *Contact Rate* increases to 4 contacts/fish/day. On day 400, the *Contact Rate* returns to 2 contacts/fish/day. (*Hint*: Use two different STEP functions and a constant value).
 - The *Resistance Time* begins at 0 and then increases by 0.5 days per day of simulation. (*Hint*: Use the RAMP function. You do not have to specify a time value).
- Open the STELLA® model **CHAP3b.STM**. This model is in “flight simulator mode” and allows you to make real-time changes to several variables. (See the appendix for a description of the “flight simulator” mode of simulation in STELLA®). Use the slider bar controls to make adjustments as the model runs. Run the model several times, each time simulating a step change in one variable. Observe the impact on system behavior, then briefly summarize what role you think each variable plays in the behavior of the overall system.
- Use the model **CHAP3b.STM** and STELLA’s® **Sensi Specs** to help you perform a sensitivity analysis on the remaining seven exogenous variables in the Fish Disease Model. Use a +/- 50% range for each variable. Produce a brief write-up for each variable that includes:
 - A listing of the three values you used for the sensitivity analysis.
 - A graph showing the results of the sensitivity runs for that variable.
 - A brief paragraph summarizing how the system responded to the changes

in the exogenous variable and why you think the system responded that way. You should also state clearly whether you consider the variable to exert high leverage on the system, low leverage, or neither. Then state what implications your analysis has for taking corrective action against disease X.

Section 3.6

12. Modify the model **CHAP3a.STM** to model the first intervention (repeated capture and removal of infected fish). Assume that this policy is instituted 50 days after the beginning of the simulation and that it results in doubling the *Fatality Rate*. Briefly summarize what impact this intervention has on the system. Would you recommend this approach to curbing the disease? Why or why not?

13. Modify the model in **CHAP3a.STM** to run the “best case” analysis of the second intervention (introducing a more resistant strain of fish for which the infectiousness of disease X is 50% lower). Based on the results of the “best case” analysis, state whether you would recommend any further evaluation of the second intervention.

14. Modify the **CHAP3a.STM** system diagram to show how you would simulate a gradual introduction of 40 of the more resistant fish per year. (Hint: you will need to add a new series of reservoirs to keep track of the more resistant fish.)

15. Formulate one other realistic intervention for addressing the disease X epidemic. Briefly describe the intervention. Modify **CHAP3a.STM** and make some simulation runs to evaluate your idea.

3.8 Suggested Readings

Edelstein-Keshet, L. 1988. *Mathematical Models in Biology*. New York: Random House.

Hethcote, H.W. 1976. Qualitative Analyses of Communicable Disease Models. *Mathematical Biosciences* 28:335–56.

See the World Wide Web page from the American Oceans Campaign at www.american-oceans.org/issues/fishkill.htm for a brief introduction to the *pfiesteria piscicida* phenomenon.

3.9 Appendix: Modeling System Perturbations in STELLA®

STELLA® provides two primary ways of introducing perturbations into a system model. These are:

- Use of predefined functions that “fire” at specified points in time
- Real-time manipulation of system variables while running the system in flight simulator mode

Both approaches can be useful, depending on the context. This appendix briefly describes how to implement each.

3.9.1 Use of Predefined Functions

We will focus attention on the three types of perturbations described in this chapter (i.e., the PULSE, STEP, and RAMP). Each of these can be modeled in STELLA® by using functions having the same names. These functions can be used individually or in combination as part of the equation defining a flow or converter in the system.

1. The PULSE function. This function causes the value of a flow or converter to “spike” at a specified point in time. The format for the PULSE function is: PULSE (*volume, first time, interval*), where *volume* dictates size of the pulse (the volume is equal to the height of the pulse, assuming it lasts for one full-time unit), *first time* is the time of the first occurrence of the pulse, and *interval* is the number of time units between successive pulses. The user must specify the volume. The values of first time and interval can be left unspecified. If first time is not specified, however, then the pulse will occur at the beginning of the simulation run. If the interval is not specified, then the pulse will occur at each Δt step in the simulation. The height of the pulse is equal to $\text{volume}/\Delta t$. Hence, if $\Delta t = 1$, then the pulse height is the same as the *volume*. If $\Delta t < 1$, then the pulse height will be greater than the volume.

2. The STEP function. This function causes the value of a flow or converter to make a one-time step change at a specified point in time. The format for the STEP function is: STEP (*height, time*), where (1) both the height and time values must be specified, (2) the height value defines the size of the step, and (3) the time value specifies the time at which the step is to occur. For example, if a quantity in the model is specified with the value STEP (10,23), then that quantity will increase by 10 units at the twenty-third time unit.

3. The RAMP Function. The RAMP function increases or decreases at a constant rate, beginning at a prespecified time point. The format for the RAMP function is RAMP (slope, time), where (1) the slope defines the rate at which the RAMP changes (in units of change per unit time), (2) the time value defines the starting time at which the charge begins and, (3) the value of the slope must be specified, but the time value is optional. When the time is not specified, the RAMP begins at the outset of the simulation. Prior to the starting time, the RAMP function is set to zero.

3.9.2 Use of the Flight Simulator Mode

By running a STELLA® model in the “flight simulator” mode, the user is allowed to make real-time adjustments to model variables as the simulation progresses. The effects of these adjustments can be seen immediately. In this way, perturbations can be introduced and studied. These changes, however, cannot be introduced with the same level of accuracy and timing as when using the predefined functions described earlier. Nonetheless, the flight simulator mode provides a powerful tool for studying system dynamics because the user can introduce a given perturbation and then immediately observe the system response.

To run a model in the flight simulator mode, select *Time Specs* under the **Run** menu. Then change the *Interactive Mode* from *Normal* to *Flight Sim* in the resultant dialogue box. Any subsequent runs will allow dynamic, real-time changes to the system during the run.

In order to make these real-time adjustments, it is necessary to add some controllers (e.g., slide bars) that correspond to the variables you wish to adjust during the run. These controllers are found on the tool bar in the top level of the STELLA® modeling environment. After placing the icon for a controller onto the workspace, just double click on it. This will provide a dialogue box from which you can select the variable to which that controller will be connected. Models in later chapters are designed primarily for flight simulator mode.

4

Modeling Predator–Prey Systems

Chapter Objectives—

After you finish this chapter, you should be able to:

- 1. Describe qualitatively and mathematically the growth of populations with and without carrying capacity constraints.*
- 2. Describe qualitatively and mathematically the Lotka-Volterra predator-prey model.*
- 3. Explain why predator-prey relationships often exhibit oscillatory behavior.*
- 4. Manipulate variables in a dynamic predator-prey model and explain the outcomes from these perturbations.*
- 5. Build or enhance a predator-prey model to include multiple predators or multiple prey.*

4.1 The Problem

Any child who has ever witnessed a hungry spider attacking a helpless insect that is caught in its web has observed first hand the brutal reality of predator-prey systems. We live in a world where living things consume other living things for survival. In order for this to occur, there must necessarily be a creature that captures and devours (the predator) and a creature that is captured and devoured (the prey).

In a stable ecosystem, both predator and prey live in a symbiotic balance, where the size of each population is regulated by the size of the other. Humans can interfere with these ecosystems, however, in ways that tilt the balance between predator and prey to destructive levels. For example, human activity may destroy the habitat of a particular prey species, thereby destroying the species altogether. Along with this species loss is the loss of any predator species that depended upon this prey for survival.

To our credit, humans have also recognized the importance of maintaining diverse and rich natural ecosystems. As such, we have begun to repopulate natural areas with once-native predator species that had become extinct due to human activities. Just as the removal of a prey species from their habitat has an impact on the entire ecosystem, the introduction of predator species into an area also impacts the ecosystem and the populations of other species within that ecosystem.

This chapter will introduce you to the rich relationships evident in a predator-prey system. As you might expect, these relationships cause system

behavior patterns that are similar to those discussed in Chapter 2. You will see elements of exponential growth, logistic growth, and oscillation in the predator–prey relationship. This chapter will illustrate how these behavior patterns are combined to model real-life predator–prey systems.

Our discussion begins with the description of an imaginary problem that provides the context around which predator–prey models will be discussed throughout the chapter. You will ultimately build and explore a model that analyzes this problem. The problem is common to wildlife managers in many areas of the world. It involves the introduction of a predator species into a wildlife refuge.

Problem 4.1 You are a wildlife manager conducting a program to reintroduce red wolves into a major wildlife refuge. Although once native to the area, these wolves were eliminated decades ago by hunters, farmers, and developers. The wolves' primary source of food is deer, of which the refuge has a large population. You would like to explore how the deer and wolf populations might interact and the impacts this interaction might have on the future populations of both species.

4.2 Background Information

4.2.1 Modeling Ecosystem Populations

We learned in Chapter 1 that the size of a population P at a future time $(t + \Delta t)$ is given by the simple difference equation

$$P(t + \Delta t) = P(t) + B \cdot \Delta t - D \cdot \Delta t \quad (4.1)$$

where P = the population

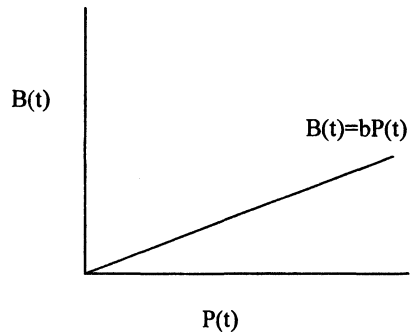
B = the number of births per time unit

D = the number of deaths per time unit

For many species, the number of births and deaths (B and D in Equation 4.1) are a function of *population density*, which is defined as the number of individuals of a given species divided by the area under study. If the area under study is fixed, then births and deaths will be a function of the *size* of the population in this fixed area. In particular, the number of births tends to increase as the population in a fixed area increases (i.e., as the *population density increases*). This happens because a higher population density provides more of an opportunity for mating among individuals in the population. It is easier for individuals to find a mate and produce offspring. (The exception may be where population density is so high, and thus resources so scarce, that individuals choose not to mate or are too unhealthy to do so. This case will be discussed later in the chapter). Thus, we would expect,

$$B(t) = bP(t) \quad (4.2)$$

FIGURE 4.1. Linear relationship of births and population.



where b is a positive coefficient representing a *birth rate* in units of “births per capita per time,” and $P(t)$ is the number of individuals in the population at time t . This relationship is shown in Figure 4.1. Here we show the relationship to be linear, but we will see later that we can relax this restriction and explore situations where births take on a variety of functional forms with respect to population size.

We may also expect deaths to increase as the population within a given area increases. This is due to the fact that there will likely be more *intraspecies competition* for limited resources (e.g., food and shelter). Individuals will have to compete with other individuals to obtain the resources needed to survive. In addition, diseases might spread more easily under conditions of high population density. This case is described mathematically as

$$D(t) = dP(t) \quad (4.3)$$

where d is a positive linear death rate constant with the units of “deaths per capita per unit time.” This relationship is shown graphically in Figure 4.2.

The case in which the number of births and deaths are proportional to the size of the population was described in Chapter 2 as an example of the **exponential growth** behavior pattern. The difference equation for this system is given by:

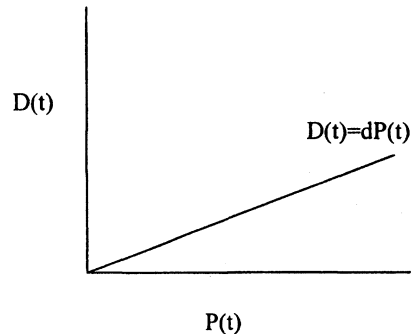


FIGURE 4.2. Linear relationship of deaths and population.

$$\begin{aligned}
P(t + \Delta t) &= P(t) + B \cdot \Delta t - D \cdot \Delta t \\
&= P(t) + b \cdot P(t) \cdot \Delta t - d \cdot P(t) \cdot \Delta t \\
&= P(t) + (b - d) \cdot P(t) \cdot \Delta t
\end{aligned}$$

Following the same reasoning leading up to Equation (2.10), we can derive the equation describing the rate at which the population $P(t)$ changes over time. This rate equation is given by Equation (4.4).

$$\begin{aligned}
P(t + \Delta t) - P(t) &= (b - d)P(t)\Delta t \\
\frac{P(t + \Delta t) - P(t)}{\Delta t} &= (b - d)P(t) \\
\lim_{\Delta t \rightarrow 0} \frac{P(t + \Delta t) - P(t)}{\Delta t} &= (b - d)P(t) \\
\frac{dP}{dt} &= (b - d)P \tag{4.4}
\end{aligned}$$

Equation (4.4) shows that the rate at which the population changes over time (i.e., $\frac{dP}{dt}$) is proportional to the size of the population. The constant of proportionality is $(b - d)$, which is the difference between the birth and death rates. If the birth rate is greater than the death rate (i.e., $b > d$), then the right-hand side of Equation (4.4) will be positive. This means that the population will be increasing in size. If $b < d$ instead, then we will have an ever-decreasing population (and ultimate extinction).

Equation (2.11) gives the equation for the size of an exponential population at time t . Note that the quantity k in that equation corresponds to the difference between the birth and death rates $(b - d)$. That is,

$$P(t) = P_0 e^{(b-d)t}$$

We will now show that the equation for $P(t)$ given earlier is in fact the solution to the rate equation given in Equation (4.4). This can be accomplished by integrating Equation (4.4) and simplifying to find the expression for $P(t)$. This will be shown shortly.

$$\begin{aligned}
\int_0^t \frac{dP}{dt} dt &= \int_0^t (b - d)P(t) dt \\
\int_{P_0}^{P(t)} \frac{dP}{P} &= \int_0^t (b - d) dt \\
\ln P(t) - \ln(P_0) &= (b - d)t \\
\ln \frac{P(t)}{P_0} &= (b - d)t \\
P(t) &= P_0 e^{(b-d)t} \\
&= P_0 e^{\lambda t}
\end{aligned}$$

where $\lambda = (b - d)$. As expected, we obtain an exponential growth/decay function, where the net growth rate is equal to $(b - d)$, the difference between the birth and death rates. The quantity λ represents the **net growth rate** of the population.

The model titled *CHAP4a.STM* on your CD-ROM can be used to illustrate these equations better. This model simulates the growth of a population of deer over time. The model is set up to conduct calculations similar to those in the preceding difference equations over a period of 25 years, with $\Delta t = 0.25$ years. When you open the model, you will see a schematic of the system diagram on top, a set of “slider bars” underneath that diagram, and a graph near the bottom that shows population growth over time. Default settings for birth rate and death rate are shown in the slider bars, with default values of 0.10 births/capita/year and 0.08 deaths/capita/year, respectively. The graph shows the population for the 25-year period for the default settings.

CHAP4a.STM is set to run in “flight simulation” mode. This allows you the opportunity to modify variables while the model is running and to view the response of the model to those modifications. First, run the model without changing any variables by clicking on **Run** in the top toolbar, and then clicking on **Run** in the drop-down menu. You will notice the graph increases exponentially at an annual rate of $b - d$ (or $0.10 - 0.08 = 0.02$). Now, run the model again, but this time gradually adjust either the birth rate or the death rate constants. What happens? Does this make sense?

4.2.2 Population Growth Under Carrying Capacity Constraints

We have so far discussed the growth of a population as if it occurred independently of other populations or resources within the ecosystem. When this is the case, the population will grow (or decay) exponentially. We will now modify the simple exponential growth-decay model to account for conditions where the available resources constrain the growth of a population. The resulting model will resemble the **logistic growth** model described in Section 2.4.

Consider an ecosystem that supports a population $N(t)$. Because the population depends on the resources in the ecosystem for survival, we will assume that $N(t)$ cannot exceed a certain size. In Chapter 2, we referred to this maximum sustainable size for $N(t)$ as the **carrying capacity** of the ecosystem. Let us suppose that the carrying capacity is equal to a fixed value K .

If the birth rate (b) is larger than the death rate (d), the population will grow and eventually approach the carrying capacity K . As this happens, individuals in the population begin to compete for food, shelter, and other resources. This is called **intraspecies competition**. Some individuals die because they cannot find the resources that were once plentiful. Others

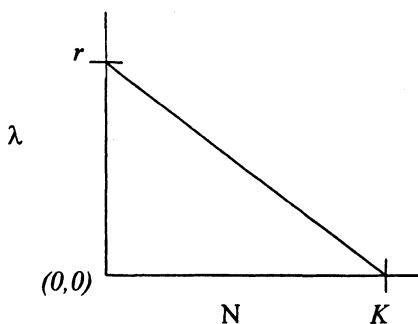


FIGURE 4.3. Net growth rate as a function of population.

catch contagious diseases that are more readily spread in more densely populated areas. Hence, the death rate begins to increase toward a value equal to the birth rate (i.e., $d \rightarrow b$). This causes λ to move closer to zero. When λ finally reaches zero (i.e., $b = d$), then the net growth rate is zero and the population “levels off” at the carrying capacity of the system.¹

The preceding discussion translates mathematically into a functional relationship between the net growth rate (λ), the current population, and the carrying capacity of the ecosystem. Consider the case where the birth rate is constant and the death rate is a function of population and carrying capacity, such that

$$b = r \quad \text{and} \quad d = \frac{rN(t)}{K}$$

Our net growth rate would then be as follows.

$$\lambda = b - d = r - \frac{rN(t)}{K} = r \left(1 - \frac{N(t)}{K} \right) \quad (4.5)$$

In this case, if the size of the population is far below the carrying capacity K , then the net growth rate, λ , will be close to r and the population will increase exponentially. As N increases toward the carrying capacity, however, the net growth rate decreases toward 0. This is shown in Figure 4.3.

Following the discussion leading up to Equation (2.16) in Chapter 2, we can write down the rate equation for the growth of the population $N(t)$.

$$\frac{dN(t)}{dt} = r \left(1 - \frac{N(t)}{K} \right) N(t) \quad (4.6)$$

Chapter 2 discussed the analytical solution to this rate equation. Note that the rate at which the population size changes is high when $N(t) \ll K$. In that

¹ Note that λ may also decrease if birth rates decline with population growth, which is often the case when populations reduce their reproduction rate during periods where resources are in short supply (Southern, 1970).

case, the model follows an exponential growth curve with a growth rate close to r . As $N(t) \rightarrow K$, however, the rate of change goes to zero $\left(\frac{dN(t)}{dt} \rightarrow 0\right)$. Thus, the population begins to level off as shown earlier in Chapter 2.

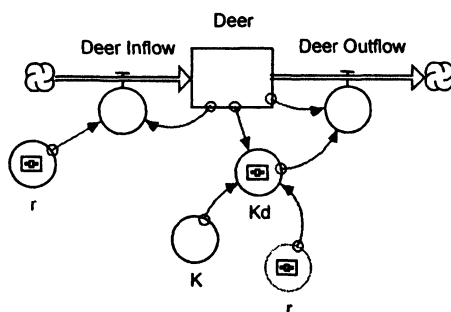
To gain a better understanding of this type of behavior pattern, open the model titled **CHAP4b.STM** on your CD-ROM. This model is a variation of the **CHAP4a.STM** model seen previously. This model can be used to demonstrate the logistic growth example discussed in this section. Notice that both the birth rate and the death rate in this model are functions of r , a parameter related to the net growth rate as shown earlier. The birth rate is defined as r , whereas the death rate is defined as the function $rN(t)/K$. The values for r and K are shown in the slider bars underneath the model schematic, with default values of 0.20 individuals/capita/year and 500 individuals, respectively. The graph shows the population for the 25-year period for the default settings. Like **CHAP4a.STM**, this model is set to run in “flight simulation” mode. Run the model with and without changing any parameter values. Explain your results.

4.2.3 Coupled Predator–Prey Populations: The Lotka-Volterra Model

The assumption of a population, such as deer, that is *uncoupled* to other species in our ecosystem is clearly unrealistic. Deer, and most other animals in an ecosystem, are either predators or prey, and thus are necessarily connected (i.e., *coupled*) to other populations. We clearly expect prey populations (like deer) to be influenced by the number of predators looking for a hearty meal.

Two of the first researchers to explore the mathematics behind predator–prey relationships were Alfred J. Lotka (1925) and Vittora Volterra (1926). In written works related to animal ecology, these two researchers independently described predator–prey relationships mathematically, in which the populations of predator and prey were intimately connected.

Before describing the mathematics, let us think intuitively about the dynamics between a population of prey $[N(t)]$ and a population of predators $[P(t)]$. Let us also assume we have a fixed area ecosystem under study (so that population sizes are reflective of population densities) and that the predator’s *only* food supply is the population of prey that exists in the ecosystem. [(Such predators are called **monophagus**, derived from the Greek *mono* (meaning “one”) and *phagus* (meaning “food”). Some predators, of course, may depend on several different types prey for survival (i.e., they may be **polyphagus**). Even some animals that have a strong preference for one type of prey will show polyphagus characteristics when that preferred food supply is scarce. For the purposes of illustrating predator–prey dynamics, however, the monophagus assumption will do).] We will also assume no

FIGURE 4.4. System diagram for logistic growth of N .

immigration or emigration of predator or prey species into or out of this ecosystem. Some of the end-of-chapter exercises will provide an opportunity for you to explore systems in which these assumptions are relaxed.

Let us first look at the prey population $[N(t)]$ and consider its dynamics. One would first suppose that this population would behave according to the logistic behavior pattern discussed in Section 4.2.2. This behavior represents a population whose growth is *density dependent*. The introduction of a predator population, however, modifies the logistic behavior and introduces some surprising patterns into the growth of the prey population.

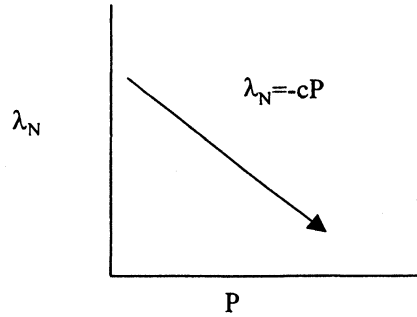
We will now reintroduce our net growth variable. We will call it λ_N this time because it corresponds to the population $N(t)$.

$$\lambda_N = r - \frac{rN(t)}{K} = r \left(1 - \frac{N(t)}{K} \right) \quad (4.7)$$

As discussed in Section 4.2.2, as $N(t)$ increases, we expect λ_N to decrease, due to intraspecies competition. Along those lines, as $N(t)$ decreases, thereby leading to less intraspecies competition, λ_N then increases. Thus, we have a system that regulates itself toward a stable size near the *carrying capacity* of the ecosystem. Such a system is depicted in Figure 4.4.

Let us now see what happens to the growth (or decline) of the prey population whenever we add a population $[P(t)]$ of hungry predators to the system. Common sense suggests that the larger the population of predators, the more rapidly the prey are hunted and killed. In order to account for this relationship, the net growth rate λ_N in Equation (4.7) can be modified by adding another term that describes the impact of the predators on the growth of the prey. Figure 4.5 shows how λ_N might decrease as $P(t)$ increases and Figure 4.6 shows how the population of predators (P) might be related to the death rate of the prey population.

Notice from Figure 4.5 that we identified a negative, linear relationship between λ_N and P . This relationship assumes that an increase in predators leads to a higher number of prey killed *per predator* per time period. In the

FIGURE 4.5. Growth rate for N versus predator population, P .

spirit of density dependence we are assuming that more predators lead to more efficient hunting! This is different than assuming that predators hunt down a constant number or a constant proportion of prey per time unit (density independent). We will now add the term $-cP$ to λ_N to represent this relationship. Thus, following the form in the preceding equations, we have:

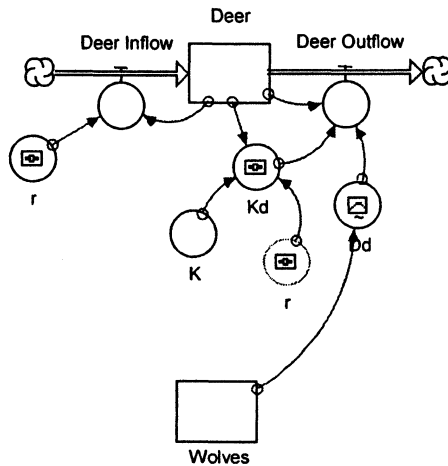
$$\lambda_N = r \left(1 - \frac{N(t)}{K} \right) - cP(t)$$

$$N(t + \Delta t) = N(t) + \lambda_N N(t) \Delta t$$

$$N(t + \Delta t) = N(t) + r \left(1 - \frac{N(t)}{K} \right) N(t) \Delta t - cP(t) N(t) \Delta t$$

$$\frac{N(t + \Delta t) - N(t)}{\Delta t} = r \left(1 - \frac{N(t)}{K} \right) N(t) - cP(t) N(t)$$

$$\frac{\Delta N}{\Delta t} = r \left(1 - \frac{N}{K} \right) N - cPN$$

FIGURE 4.6. System diagram for predator impact on N .

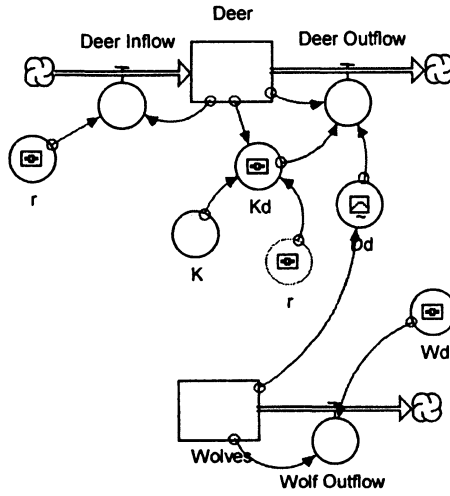


FIGURE 4.7. System diagram for predator death without prey.

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta N}{\Delta t} = \frac{dN}{dt} = r \left(1 - \frac{N}{K} \right) N - cPN \quad (4.9)$$

One can think of the $-cPN$ in Equation (4.9) as a term that converts the predator population to a certain number of prey killed (Smith and Smith, 1998). The values for c would be dependent on the efficiency of the predators in capturing and killing prey.

Let us now explore the population for the predator species, P . When we consider P in the absence of prey, we expect to see the exponential extinction of P altogether (they have nothing to eat!). Thus, there will be an outflow to the predator population, as shown in Figure 4.7. Introducing λ_p as the net growth of the predator species and ignoring the density-dependent term (this will be considered later), we have the following equations:

$$\begin{aligned} \lambda_p &= -W_d \\ P(t + \Delta t) &= P(t) + \lambda_p P(t) \Delta t \\ P(t + \Delta t) &= P(t) - W_d P(t) \Delta t \\ \frac{P(t + \Delta t) - P(t)}{\Delta t} &= -W_d P(t) \\ \frac{\Delta P}{\Delta t} &= -W_d P \\ \lim_{\Delta t \rightarrow 0} \frac{\Delta P}{\Delta t} &= \frac{dP}{dt} = -W_d P \end{aligned}$$

Note the negative sign for W_d , representing no opportunity for increasing population of predators without prey species. This rate equation simply represents the exponential decay model from Chapter 2.

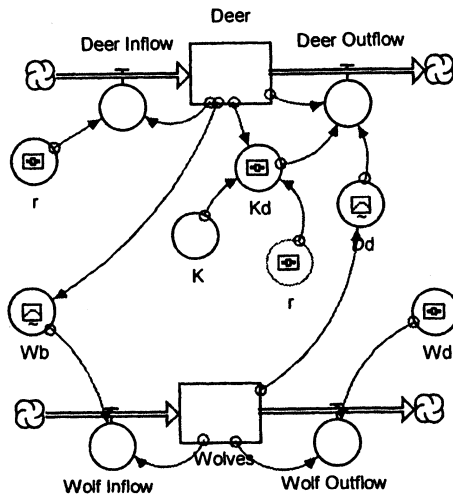


FIGURE 4.8. System diagram including predator population growth.

When we add prey to this model, we would expect that the population of predators would at least have a chance of increasing as the population of prey increases. High prey densities mean that predators will likely have a higher probability of success in their search for food. In addition, predator species will likely mate more often with such abundant resources at hand. We will have a situation shown in Figure 4.8, whereby we include an inflow to the predator population due to the presence of prey. The relationship between this growth rate for P and the population N is shown in Figure 4.9.

As N increases, the growth rate of P should also increase. We will assume a positive, linear relationship, which indicates that the more prey that are available, the more predators will be born per time period. Based on these assumptions, we will have a new set of equations that looks like the following:

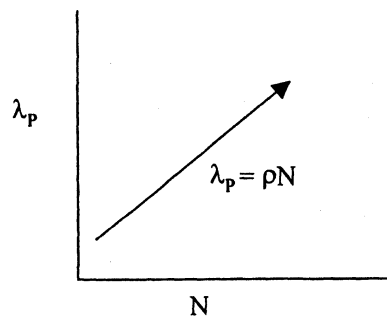


FIGURE 4.9. Relationship between predator growth rate and prey, N .

$$\begin{aligned}
\lambda_p &= -Wd + \rho N(t) \\
P(t + \Delta t) &= P(t) + \lambda_p P(t) \Delta t \\
P(t + \Delta t) &= P(t) + [-WdP(t) + \rho N(t)P(t)] \Delta t \\
\frac{P(t + \Delta t) - P(t)}{\Delta t} &= -WdP(t) + \rho N(t)P(t) \\
\lim_{\Delta t \rightarrow 0} \frac{\Delta P}{\Delta t} &= \frac{dP}{dt} = -WdP + \rho NP
\end{aligned} \tag{4.10}$$

Consider the term ρNP in Equation (4.10). This term converts the population of prey to an incremental population increase of predators. The values of ρ depend on the predator hunting efficiencies and how well predators translate prey kills into new predator individuals.

Equations (4.9) and (4.10) establish our dynamic predator–prey system. Figure 4.8 shows the complete diagram for this system. In Figure 4.8 the inflows to the deer stock are driven by r . The outflows are driven by two parameters. The first is the K_d value, which is simply the death rate due to the carrying capacity constraint. K_d is defined as $r\left(\frac{\text{Deer}}{K}\right)$, where K is the carrying capacity. The other outflow from the deer population is driven by a death rate identified as D_d . This is the death rate due to the predatory wolves. This model identifies a graphical linear relationship between the wolf population and the death rate, similar to that shown in Figure 4.5 (hence, the small graphic icon in the center of the D_d circle). You will be able to modify this relationship and explore the impact on the model later in the chapter.

For the wolf population, there is an outflow driven by W_d , which represents the decline of the wolf population by assuming that there is no available prey. With prey, we have an inflow into the wolf population driven by W_b , which is a graphical linear relationship between the prey population and the birth rate. As with D_d , the reader will be able to modify the relationship between prey population and wolf inflow rates later in the chapter.

4.3 Difference Equations and the Steady-State Solution

In this section, we explore the difference equations and steady-state solutions for each of our stock variables in the model. Recall in Chapter 1 that the reason we want to examine these rate equations and steady-state solutions is to help us understand the impacts to the system under various model perturbations. We want help in predicting whether a perturbation to the model will result in stability or instability. The basis for the following equations is in Section 4.2. We translate each of the equations for our stock variables into differential equations as follows:

For our deer population, $N(t)$:

$$N(t + \Delta t) = N(t) + r \left(1 - \frac{N(t)}{K} \right) N(t) \Delta t - D_d N(t) \Delta t$$

where, $D_d = cP(t)$ from Figure 4.5.

$$\frac{N(t + \Delta t) - N(t)}{\Delta t} = r \left(1 - \frac{N(t)}{K} \right) N(t) - cP(t)N(t)$$

$$\frac{\Delta N}{\Delta t} = r \left(1 - \frac{N}{K} \right) N - cPN$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta N}{\Delta t} = \frac{dN}{dt} = r \left(1 - \frac{N}{K} \right) N - cPN$$

For our wolf population, $P(t)$:

$$P(t + \Delta t) = P(t) + W_b P(t) \Delta t - W_d P(t) \Delta t$$

where, $W_b = \rho N(t)$ from Figure 4.9.

$$\frac{P(t + \Delta t) - P(t)}{\Delta t} = \rho N(t)P(t) - W_d P(t)$$

$$\frac{\Delta P}{\Delta t} = \rho N(t)P(t) - W_d P(t)$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta P}{\Delta t} = \frac{dP}{dt} = \rho NP - W_d P = P(\rho N - W_d)$$

Notice that each of these differential equations show a rate of change that is dependent upon the stock variable itself—indicating some relationship to our exponential functional forms seen in Chapter Two. Note also that the differential equation for N has P as a variable, and the differential equation for P has N as a variable.

Steady-state conditions for this problem are complex. Recall that the definition of steady-state is that point at which the rate of change of the stock variable equals zero (i.e., where $dN/dt = 0$ and $dP/dt = 0$). Our steady-state conditions simplify if we remove the density-dependent term (set $K_d = 0$) from our deer population rate equation. This is similar to making an assumption that the carrying capacity (K) for the deer are infinite. We can thus write the following two sets of steady-state equations.

For our wolf population, P :

$$\frac{dP}{dt} = 0 = P(-W_d + \rho N) \quad (4.11)$$

For our deer population, N :

$$\frac{dN}{dt} = 0 = N(r - cP) \quad (4.12)$$

These equations are satisfied when either:

$$P = 0 \quad \text{and} \quad N = 0 \quad (4.13)$$

or

$$P = \frac{r}{c} \quad \text{and} \quad N = \frac{W_d}{\rho} \quad (4.14)$$

In the first case, Equation (4.13), we have a deer population that is extinct, and thus a wolf population that also becomes extinct. This is obviously a stable steady-state situation because without any deer or wolf populations, it is impossible for more deer or wolves to be born.

In the second case, we can achieve a steady state at the point where our wolf population equals r/c and our deer population equals W_d/ρ . The response of each population to these parameters is interesting (and often nonintuitive) [see Haberman (1977) for a nice discussion]. For example, suppose we increase r by somehow increasing the fecundity of the deer population. These equations tell us that this leads to a steady-state system that has an increase in wolves, but not an increase in deer (i.e., the wolf population increases to consume the additional deer that are born).

Now suppose we increase the death rate of the wolves (by increasing W_d). These steady-state equations tell us that the equilibrium wolf population is unaffected. Instead, the deer population increases, thus increasing the production of more wolves to take the place of those that die. An end-of-chapter exercise asks the reader to consider the population impacts due to variations in each of these parameters.

4.4 Modeling the Dynamic Deer–Wolf System

We are now prepared to analyze the problem introduced at the beginning of this chapter—but first we need some more detailed information about the problem.

We are asked to explore the potential impacts on a deer population after the introduction of a population of wolves. We would, of course, expect qualitatively that the introduction of this predator species would place a downward pressure on the deer population. Based on the discussion of previous sections, however, this downward pressure may be autoregulating (i.e., if the deer population gets too low, the wolf population will likely die back as well, offering an opportunity for the deer population to increase in size).

These qualitative expectations are shown in a Lotka–Volterra *phase space diagram* drawn in Figure 4.10. In this figure, the Y-axis represents the wolf (predator) population, and the X-axis represents the deer (prey) population. Lines are drawn to segment the diagram into four quadrants (the horizontal line is called the *prey isocline*; the vertical line is called the *predator isocline*) (Crawley, 1992). These isoclines are merely the points at which each population is in equilibrium (based on the preceding steady-state solution).

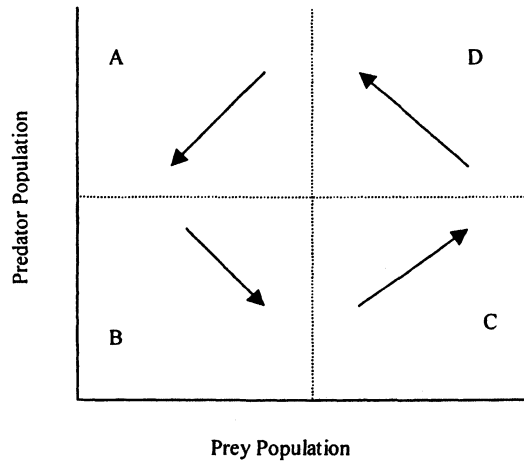


FIGURE 4.10. Lotka-Volterra phase-space diagram.

The direction of the arrows in this figure can help us qualitatively think about what will occur after we introduce this wolf population. Let us begin in quadrant C. In quadrant C, the prey population is relatively high and the predator population is relatively low. The high prey population allows the predators to begin to grow. On the other hand, because predator populations are still relatively low, the prey population also continues to grow. Thus, we have an arrow indicating movement toward the “northeast” in our diagram. We move to the “northeast” until we cross over the prey isocline.

Once in quadrant D, we have a situation where the predator population has increased to a point where it begins to reduce the prey numbers. Because the prey numbers are still relatively high, however, we continue to see increases in our predator population. Thus, we have an arrow indicating movement to the “northwest” in our diagram.

Entering quadrant A, we have a prey population that is on the decline and a predator population reaching its maximum. At this point, there are still enough predators to cause a continual reduction in prey populations. As prey population declines, however, predator population also begins to decline. Thus, we have an arrow indicating movement to the “southwest” in our diagram.

Finally, we enter quadrant B with our prey population at a low. Here we have a situation where predator species continue to decline because of the low prey population. This reduction in predators allows the prey to begin to recover. Because there still are relatively few prey, predators continue to decline in numbers. Thus, we have an arrow moving to the southeast of our diagram, where we once again enter quadrant D and the cycle continues. Note that if the predator or prey population ever crosses the Y- or X-axis, then we come to a point where either the predator or prey becomes extinct.

Let us now suppose that based on field studies, you have determined each of the critical parameters for your model. You arrive at the following inputs for the refuge you are studying:

- Initial deer population = 500
- $r = 0.25$
- $K_d = 0$ (we will assume no density dependence for now)
- $D_d = c(\text{Wolf}) = 0.005(\text{Wolf})$
- Wolf population to be introduced = 25
- $W_d = 0.30$
- $W_b = \rho(\text{Deer}) = 0.001(\text{Deer})$

Using these parameter values, we should be able to model the system and explore options that you may take in managing these deer and wolf populations. Now, open the model entitled *CHAP4c.STM*.

This model is identical to Figure 4.8. Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** on the drop-down menu. The graph in the model shows the populations of wolves and deer over time. The model is set up to conduct calculations similar to those in the previous difference equations over a period of 25 years, with $\Delta t = 0.25$ years. This model is in “flight simulation” mode, which allows the user the opportunity to modify variables while the model is running. In this model, you can use the slider bars to change r , W_d , and K_d . Note that you can set K_d to a value or you can click on the small box in the left-hand corner of the slider bar, which will force K_d to use the equation established in the model (i.e., $K_d = r(\text{Deer})/K$). You can also change the values of D_d and W_b by clicking on the graphs of these two variables and drawing different functional relationships using your mouse. On the other hand, you can enter numerical data in the table to change the relationship.

Run the model several times, and adjust the parameters to explore the results. In particular, run the model while only adjusting one variable at a time. (Use some of the techniques discussed in Chapter 3 to understand the model better). Try different graphical relationships for D_d and W_b . Then, consider the following questions:

- Does the model achieve an equilibrium point? If no, is there anything about the model that makes you feel as though it is “stable”?
- When you changed the value of K_d from 0 to another value, what happened? Can you explain what you are seeing?
- For each simulation you ran, explain what you did and what your results were. Is there any variable that seems to “drive” this system?
- Exploring this model should give you a better understanding of the dynamics of predator–prey relationships. Based on your new understanding, is there anything to be concerned about with an introduction of wolves into this wildlife refuge?

4.5 Exercises

1. How might one expect births and deaths of a given population to be related to population density? Provide an example of this phenomenon using a population of a species of your choice.

2. In the derivation of Equation (4.10), we assumed that the net growth rate (λ_p) of predators is *density independent* growth rate because it does not depend on the population of predators. Derive Equation (4.10) under conditions of density dependence; “Hint: In the expression for d_p in the derivation of Equation (4.10), replace $-W_d$ with the expression $-\beta \frac{P(t)}{\alpha}$,

where α represents the carrying capacity for the predators, and where β is a constant multiplier. Then proceed through the derivation.

3. Derive the rate Equation (4.6). Use information in Chapter 2 to assist you.

4. Using Equation (4.14), explain the impacts of a shifting increase or decrease in each parameter on deer and wolf populations.

5. Explain in your own words how the Lotka-Volterra phase-space diagram in Figure 4.10 operates. What happens if predator populations “push” prey to a point where the arrow in quadrant A touches the Y-axis? What happens if low prey populations “push” predators to a point where the arrow in quadrant B touches the X-axis?

6. Based on the phase-space diagram (Figure 4.10), draw a graph that depicts each population over time. Start in Quadrant A, where we have arbitrarily low starting population for deer and a high population for wolves, then move through each of the quadrants as we did earlier, this time plotting your deer and wolf populations on the Y-axis of your graph, with time on your X-axis. You do not need to be precise about numbers. (you cannot be without other information!), but you should be able to generate the general plot that is expected from our model.

7. What are some of the weaknesses of the Lotka-Volterra model? Make a list of things that the Lotka-Volterra model does not address.

8. Open the model **CHAP4a.STM**. Modify this model by changing the linear birth and death rate constants to more complicated functions, either mathematically or graphically. Try an increasing or decreasing function for one of your variables, while keeping the other one constant. Run your model and explain the results. Now try a model where both variables are increasing or decreasing simultaneously. Run your model and explain the results.

9. Modify the model discussed in this chapter by adding a third population of species. A good population to add is a producer species (e.g., plant) that your prey population depends upon for survival. Establish all the relationships between this new population and your wolf and deer populations. Run the model and explore your results.

4.6 References Cited and Suggested Readings

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5

Modeling Surface Water Contamination

Chapter Objectives—

After you finish this chapter, you should be able to:

- 1. Describe various types of surface water pollution, their sources, and their impacts.*
- 2. Describe qualitatively and mathematically the concept of **mass-balance**, and apply this concept to solve mass-balance problems.*
- 3. Describe the flow of oxygen into and out of water in an aquatic ecosystem, highlighting the reservoirs and flows of this oxygen.*
- 4. Construct a systems diagram that illustrates how oxygen may be depleted from an aquatic system due to inputs of organic matter.*
- 5. Through manipulation of a systems model, identify the driving variables of the **DO-sag curve** and describe this curve's response to perturbations of those variables.*

5.1 The Problem

Surface water contamination is a nefarious problem throughout the world. Over the past several decades, thousands of miles of streams and rivers, and millions of acres of lakes and ponds have exhibited declining water quality due to industrial effluent, agricultural runoff, and other human activities.

One of the more notorious manifestations of surface water contamination is the reduction of dissolved oxygen. This reduction is due to several factors, although one of the most prominent is the dumping of organic waste into our waterways. As this waste decomposes, dissolved oxygen in the water is consumed, thus leaving the water “oxygen depleted.” Such oxygen-depleted water may not be able to support aquatic life, which depends upon oxygen for survival. This results in a dead body of water that cannot support the flora and fauna that are part of the aquatic ecosystem.

In this chapter, you will examine how water pollution is transported in a surface water environment, and how this pollution affects dissolved oxygen over time and distance. In particular, you will address the following scenario.

Problem 5.1 A municipal wastewater treatment facility is planning to locate upstream on a popular fishing river. The wastewater facility will continuously discharge wastewater into this river. The effluent will have high biochemical oxygen demand (BOD) and low dissolved oxygen (DO) concentrations. You want to determine the impact of the effluent on the DO content of the river (and ultimately on the river's fish population). You also wish to evaluate the potential impact of corrective measures the facility can take to address potential DO problems.

5.2 Background Information

5.2.1 Water Pollution

Water is a resource that nearly all living things need. About 97% of the world's water is found in the oceans. The remainder consists of freshwater, which is found in ice caps and glaciers (1.984%), groundwater (0.592%), lakes (0.007%), soil moisture (0.005%), atmospheric water vapor (0.001%), biota (0.0001%), and rivers (0.0001%) (Miller, 1994). Although our accessible freshwater makes up a very small percentage of our earth's total water supply, it is this water that we depend upon for survival.

We pollute our surface water resources a number of ways. One taxonomy categorizes water pollution based on its source. In this categorization, we have *point sources* and *nonpoint* (or *area*) *sources*. Point sources are those in which we can directly identify the point of discharge, such as a discharge pipe from a waste treatment facility or industrial processing plant. *Nonpoint sources* are those that do not have a directly identifiable discharge point, such as agricultural runoff and urban runoff.

Another taxonomy categorizes water pollution with respect to the ecological impacts it imparts on aquatic ecosystems. A third categorizes pollution by the general types of pollution (e.g., nutrients, organics, toxics, etc.). Table 5.1 provides an example of this categorization, with additional columns addressing impacts and sources for each type of pollutant.

5.2.2 Dissolved Oxygen and Oxygen Depletion

A major concern in many aquatic ecosystems is the availability of *dissolved oxygen* in the water. *Dissolved oxygen* (DO) is the amount of molecular oxygen (O_2) dissolved in water. Dissolved oxygen is one of the most important criteria in determining the quality of water because oxygen is required for most aquatic life.

The amount of DO that water can hold is a function of Henry's constant and partial pressures based on Henry's Law:

TABLE 5.1. Types, impacts, and sources of water pollution.

Type	Impact	Source examples
Heat (thermal pollution)	Reduces dissolved oxygen in water via solubility relationship	Power plants; industrial facilities
Silt and sediment	Increases turbidity; lowers flow; interferes with fish spawning; interferes with light penetration which reduces photosynthetic activity of bottom-dwelling plants	Erosion from agriculture, construction, mining, and forestry; urban storm runoff
Nutrients (namely, nitrogen and phosphorus)	Promotes algae growth, which reduces DO through decomposition and light reduction to bottom-dwelling plants (inhibits photosynthetic activity)	Fertilizers; municipal and domestic wastewater effluent
Organic pollution	Reduces dissolved oxygen in water via decomposition	Agricultural runoff; municipal wastewater effluent
Microorganisms	Transmits infectious disease	Agricultural runoff; municipal wastewater effluent
Acids	Increases stress to pH-sensitive aquatic life; may increase heavy metal concentrations in water due to solubility relationships	Acid mine drainage; acid precipitation
Various organic and inorganic chemicals	Increases toxicity; reduces aesthetic quality of water (e.g., color and smell)	Industrial discharges; municipal wastewater effluent

$$C_{O_2} = K_{O_2} \cdot P_{O_2}$$

where C_{O_2} is the concentration of dissolved oxygen in the water [milligrams per liter (mg/L)], K_{O_2} is Henry's constant [milligrams per liter per atmosphere (mg/L/atm)], and P_{O_2} is the partial pressure of oxygen at the atmospheric–water boundary [atmospheres (atm)].

Henry's constant can be experimentally determined for any type of gas–liquid interface. For oxygen and water, K_{O_2} has an inverse relationship with temperature. For example, K_{O_2} has a value of 61.2 mg/(L·atm) for water at 5°C, but has a value of 40.2 mg/(L·atm) for water at 25°C. This fact, along with Henry's Law, stated earlier, implies that the warmer the temperature of the water, the less DO the water can contain.

If we know the value for K_{O_2} and the partial pressure of oxygen on the liquid surface, we can use Henry's law to calculate the DO expected in a given body of water. For example, we know that O_2 makes up approximately 21% of the composition of the atmosphere. The partial pressure of O_2 in air at one atmospheric pressure (1 atm), therefore, is 0.21 atm. If the tempera-

ture of the water is 5°C, then $K_{O_2} = 61.2 \text{ mg}/(\text{L} \cdot \text{atm})$, and the expected concentration of DO at 1 atm is

$$C_{O_2} = \left(61.2 \frac{\text{mg}}{\text{L} \cdot \text{atm}} \right) (0.21 \text{ atm}) = 12.9 \frac{\text{mg}}{\text{L}}$$

The preceding value is under ideal conditions and gives the concentration of oxygen that can be dissolved in the water under saturated conditions. We will refer to this saturation level of DO as DO_{sat} .

In reality, there are many complex interactions that occur within the water environment that cause DO levels to vary above or below DO_{sat} . For example, waterborne plants undergoing photosynthesis take carbon dioxide (CO_2) out of the water and release O_2 . In contrast, creatures such as fish undergoing respiration take O_2 out of the water and release CO_2 . Figure 5.1 summarizes the inflows and outflows of O_2 and CO_2 in a body of water and the atmosphere.

Oxygen is also removed from aquatic systems through the decomposition of organic matter. When organic matter decomposes in an oxygen rich environment (i.e., *aerobic* decomposition), the following general reaction takes place:



From the preceding discussion, we see that the addition of organic waste into an aquatic system can reduce DO levels through the decomposition of this waste. The amount of oxygen required to decompose a certain amount of organic waste is its **biochemical oxygen demand** (BOD) and is measured in concentration units (typically, milligrams per liter). Values for BOD are often used to characterize the general water quality of a given sample of water

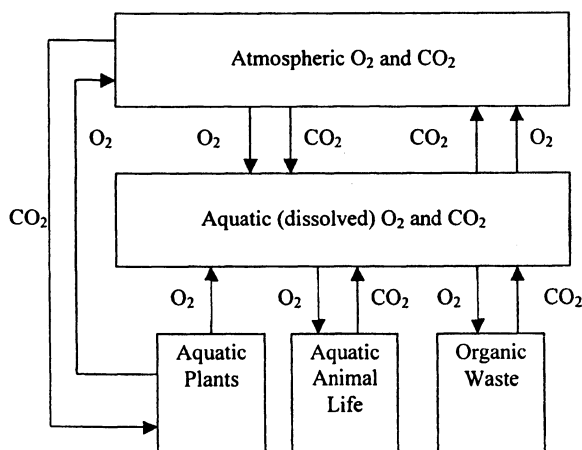


FIGURE 5.1. Inflows and outflows of oxygen in an aquatic environment.

because it is an indirect measure of the amount of organic impurities in the water. We often use the term **ultimate BOD** (BOD_{ult} , sometimes referred to as L_0) to represent the *total* amount of oxygen (milligrams per liter) that would be required to oxidize *all* the organic waste in a given water sample.

Discharging nutrients into an aquatic system can also reduce DO levels below the DO_{sat} value. Nutrients such as nitrogen (N) in the form of nitrates (NO_3^-) and phosphorus (P) in the form of phosphates (PO_4^{2-}), lead to excessive plant and algae growth. As algae and plants grow and cover the water surface, they reduce the amount of sunlight that reaches plants on the floor bottom. This leads to less photosynthesis and therefore lower oxygen production by plants in the water. In addition, algae and plants decompose as they die, thereby consuming even more oxygen.

In general, water with DO values below 5 or 6 mg/L at 20°C is considered polluted. Such low DO values can place sensitive aquatic life at risk. DO values in the 7–10 mg/L range are more desirable and are what we would expect in an unpolluted body of water at 20°C.

5.2.3 Mathematical Relationships and Our Systems Diagram

For a given body of water, predicting the impact on DO levels due to organic effluent is a common activity of hydrologists, water chemists, and environmental regulators. Dissolved oxygen predictions are used to determine whether a certain activity should be allowed along a body of water. For example, in the United States when an industry wishes to build a facility along a river and discharge organic waste into the river, the industry must obtain a permit. Part of the application process for obtaining a permit involves modeling the impacts the organic waste will have on DO levels downstream from the proposed site.

In order to conduct such estimates, we need to consider all the inflows and outflows of O_2 in the water as it travels downstream. As we will find out, these flows are a function of the amount of DO and BOD in the water itself, as well as the volumetric flow of the river. We therefore need to acquire the following information.

- DO of the river before mixing with the industrial effluent
- DO of the industrial effluent
- BOD of the river before mixing with the industrial effluent
- BOD of the industrial effluent
- Volumetric flow of the river before mixing with the effluent
- Volumetric flow of the industrial effluent.
- The DO_{sat} level for the river before mixing with the effluent

With this information, we can determine how DO levels in the river will change downstream of the industrial discharge point. The analysis usually involves two major steps:

1. Establish the initial values for DO and BOD immediately after the effluent mixes with the river. The application of mass-balance concepts generates these initial conditions.

2. Predict the expected downstream DO levels. These levels will be determined by an inflow process that we will call **reoxygenation** (or **aeration**), and an outflow process that we will call **deoxygenation**. As the water travels downstream, it absorbs O_2 from the atmosphere (i.e., **reoxy- genation**), thereby increasing the DO levels. At the same time, the **deoxygenation** process consumes DO through the decomposition of organic waste.

Suppose we want to analyze the impact of a proposed new organic discharge into a river (possibly from a waste treatment facility). For Step 1, we can establish initial values of DO and BOD by taking a field sample of the surface water and measuring its DO level and BOD level using various lab techniques (Vesiland et al. 1990). We can use DO and BOD estimates from our expected waste discharge to obtain DO and BOD values at the point where the discharge mixes with the water. In order to do this, we will make two assumptions:

- The pollutant discharge mixes uniformly throughout the original body of water (the **uniform mixing** assumption).
- The total mass of DO after mixing is equal to the sum of the masses of DO in the original water body and the pollutant discharge (the **mass-balance** assumption).

We can now calculate the DO for the body of water immediately after mixing with the discharge. For a standing body of water, the **mass-balance equation** is:

$$DO_s V_s + DO_p V_p = DO_a (V_s + V_p)$$

where DO_s = the DO value of the water body (milligrams per liter) before mixing

V_s = the volume of the water (liters) before mixing

DO_p = the DO of the pollutant discharge (milligrams per liter)

V_p = the volume of the pollutant discharge (liters)

DO_a = the DO of the water after the pollutant has been added (milligrams per liter)

One can use this equation to solve for DO_a .

For flowing water (e.g., a river or stream), one can use flow rates (Q , in liters per second) instead of volume. The equation becomes:

$$DO_s Q_s + DO_p Q_p = DO_a (Q_s + Q_p)$$

where Q represents flow (liters per second) and replaces V (liters). The same mass balance principles can be applied to calculate BOD values immediately

after mixing (i.e., simply replace the preceding DO variables with BOD variables).

Both DO and BOD can clearly be represented as reservoir variables in a system diagram, each representing respective characteristics of a parcel of water as it moves downstream. The initial values of these reservoirs are based on the preceding calculations. As the parcel of water travels downstream, we can observe how the *reoxxygenation* and *deoxygenation* processes change the levels of the DO reservoir (and ultimately the level of the BOD reservoir) over time. In doing so, we can determine how the BOD and DO levels of the stream will change downstream of the discharge point.

The consumption of organic waste occurs as organic matter is decomposed on its trip down river. This consumption is measured by a decrease in the BOD value of the water. Because BOD is a measure of the total amount of oxygen needed to decompose a given amount of waste in a parcel of water, this value will decrease as that waste is consumed.

The rate at which BOD decreases is itself a function of the amount of BOD present. When there are large amounts of BOD in the water, bacteria have little trouble finding this waste. Hence, the decomposition of organic waste occurs rapidly. The rate at which the BOD level drops is also rapid. As the organic matter is consumed, however, it is more difficult for bacteria to find and consume the remaining organic matter. Hence, the rate of consumption drops, and the rate at which BOD declines also drops.

You have seen similar cases in earlier chapters whereby we can model such behavior using an exponential decay model. This system infrastructure for the BOD case is shown in Figure 5.2, where we use k_1 as the exponential decay constant.

The outflow shown in Figure 5.2 can be described mathematically as

$$\text{BOD}_{\text{out}}(t) = k_1 \text{BOD}(t) \quad (5.1)$$

As waste is consumed, the DO of the water is also being consumed. In fact, by definition the amount of DO consumed is identical to the amount of BOD consumed. (To convince yourself of this, see end-of-chapter Exercise 1). Hence, we must define the outflow from the DO reservoir to be identical to the outflow from the BOD reservoir. As shown in Figure 5.3, where we draw a connector between the BOD reservoir and the DO outflow, we can modify our systems diagram to depict this.

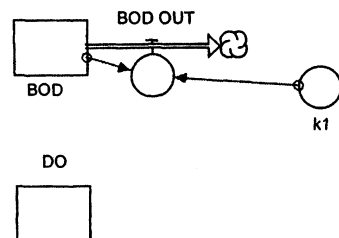


FIGURE 5.2. System infrastructure for BOD decay.

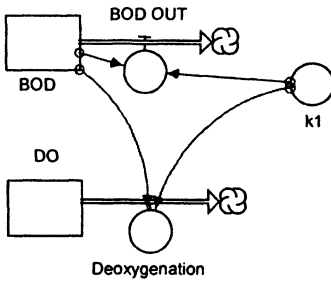


FIGURE 5.3. System infrastructure for deoxygenation.

As you might guess, the DO outflow is identical to the BOD outflow; that is,

$$DO_{out}(t) = k_1 BOD(t) \quad (5.2)$$

For this reason, we call k_1 the system's **deoxygenation coefficient**. This coefficient is measured in inverse time units (typically days^{-1}). This constant is measured empirically and suggests a rate at which DO is consumed through decomposition of organic waste. Note that this equation does not account for deoxygenation from other causes (e.g., animal respiration).

At the same time that DO is consumed through organic decomposition, oxygen is also added back into the water through **reoxygenation** (i.e., aeration). The rate at which this occurs depends on how much more oxygen the water could theoretically hold. This is referred to as the **oxygen deficit**. Because DO_{sat} is the theoretical saturation level of oxygen that the water can hold and DO is the amount of oxygen currently in the water, the **oxygen deficit** is equal to $DO_{sat} - DO$. If the oxygen deficit is large, then the water will have a greater propensity to absorb oxygen from the atmosphere. If the oxygen deficit is small, then atmospheric oxygen will be absorbed much more slowly. In order to illustrate this phenomenon, imagine a bone-dry sponge that is placed in a small pool of water. At first, the sponge is desperately dry and rapidly draws water into its open pores. As the sponge fills with water (i.e., becomes saturated), the rate at which it absorbs water from the pool decreases. At its saturation point, the sponge has no more room for additional water, and the rate of absorption falls to zero.

We can depict this reoxygenation process graphically by adding an inflow to our DO reservoir. The value for this inflow will be determined by the value of DO_{sat} , the current DO level, and a rate constant that defines how quickly the DO flows into the water. These system components are reflected in Figure 5.4, where we have identified k_2 as our reoxygenation rate constant.

The reoxygenation flow at time t mathematically has a value $DO_{in}(t)$ as follows:

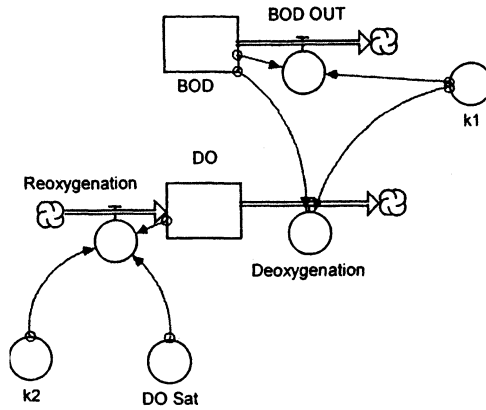


FIGURE 5.4. System infrastructure for reoxygenation.

$$DO_{in}(t) = k_2(DO_{sat}(t) - DO(t)) \quad (5.3)$$

where k_2 is called the **reoxygenation coefficient**. This coefficient is also measured in inverse time (typically days^{-1}). The value of k_2 is often a function of the amount of surface water exposed to the atmosphere, with higher exposure (e.g., whitewater rapids) having higher k_2 values, and lower exposure (e.g., stagnant pond) having lower k_2 values. (Note that some reoxygenation may also occur due to the photosynthetic activities of waterborne plants within the parcel of water. We will ignore those impacts for now and provide an end-of-chapter exercise to incorporate them into the model).

Finally, the BOD of the flowing parcel of water may also have an inflow both from the organic effluent as well as from the natural death and decay processes of organic plants and animals within the parcel. This inflow is independent of current DO and BOD levels, and it depends more on which type of ecosystem we are studying. We can add this system inflow to our diagram, as shown in Figure 5.5. For our purposes, we will consider this to be a constant inflow defined mathematically as:

$$BOD_{in}(t) = A$$

Thus Figure 5.5 displays the complete system diagram for describing the changes in DO levels in a stream over time.

5.3 Difference Equations and Relationships

Our systems diagram and the mathematics behind it lead to the following difference equations for the DO and BOD reservoirs.

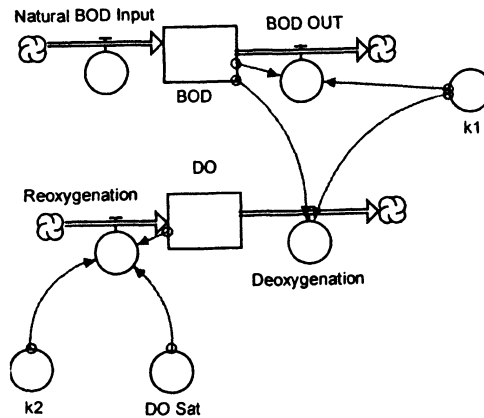


FIGURE 5.5. Complete systems diagram for DO sag model.

$$DO(t + \Delta t) = D(t) + k_2 \cdot [DO_{\text{sat}} - DO(t)] \cdot \Delta t - k_1 \cdot BOD(t) \cdot \Delta t$$

$$BOD(t + \Delta t) = BOD(t) + A \cdot \Delta t - k_1 \cdot BOD(t) \cdot \Delta t$$

The term involving k_2 corresponds to the *reoxygenation* inflow. From the discussion at the end of section 5.2.3, we know that the size of this inflow is large if the *oxygen deficit* [i.e., $DO_{\text{sat}} - DO(t)$] is large, and it is small if the *oxygen deficit* is small. Thus, if the DO after mixing with the pollution discharge is well below the saturation level, the *reoxygenation* inflow will initially be large, and will tend to drive DO levels back toward DO_{sat} . As time passes and the DO in the stream increases toward DO_{sat} , however, the *reoxygenation* inflow will scale back toward zero.

The term involving k_1 in the preceding difference equation corresponds to the *deoxygenation* outflow in Figure 5.6. Based on the discussion in Section 5.2.3, this outflow is equal to the outflow from the BOD reservoir. If the BOD reservoir is large, then the demand for oxygen to fuel the decomposition of organic waste is high. DO will therefore be consumed at a rapid rate. As organic waste is consumed, BOD levels will decrease, and the rate at which DO is consumed will decrease correspondingly.

Thus, the behavior of the DO reservoir over time depends on the relative sizes of the reoxygenation and deoxygenation processes. This behavior translates into the familiar **DO Sag Curve**, first discussed by Streeter and Phelps (1925). Components of the DO Sag Curve are presented in Figure 5.6. Examine the figure and identify those time intervals where the *reoxygenation* inflow is largest and when it is smallest. Repeat this analysis for the *deoxygenation* outflow.

The rate equation for the *BOD* reservoir is derived in the usual manner as:

$$\begin{aligned}
 \text{BOD}(t + \Delta t) &= \text{BOD}(t) + A \cdot \Delta t - [k_1 \cdot \text{BOD}(t)] \cdot \Delta t \\
 \text{BOD}(t + \Delta t) - \text{BOD}(t) &= A \cdot \Delta t - [k_1 \cdot \text{BOD}(t)] \cdot \Delta t \\
 \frac{\text{BOD}(t + \Delta t) - \text{BOD}(t)}{\Delta t} &= A - k_1 \cdot \text{BOD}(t) \\
 \lim_{\Delta t \rightarrow 0} \frac{\text{BOD}(t + \Delta t) - \text{BOD}(t)}{\Delta t} &= A - k_1 \cdot \text{BOD}(t) \\
 \frac{d\text{BOD}}{dt} &= A - k_1 \cdot \text{BOD}(t)
 \end{aligned}$$

The rate equation for the DO reservoir is given by

$$\begin{aligned}
 \text{DO}(t + \Delta t) &= \text{DO}(t) + k_2[\text{DO}_{\text{sat}} - \text{DO}(t)] \cdot \Delta t - k_1 \cdot \text{BOD}(t) \cdot \Delta t \\
 \text{DO}(t + \Delta t) - \text{DO}(t) &= k_2[\text{DO}_{\text{sat}} - \text{DO}(t)] \cdot \Delta t - k_1 \cdot \text{BOD}(t) \cdot \Delta t \\
 \frac{\text{DO}(t + \Delta t) - \text{DO}(t)}{\Delta t} &= k_2[\text{DO}_{\text{sat}} - \text{DO}(t)] - k_1 \cdot \text{BOD}(t) \\
 \frac{d\text{DO}}{dt} &= k_2[\text{DO}_{\text{sat}} - \text{DO}(t)] - k_1 \cdot \text{BOD}(t)
 \end{aligned}$$

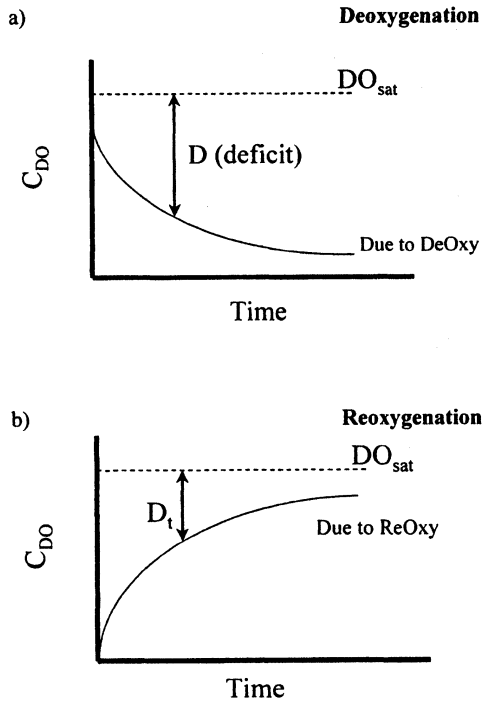


FIGURE 5.6. Components of the DO sag curve. (a) Deoxygenation. (b) Reoxygenation.

The rate equations expressed here represent a coupled system of differential equations.

The steady-state solution to this problem occurs where the change of BOD and DO with respect to time is equal to zero (i.e., where the inflows equal the outflows). Hence, a steady state is achieved whenever the following two conditions hold:

$$\begin{aligned}\frac{dBOD}{dt} &= 0 = A - k_1 BOD, \text{ and} \\ \frac{dDO}{dt} &= 0 = k_2 [DO_{\text{sat}} - DO(t)] - k_1 \cdot BOD(t)\end{aligned}$$

It can be shown that these conditions are satisfied whenever the BOD and DO reservoirs reach the following steady-state levels.

$$\begin{aligned}\overline{BOD} &= \frac{A}{k_1}, \text{ and} \\ \overline{DO} &= DO_{\text{sat}} - \frac{A}{k_2}\end{aligned}$$

Consider these steady-state solutions in detail. Notice that our steady-state DO value will never reach the DO_{sat} value as long as there is some continuous flow of BOD into the system (i.e., A is nonzero). If the *reoxygenation coefficient* k_2 is large, however, then the dissolved oxygen in the stream will eventually recover to levels that are very close to DO_{sat} .

5.4 Modeling the Dynamic DO System

We are now prepared to explore the problem introduced at the beginning of this chapter. First, however, let us examine the DO sag model without the complications of the added discharge from the wastewater facility. Let us suppose you obtain some data on the river as follows:

- Initial BOD = 3.33 mg/L
- Initial DO = 8.5 mg/L
- $DO_{\text{sat}} = 11.0 \text{ mg/L}$
- $k_1 = 0.30$
- $k_2 = 0.40$
- Natural BOD inflow = 1.0 mg/L

Using these parameter values we can begin to better understand the impacts of various perturbations to the system. Open the model entitled **CHAP5a.STM**. This model is identical to Figure 5.5. Before running the model, you should be able to determine mathematically whether the preceding parameter values represent a steady-state condition.

Now, run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** on the drop-down menu. The graph in the model shows the concentrations of BOD and DO over time. The model is set up to conduct calculations similar to those in the previous difference equations over a period of 25 days, with $\Delta t = 0.25$ days. This model is in “flight simulation” mode, which allows the user the opportunity to modify variables while the model is running. In this model, you can use the slider bars to change k_1 , k_2 , and DO_{sat} while the model is running. You can also use the dials to adjust the initial values for BOD and DO (note that you cannot adjust BOD and DO values while the model is running because these values are determined endogenously within the model). Finally, you can click on the table icon to see the actual BOD and DO values for each day of the run.

Run the model several times, and adjust the parameters to explore the results. In particular, run the model while adjusting only one variable at a time. Use the model analysis strategy outlined in Chapter 3 as a method for analysis. Then, consider the following questions:

- If you change a parameter and then let the model continue to run, does the DO value move toward an equilibrium point? Is it a “stable” model?
- When you change the value of DO_{sat} , what happens? Can you explain what you are seeing?
- For each simulation you ran, explain what you did and what your results were. Is there any variable that seems to “drive” this system?

Now, consider the problem presented at the beginning of this chapter. We are asked to determine the impact of a wastewater treatment facility on this river’s downstream DO concentrations. The facility will discharge wastewater directly into the river. The wastewater will have high BOD values and low DO values, which will initially increase the BOD and lower the DO of the river water. We are interested in what the DO value of the water will be as it flows toward popular fishing areas of the river downstream.

There are several ways to approach this problem. The most straightforward is to modify the previous model (*CHAP5A.STM*) to incorporate the new “initial” DO and BOD values that will be created by the mixing of the organic effluent with the river water. In order to determine what these new values should be, we need to have data on the river flow and discharge characteristics. Also, in order to determine the impact of this discharge at a certain distance downstream, we need to know the velocity of the river. Let us suppose we have the following data for this problem:

- Q_r (river flow before mixing) = 300,000 L/min
- BOD_r (river BOD before mixing) = 3.33 mg/L
- DO_r (river DO before mixing) = 8.5
- Q_e (effluent flow before mixing) = 100,000 L/min

- DO_e (effluent DO before mixing) = 2 mg/L
- BOD_e (effluent BOD before mixing) = 40 mg/L
- V_r (velocity of the river after mixing) = 20 m/min
- d (distance from discharge to fishing site) = 40 km

Your first step will be to determine the BOD and DO values of the stream immediately after the discharge flows into the river. You can use the mass-balance equations presented earlier in this chapter to do this. You will also want to determine how long it takes before the discharge reaches the fishing site.

Based on the preceding parameters, modify the initial BOD and DO values for the model *CHAP5A.STM*. Using the same default values for k_1 , k_2 , DO_{sat} , and *Natural BOD Inflow*, run the model and answer the following questions:

- What will the DO concentrations of the water be at the time that the discharge-river mixture reaches the fishing location? Are these concentrations on a downward, upward, or flat trajectory (i.e., is the problem getting worse, better, or remaining the same)?
- Do you think the problem would be worse in the summer or winter months? Why? How could you explore the potential seasonal impacts of the wastewater facility using this model?
- Describe at least two technologies and two policies one could employ to increase the DO levels at the fishing location.
- Modify the model to incorporate one technology from the preceding bullet. Run your new model and discuss the results.
- Modify your model to incorporate one policy from the fourth previous bullet. Run your new model (either with or without the technology modification) and discuss the results.

5.5 Exercises

1. Identify point and nonpoint sources of water pollution in your geographic area. Note the source, the type of pollutant, and the possible effects on your region's waterways.

2. A 1 L water sample is taken from a stream and has a BOD reading of 20 mg/L. After several days a new BOD reading is taken and a value of 10 mg/L is obtained. How much DO (in milligrams per liter) was consumed in the water over these several days?

3. A 1 million L body of water with a DO of 8 mg/L uniformly mixes with a 200,000 L body of water with a DO of 2 mg/L. What is the new DO of this mixture?

4. A stream flows at a rate of 500 L/min. The water of this stream has a DO value of 8 mg/L. An industrial facility discharges water with a DO value

of 3 mg/L at a rate of 60 L/min into the stream. Assuming uniform mixing, what is the DO value of the water at the point where these two flows meet?

5. We have assumed a constant flow of both the tributary and the facility discharge for the scenario in this chapter. Dissolved oxygen analyses are often done using “worst-case” conditions, where stream flow is very low and wastewater effluent flow is high. Develop some worst-case conditions for the problem in this chapter. Modify and run the model, and discuss your output.

6. We have assumed constant values for our deoxygenation and reoxygenation constants throughout the time period in which water flows from the wastewater site to the lake. Modify and run the original model so that this assumption does not hold, and discuss your results.

7. Identify and describe any feedback mechanisms in the original or revised model. Are they positive or negative?

8. The problem as modeled gives us a DO profile over time. Modify the model so that you can observe a graph that depicts DO versus distance?

9. Populations of aquatic species can be adversely affected by decreases in DO values. Modify the original model to include a population of a particular aquatic species that depends upon DO for survival. Evaluate how the population may be affected for a given segment of the river.

10. Think of another system that might exhibit the same characteristics as the DO sag model. Using what you learned in this chapter, build a model that depicts your new system.

5.6 References Cited and Suggested Readings

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6

Matter Cycling in Ecosystems

MUFASA: Everything you see exists together in a delicate balance, from the crawling ant to the leaping antelope.

SIMBA: But, Dad, don't we eat the antelope?

MUFASA: Yes, Simba. But let me explain. When we die our bodies become the grass and the antelope eat the grass, and so we are all connected in the great circle of life.

—FROM WALT DISNEY'S THE LION KING

Chapter Objectives—

After you finish this chapter, you should be able to:

1. *Explain the role of matter cycles in ecosystems and identify the primary matter cycles and how matter is processed through these cycles.*
2. *Explain the difference between donor-controlled and receptor-controlled flows, and be able to hypothesize on the type of flow that exists given a particular flow within a matter cycle.*
3. *Manipulate a model identifying the flow of matter through an ecosystem in order to explore the impacts of perturbations on that system.*
4. *Develop a systems model that accurately represents the flow of matter through an ecosystem.*

6.1 The Problem

Matter cycles provide the mechanism by which nutrients and other basic building blocks of life are made available to living organisms. Without matter cycles, life would not exist—it is the cycling of matter that allows the lion to feed off the antelope, the antelope to feed off the grass, and the grass to ultimately feed off of both creatures.

In this chapter we will explore the dynamic flow of matter among various physical states. The type of matter we focus on is *phosphorus*, a required nutrient for plant and animal growth. We will examine the phosphorus cycle in a lake ecosystem and the impacts of perturbations on this cycle [see Harte (1992) for a similar problem]. In particular, the scenario we explore is as follows:

Problem 6.1 A lake ecosystem, home to many phosphorus-limited algae species, is threatened with a potential influx of inorganic phosphorus

runoff from a local construction project. This phosphorus may enhance algae growth in the lake, thus reducing dissolved oxygen and increasing turbidity in the water. This will in turn create problems for the aquatic animals and bottom-dwelling plants living in the lake. A local planning authority has asked you to explore the impact on the lake's phosphorus cycle due to this potential inflow of inorganic phosphorus.

6.2 Background Information

6.2.1 Matter Cycling

Matter cycling (sometimes referred to as “biogeochemical cycling”) is the process by which nutrients and compounds are circulated throughout our ecosystem. Thriving ecosystems depend on the circulation of this matter; otherwise, living organisms would not be able to access the nutrients needed for survival.

Several processes drive matter cycling: solar energy, gravity, and the producer–consumer–decomposer chain. Solar energy is the main driver. The sun provides energy to plants (i.e., *producers*) that use this energy in the process of photosynthesis. In photosynthesis, these plants combine light, carbon dioxide, and basic inorganic nutrients to form more complex carbon compounds used to store energy for growth. Animals (i.e., *consumers*) eat producers and break down the complex compounds in order to derive energy and nutrients for sustaining life. When these consumers die, *decomposers* attack. Decomposers break down dead consumers and producers into basic nutrients and release these nutrients back into the environment. Once in their simplest forms, these nutrients are now ready for uptake once again by producers. Gravity affects these cycles by physically driving the matter flows (e.g., ensuring that a dead consumer really does fall to the ground where decomposers can get at it!). Figure 6.1 shows

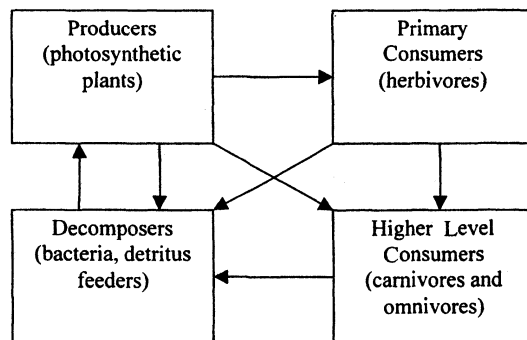


FIGURE 6.1. Matter flows for producer–consumer–decomposer web.

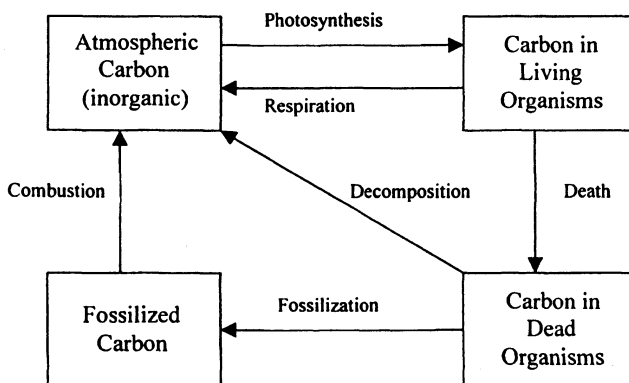


FIGURE 6.2. The atmospheric-terrestrial carbon cycle.

the producer-consumer-decomposer relationships for a simplified ecosystem. In this figure, the arrows represent matter flows between four different matter stocks (where each stock represents a population of living organisms).

The primary matter cycles existing in our environment include (Miller 1994) carbon cycle, nitrogen cycle, phosphorus cycle, sulfur cycle, and hydrologic cycle. These cycles are quite complex, with each containing flows that transform matter into various chemical and physical states. For example, matter can take the physical form of solid, liquid, or gas. Matter can also take on various chemical states (e.g., we can have sulfur in the form of H_2S or SO_2). Each of these cycles is depicted in Figures 6.2 through 6.6. These figures are simplified, but they still show the basics of each cycle. The boxes represent stocks or reservoirs of matter; the arrows represent flows between these reservoirs.

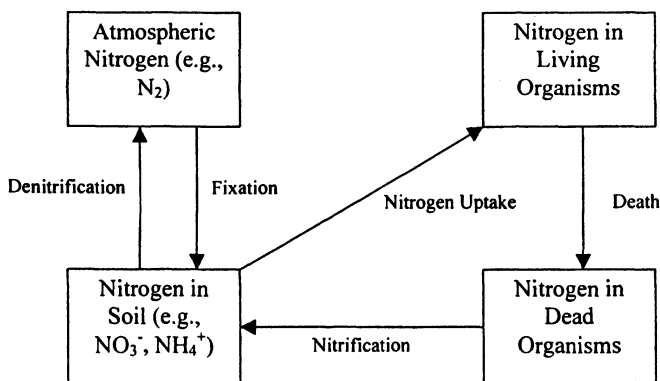


FIGURE 6.3. The nitrogen cycle.

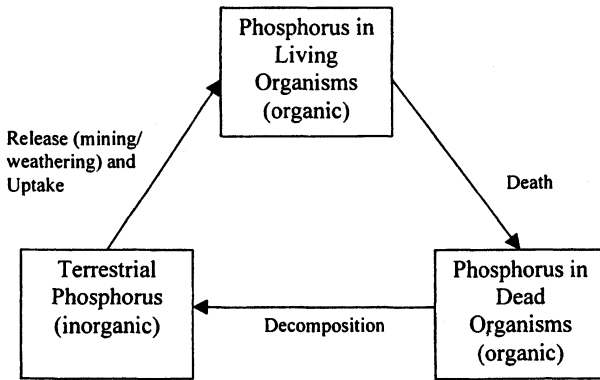


FIGURE 6.4. The phosphorus cycle.

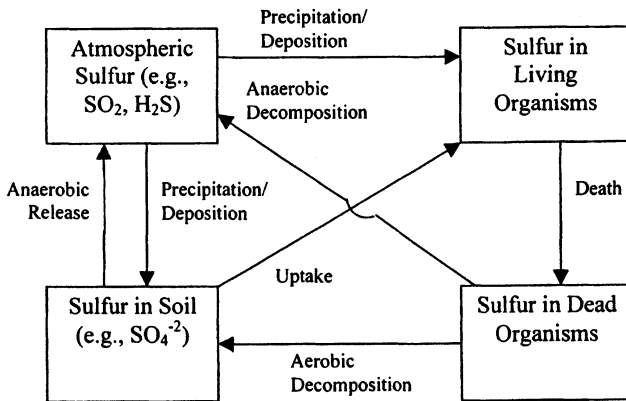


FIGURE 6.5. The sulfur cycle.

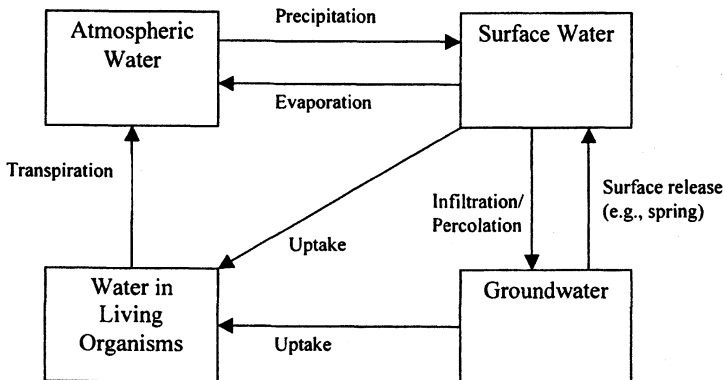


FIGURE 6.6. The hydrologic cycle.

You should notice that there are some common components for each of the nutrient cycles (C, N, P, S). All have some living stock, whereby the nutrient exists in an organic, living state. For each nutrient cycle, there is always a death process that moves matter from a living to a dead state. Finally, there is always a decomposition process by which bacteria break down dead matter into a basic inorganic state.

Although the cycles are similar, each cycle is somewhat unique. For example, although carbon, nitrogen, and sulfur compounds can exist in the atmospheric (i.e., gaseous) state, phosphorus cannot; thus, we do not expect much matter transport via the atmosphere for phosphorus as we would for other cycles (e.g., phosphorus particles can be transported by wind, but only in small quantities). Another unique aspect of each cycle is the way in which producers can access the nutrient. For example, in the carbon cycle producers can directly access carbon via CO_2 in the gaseous state (i.e., through photosynthesis); however, producers cannot do the same with gaseous N_2 . Nitrogen must undergo a process called *nitrogen fixation*, whereby bacteria convert the nitrogen into compounds that can be used by producers [e.g., nitrate ions (NO_3^-) or ammonium ions (NH_4^+)].

6.2.2 Mathematical Relationships and Our Systems Diagram

In this section, we will discuss the generic relationships that can be used to define a matter cycle system and use these relationships to build a systems diagram. These relationships provide information on *how* (described in terms of quantity and rate) matter flows among its reservoirs.

When we consider matter cycles in which the matter is *conserved* (i.e., the total matter in the system is constant), then our matter reservoirs need to be connected by inflows and outflows. We will not have matter emanating from or going to nowhere; instead, the inflow to one reservoir will inevitably be the outflow from another reservoir, and vice versa. This description implies that our matter reservoirs need to be connected to one another by flow constructs. Such a system is shown in Figure 6.7 for a generic type of matter that exists in three different forms.

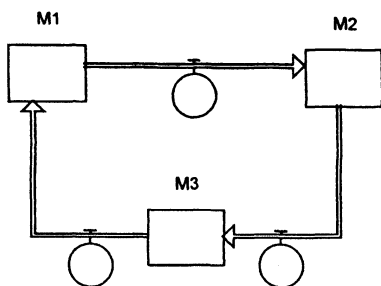


FIGURE 6.7. Generic matter reservoirs for a conserved system.

The rate at which matter flows from one reservoir to another is determined by the type of physical process that drives this flow. There are typically three kinds of flow processes: **donor-controlled**, **receptor-controlled**, or both **donor and receptor-controlled** (Harte 1992). *Donor-controlled* matter flows are those in which the flow rate is determined by the amount of matter in the reservoir *from which* the flow is emanating. *Receptor-controlled* matter flows are those in which the flow rate is determined by the amount of matter in the reservoir *to which* the flow is going.

From a systems diagram perspective, these three kinds of flows are shown in Figure 6.8. Notice that each flow is related to the donor-reservoir, the receptor-reservoir, or both. Note as well that each flow is related to a rate constant converter that helps define the rate at which the flow occurs (k_1 , k_2 , and k_3).

We can use donor-controlled or receptor-controlled rate equations (or both) to describe mathematically, the flow of matter from one stock (e.g., M_i) to another stock (e.g., M_j). We can write the flow equations a number of different ways; however, one set of possibilities is:

$$F_{M_1M_2} = k_1 M_1 \text{ (for donor-controlled flow)}$$

$$F_{M_2M_3} = k_2 M_3 \text{ (for receptor-controlled flow)}$$

$$F_{M_3M_1} = k_3 M_3 M_1 \text{ (for donor- and receptor-controlled flow)}$$

where $F_{M_iM_j}$ represents a flow from stock M_i to stock M_j , and the k 's represent some rate constant. Note the units of the rate constants in these examples. If flow is described as units of matter per time, then the rate constant must be in inverse time units (e.g., days^{-1}) for the donor- and receptor-controlled flows. Rate constant units for the last case (donor- and receptor-controlled flow) must be in $(\text{matter}^{-1})(\text{time}^{-1})$.

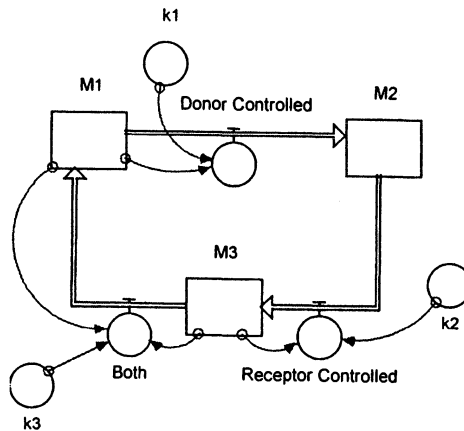


FIGURE 6.8. Diagram depicting three types of flows.

The donor-controlled flow equation and the receptor-controlled flow equation are relatively straightforward. In the donor-controlled equation, the flow of matter equals some rate constant times the amount of matter in the donor stock. For example, this might be the case for the flow of living matter to dead matter. In such a case, it is likely that the amount of living matter will define the amount of matter that “becomes dead” in each period. A small amount of living matter would likely mean that only a small amount is available to “die.” On the other hand, a large amount of living matter would likely mean that larger amounts of matter would flow from the living to the dead stock.

In the receptor-controlled equation, the flow of matter equals some rate constant times the amount of matter in the receptor stock. This might be the case for the flow of nutrients from a dead inorganic stock to a living organic stock (i.e., the “uptake” of nutrients by living matter). In this case, it is likely that the more living matter we have, the more consumption (or uptake) will occur from the inorganic nutrient stock.

The more complicated relationship is that defined by the donor- and receptor-controlled flow. The relationship here is such that high levels of either the donor *or* receptor stock variable will lead to high flows of matter between these two stocks. This relationship can be defined in a variety of ways, not only as $F_{M_3M_1} = k_3M_3M_1$ shown here. For example, a donor- and receptor-controlled relationship can be defined as $F_{M_3M_1} = k_3M_1 + k_4M_3$. The difference in these two flow equations is that in the former (multiplicative) as one of the stocks goes to zero, the flow between the two stocks also goes to zero. In the latter (additive), however, as one of the stocks goes to zero, the flow between the two stocks does not necessarily go to zero (which leads to the impossibility that matter might flow from a stock that has no matter!). Depending on the system under study, the appropriate donor- and receptor-controlled equation can be determined; we will use the multiplicative formulation in this chapter.

The two most important questions for the modeler are: (1) What is the nature of the flow equation (i.e., is it donor-controlled, receptor-controlled, or both)? (2) What are the values of the rate constants? To answer the first question, one must clearly understand the processes of matter cycling for the system under study. For a given flow, is the matter being “pushed” out of its current state (donor-controlled), or is the matter being “pulled” out of its current state (receptor-controlled)? Some general rules of thumb may apply:

- Flow of matter from a living organic state to a dead organic state is usually a donor-controlled process. We expect that the amount of living matter will control the flow from living to dead, unless the existence of the dead organisms actually affects factors that may cause death.
- Flow of matter representing the uptake of inorganic nutrients by a living organic state is usually a donor- and receptor-controlled process. Both the

amount of nutrients available and the number of living organisms vying for those nutrients dictate this flow. Similar to the predator-prey relationships seen in Chapter 4, more nutrients will lead to higher growth of living matter, and more living matter will lead to a greater amount of uptake of nutrients for survival.

- Flow of matter from a dead organic state to an inorganic state (i.e., the decomposition of dead matter) is usually a donor-controlled process. Here, we expect the flow to be driven by the amount of dead matter that is decomposing and not by the inorganic matter to which the dead matter is flowing.

Next, the rate constants that are used in the flow equations must accurately depict the rate of transformation of matter from one state to the next. This is usually an empirical estimate that is generated by studying the stock of matter in various states over long periods of time. The following information might help us in our quest to determine these rate values:

- For flows from living to dead matter, we can often estimate rate values based on the lifespans of the living organisms.
- For flows from inorganic to living matter, we can often estimate rate values based on the growth rates of the living matter.
- For flows from dead matter to inorganic matter, we can often estimate rate values based on the rate at which the matter decomposes.

Based on the scenario presented in this chapter, we can think of a systems diagram that has phosphorus stock variables for each of the matter reservoirs, connected by flows depicting the movement of phosphorus from one reservoir to another. (We will ignore the role of the construction site for the time being). These three reservoirs should represent phosphorus in the living organic state (let us call this L), in the inorganic state (let us call this I), and in the dead organic state (let us call this D). Figure 6.9 depicts these three reservoir variables.

We now need to determine which kinds of flows we need between each of these three reservoirs. First, using our “rules of thumb” presented earlier, we might imagine that the flow of living organic phosphorus (L) to dead organic phosphorus (D) should be a donor-controlled process. It is

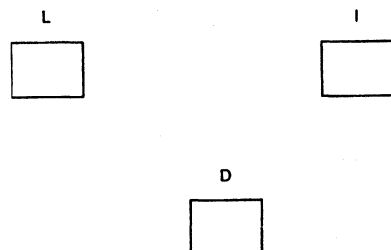


FIGURE 6.9. Three reservoirs for the phosphorus problem.

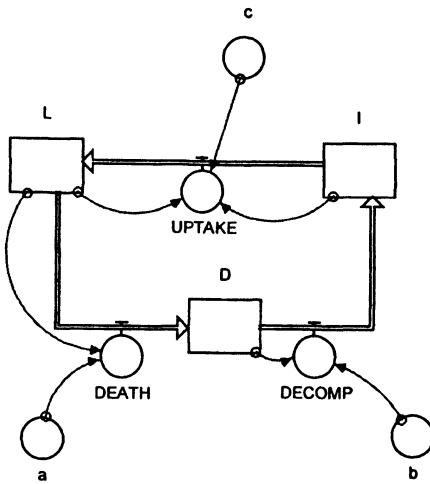


FIGURE 6.10. Reservoirs and flows for the phosphorus problem.

the amount of living matter that will ultimately determine the amount that dies. Second, the flow of phosphorus from reservoir *D* to our inorganic reservoir (*I*) will also be a donor-controlled process, for reasons outlined earlier. Finally, the flow from *I* to *L* will probably be both a donor- and receptor-controlled process, because this flow rate will depend both on the amount of phosphorus available in the inorganic state and on the amount of living matter desiring to take up this phosphorus. Thus, we can draw the complete systems diagram for the phosphorus cycle as shown in Figure 6.10, where *DEATH*, *DECOMP*, and *UPTAKE* represent our system flows.

These relationships can be defined mathematically as:

$$\begin{aligned} \text{DEATH} &= aL \\ \text{DECOMP} &= bD \\ \text{UPTAKE} &= cIL \end{aligned}$$

6.3 Difference Equations and the Steady-State Solution

It is helpful to explore the relationships depicted in this model further by writing out the difference equations (and rate equations) for each stock variable. Recall from earlier chapters that in using difference equations, we know that the value of a stock at time $t + \Delta t$ is equal to the stock at time t plus all inflows and minus all outflows. The following difference equations describe each of our phosphorus stocks.

For our stock of living organic phosphorus (*L*), we have:

$$\begin{aligned} L(t + \Delta t) &= L(t) + cI(t)L(t)\Delta t - aL(t)\Delta t \\ L(t + \Delta t) - L(t) &= [cI(t)L(t) - aL(t)]\Delta t \end{aligned}$$

$$\begin{aligned}\frac{\Delta L(t)}{\Delta t} &= (cI(t) - a)L(t) \\ \lim_{\Delta t \rightarrow 0} \frac{\Delta L}{\Delta t} &= \frac{dL}{dt} = (cI - a)L\end{aligned}\quad (6.1)$$

For our dead organic phosphorus (D), we have:

$$\begin{aligned}D(t + \Delta t) &= D(t) + [aL(t) - bD(t)]\Delta t \\ D(t + \Delta t) - D(t) &= [aL(t) - bD(t)]\Delta t \\ \frac{\Delta D(t)}{\Delta t} &= [aL(t) - bD(t)] \\ \lim_{\Delta t \rightarrow 0} \frac{\Delta D}{\Delta t} &= \frac{dD}{dt} = aL - bD\end{aligned}\quad (6.2)$$

For our inorganic phosphorus state (I), we have:

$$\begin{aligned}I(t + \Delta t) &= I(t) + bD(t)\Delta t - cI(t)L(t)\Delta t \\ I(t + \Delta t) - I(t) &= [bD(t) - cI(t)L(t)]\Delta t \\ \frac{\Delta I(t)}{\Delta t} &= [bD(t) - cI(t)L(t)] \\ \lim_{\Delta t \rightarrow 0} \frac{\Delta I}{\Delta t} &= \frac{dI}{dt} = bD - cIL\end{aligned}\quad (6.3)$$

The system defined by these equations and by the system schematic is a dynamic system. As we have seen with earlier population models, however, many dynamic systems achieve *steady-state* conditions. At steady state, the following will be true:

$$\frac{dL}{dt} = \frac{dD}{dt} = \frac{dI}{dt} = 0$$

This simply states that the inflows and outflows for each stock are equivalent, so the value of the stock remains constant (i.e., changes in the stock over time are zero). Given the preceding rate equations, we can determine the relationships among the rate constants and the steady-state stock variables using simple algebra, which we leave as an end-of-chapter exercise for the reader.

6.4 Modeling the Dynamic Phosphorus System

We are now ready to develop the model for the problem described in this chapter and to explore how the system responds to the perturbations discussed in that problem. First, however, let us examine the phosphorus cycling

model without the complications of the added inorganic inflow from the construction site. Let us suppose that you conduct some field tests and determine the following for the lake system (Harte 1992):

- Volume of Lake = 10^6 L
- L = 0.2 moles (P)
- D = 1.0 moles (P)
- I = 0.1 moles (P)
- a = $.25 \text{ yr}^{-1}$
- b = $.05 \text{ yr}^{-1}$
- c = $2.5 \text{ yr}^{-1} \text{ mole}^{-1}$

Using these parameter values we can begin to better understand the impacts of various perturbations to the system. Open the model entitled **CHAP6a.STM**. This model is identical to Figure 6.10. First, check to see whether these values represent a steady-state condition using the steady-state mathematical relationships presented earlier.

Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** on the drop-down menu. The graph in the model shows the amount of phosphorus in the dead organic (D), inorganic (I), and living organic (L) states. The model is set up to conduct calculations similar to those in the preceding difference equations over a period of 20 years, with $\Delta t = 0.25$ years. This model is in “flight simulation” mode, which allows the user the opportunity to modify variables while the model is running. In this model, you can use the slider bars to change rate constants a , b , and c while the model is running. You can also use the dials to adjust the *initial* values for D , I , and L (note that you cannot adjust D , I , and L values while the model is running because these values are determined endogenously within the model). Finally, you can click on the table icon to see the actual values for D , I , and L for each year of the run.

Run the model several times, adjusting different parameters each time. Explore your results. In particular, run the model while only adjusting one variable at a time. Then, consider the following questions:

- If you change a parameter and then let the model continue to run, do the D , I , and L values of phosphorus move toward an equilibrium point? Is it a “stable” model?
- What happens when you change the initial value of one of the phosphorus stocks? Can you explain your results?
- For each simulation you ran, explain what you did and what your results were. Is there any variable that seems to “drive” this system? Use the strategies discussed in Chapter 3 for your analysis.

Now, let us focus on analyzing the problem presented at the beginning of this chapter. We are asked to determine the impact of an increased flow of inorganic phosphorus into a lake ecosystem due to a construction project. During construction, which is expected to last 1 year, inorganic phosphorus

from the rocks and soil will be released. That phosphorus will flow into the lake via runoff, thereby adding phosphorus to our closed lake phosphorus cycle. Because there are phosphorus-limited algae at the lake, this increase in phosphorus may cause increased algae growth and all of its associated problems. Planners are interested in exploring the impacts of this potential construction on the phosphorus cycle.

To analyze this problem, we need to modify our original model by adding an inflow into our inorganic phosphorus reservoir. Open the model **CHAP6b.STM** located on your CD-ROM. This model is identical to model **CHAP6a.STM**, except it includes an additional inflow into the inorganic phosphorus stock. The quantity and duration of the inflow are controlled by the self-described exogenous variables shown. These can be modified using the “dial” functions below the model. The default value is an inflow of 0.25 moles/year lasting 1 year. The model is also designed so that this inflow does not begin until year 5. [This was done using a nested IF-THEN-ELSE statement in STELLA®. The exact statement is: *Construction = IF(Time < 5) THEN(0) ELSE(IF(Time < (5 + Duration)) THEN(Amount) ELSE(0))*].

Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** in the drop-down menu. Study the output from the model and answer the questions that follow.

- What phosphorus stock is most affected by the construction influx? Can you explain why this is the case?
- What happens in the “transition period” immediately after the construction influx? Explain qualitatively why this occurs.
- Does the model begin in steady state? Does the model achieve steady state after the construction perturbation? Does it achieve a value that you would expect, considering your steady-state equations examined earlier in this chapter?
- Modify the values of various parameters in the model using the slider and dial tools and run the model. Discuss your results quantitatively and qualitatively.

6.5 Exercises

1. In Figures 6.2 through 6.6 we show each of the matter cycles as a **closed** system. This is not always the case, because most ecosystems receive flows of matter into and out of their ecosystem. How can you modify these figures (and any models discussed in this chapter) to account for these flows?

2. Notice that in Figure 6.2 we only depict the atmospheric–terrestrial carbon cycle. Develop a matter flow diagram, similar to those described earlier, for the *aquatic* carbon cycle. You may have to do some research to determine the stocks and flows for the aquatic carbon system. Then, connect your diagram with Figure 6.2 in order to obtain an atmospheric–aquatic–terrestrial matter flow diagram of the carbon cycle.

3. Choose a matter cycle from Figures 6.2 to 6.6. For each of the flows in that figure, discuss whether the flow is donor-controlled, receptor-controlled, or both. Also, discuss whether you expect rate constants for those flows to be large or small.

4. Assume a steady-state population of 1 million humans, each with a mass of 70 kg. Also assume a functional form of the flow of “human mass” from dead to living as described earlier (i.e., as a donor-controlled flow). What rate constant would you use if you were told the life expectancy of this population of humans was 75 years? Based on the numbers given, how much “human mass” (in kilograms per year) would you expect to flow from the human population each year?

5. Using algebra, establish the relationships between L , D , and I at steady state. (*Hint:* Set each of the differential equations to zero and then solve for each variable.) Then, using your results, qualitatively discuss how changes in each of the rate constants affect your steady-state conditions. Explain why these changes make sense using your knowledge of matter cycling as a basis for your answer.

6. Modify model **CHAP6a.STM** by creating an outflow from your stock variable L at some time t (use **CHAP6b.STM** for an example of how this might be done). This outflow might represent the elimination of some living species in the ecosystem. What happens to the system? How do your new stock values change over time?

7. Modify any of the models considered in this chapter so that the rate variables are not constant, but are a function of the two stocks they influence (i.e., the donor- and receptor-). Run your model and explain your results.

8. Modify model **CHAP6b.STM** for the case in which an industrial facility locates on this lake system and contributes a continuous (but not necessarily constant) flow of inorganic phosphorus into the lake. How does this activity affect the overall balance of the phosphorus cycle?

9. As mentioned in the original scenario, increases in phosphorus flows into lakes and ponds often provide nutrients for excessive algae growth. This excessive growth is called an *algae bloom*. Algae blooms have been known to cover an entire lake surface. Algae blooms can be detrimental to the entire lake system because they prohibit sunlight from penetrating the lake surface, thus eliminating photosynthesis activities of lakebed and suspended plants. This lack of photosynthetic activity reduces the inflow of dissolved oxygen in the water. In addition, excessive algae that dies during and after algae blooms decompose on the lake bottom, further depleting the water of dissolved oxygen essential for aquatic life. Discuss how the activities conducted during this module could be used to explore algae blooms. What additions to your model are necessary to evaluate the detailed impacts of

increased phosphorus inputs into a lake or pond? If you are comfortable building models in STELLA®, create a model that links the phosphorus cycle presented here with a model of algae population.

10. The phosphorus cycle is only one of the cycles discussed earlier. Using ideas discussed in this module, construct a dynamic model for another matter cycle discussed in this chapter. You will have to conduct some research in order to determine your initial stock and flow values. Similar to what was done in this chapter, create a scenario and associated perturbations to demonstrate how your cycle responds to various human, technological, and natural phenomena.

6.6 References Cited and Suggested Readings

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7

Modeling Mobile Source Air Pollution Inventories

Chapter Objectives—

After you finish this chapter you should be able to:

- 1. Describe the role of mobile sources in urban air quality and identify the major variables that affect total mobile source inventories.*
- 2. Define a cohort model and identify situations in which cohort models are useful modeling constructs.*
- 3. Build a cohort model for an appropriate application.*
- 4. Identify technology and policy options for reducing mobile source air pollution, and understand how these options might affect the different variables of a mobile source emissions model.*

7.1 The Problem

Air pollution in urban areas is a growing concern throughout the world. Hundreds of millions of people currently live in areas with air pollution far higher than that thought to be “healthy” air. Of particular concern are the high levels of ozone, particulate matter (PM), carbon monoxide (CO), and air toxics that are ubiquitous in metropolitan areas worldwide. This pollution leads to increases in lung cancer, respiratory illness, and other chronic and acute toxic effects.

One of the primary sources of urban air pollution is light- and heavy-duty vehicles (i.e., mobile sources). Vehicles burn gasoline or diesel fuel in their engines and emit CO, oxides of nitrogen (NO_x), unburned hydrocarbons (HC), and air toxics (e.g., formaldehyde, benzene, or acetaldehyde). Each of these is either a primary pollutant or a precursor to secondary pollutants like low-level ozone. Because of trends that suggest that pollution from mobile sources is increasing as a percentage of total emissions, pollution control efforts of many government air quality agencies have shifted from stationary sources to mobile sources.

To better understand the role of mobile source emissions in urban areas we need to develop models that can help to predict total emissions from these sources. We call this conducting an “emissions inventory.” In addition, these models should enable us to test and evaluate regulatory mechanisms for mobile source pollution control. In this chapter you will investigate such a model. The scenario you will explore in this chapter is as follows:

Problem 7.1 An urban area is concerned about increasing concentrations of carbon monoxide in its urban airshed. Carbon monoxide is a poisonous gas that causes respiratory and other problems in humans. A metropolitan planning organization has asked you to develop a model that predicts total annual carbon monoxide emissions from mobile sources within the metropolitan area over the next 20 years. You are also asked to explore the effectiveness of various air quality improvement technologies and policies.

7.2 Background Information

7.2.1 Emissions from Mobile Sources

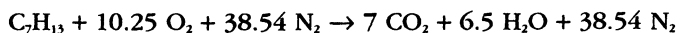
Automobile, truck, and bus populations in the United States have topped 200 million vehicles, representing more than 30% of the total number of registered vehicles in the world (Davis 1997, Tables 1.1 and 1.2). In the United States, there are about 135 million automobiles in operation, each traveling an average of 11,300 miles/year—a figure that has increased at a 2.0% annual rate from 1985 to 1995 (Davis 1997, Table 3.9). At an average fuel economy of 22.6 miles per gallon, this transportation represents the consumption of 68 *billion* gallons of gasoline annually, or approximately 4.6 million barrels of oil *per day*. (Although these numbers are for the United States, similar situations exist in other industrialized countries and are expected to arise in developing nations in the not-too-distant future).

Consuming all this petroleum does have its downside from both an energy security and environmental perspective. Energy security issues, arising from an overwhelming dependence on petroleum in our transportation sector, will not be discussed in this chapter. We will concentrate instead on the environmental impacts of burning this petroleum in our automobiles.

Most automobiles in the United States operate using four-stroke, spark ignition internal combustion engine (ICE) technology. With this technology the downward movement of a piston encased in a cylinder and the opening of an intake valve draw a fuel-air mixture into the cylinder. Upward movement of the piston compresses this mixture and a spark is applied to ignite the mixture. The mixture combusts and expands rapidly, triggering another downward motion of the piston. On its return, the piston pushes exhaust gases out of the cylinder through an exhaust valve. The components of this exhaust gas are what have created air quality problems in urban areas.

Exhaust emissions can be explored by referring to the combustion chemistry that is occurring in an ICE. Gasoline is a very complex mixture of hydrocarbons; however, we may consider an average ratio of hydrogen to carbon for a gasoline molecule to have a formula of C_8H_{18} (Masters 1998). The following reaction shows this combustion process for this average hydrocar-

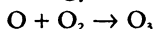
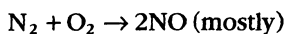
bon under ideal conditions, including air (O_2 and N_2) components of the mixture.



In this case, the stoichiometric ratio of air-fuel is equal to about 14.5. This can be calculated by taking the ratio of the mass of air (based on 10.25 moles of O_2 and 38.54 moles of N_2) needed to combust the mass in one mole of fuel (C_7H_{13}).

If the air-fuel mixture is *rich* (i.e., the ratio is less than 14.5), then there will not be enough air to burn all the fuel. Some unburned fuel (hydrocarbons) will be released through the exhaust. In addition, under rich conditions, carbon atoms cannot find the two oxygen atoms needed to form CO_2 , and will form CO instead. On the other hand, if the mixture is *lean* (i.e., the ratio is greater than 14.5), then there will be excess nitrogen and oxygen that can combine at high engine temperatures to form NO_x .

Exhaust gases of CO, HCs, and NO_x are of concern for air quality managers. Carbon monoxide is a poisonous gas that leads to respiratory problems in humans. NO_x and HCs are the precursors of low-level ozone, as demonstrated in the following (very simplified) reactions:



In addition to CO, HCs, and NO_x , automobiles also emit hazardous air pollutants, primarily benzene, formaldehyde, acetaldehyde, and 1,3-butadiene. These air toxins are starting to gain more attention in the regulatory community due to the fact that these emissions may be the highest sources of lung cancer from air pollution in urban areas. Finally, particulate matter (PM) from gasoline engines also has toxic implications, although PM is more associated with diesel engines than it is with gasoline.

Many countries have standards for emissions from mobile sources. For example, in the United States, the Clean Air Act Amendments of 1990 have identified exhaust emissions standards for new vehicles produced in model year 1995 (MY95) and beyond. These emissions standards are shown in Table 7.1. These particular standards apply to light-duty cars and trucks with gross vehicle weights of less than 3,750 pounds. In addition, some states within the United States (e.g., California) have adopted more stringent emissions standards. The values in Table 7.1 reflect the U.S. federal government's emissions requirements.

Although these emissions levels are low, the large numbers of vehicles on the road and the high annual miles traveled per vehicle mean that the mobile source contribution to air pollution in urban areas is still high. In fact, in the United States in 1995, highway vehicles contributed 64% of national CO emissions, 35% of national NO_x emissions, and 27% of national volatile organic compound (VOC) emissions (Davis 1997).

TABLE 7.1. U.S. emissions standards for new light duty vehicles.

Type	Less than 5 years or 50,000 miles	Less than 10 years or 100,000 miles
Carbon monoxide	3.4 grams/mile	4.2 grams/mile
Nonmethane hydrocarbons	0.25 grams/mile	0.31 grams/mile
Oxides of nitrogen	0.40 grams/mile	0.60 grams/mile

7.2.2 Vehicle Deterioration and Cohort Models

We can calculate emissions from a given population of vehicles using the following equation:

$$E_i = \sum_j V_j \cdot VMT_j \cdot EF_{ij} \quad (7.1)$$

where E_i = the total annual emissions of pollutant i (grams/year)

V_j = is the total number of vehicles of type j on the road during that year (vehicles)

VMT_j = the average annual miles traveled for vehicles of type j (miles/vehicle/year)

EF_{ij} = the average emissions of pollutant i for vehicle type j (grams/mile). (Note EF is often referred to as the vehicle "emission factor").

The emissions calculation in Equation (7.1) can be reflected in a simple systems diagram, as shown in Figure 7.1. This figure shows a reservoir representing a vehicle population. This reservoir has an inflow (*Vehicles Purchased*) and an outflow (*Vehicles Scrapped*). Emissions are represented by a converter that is related to an average emissions factor (EF) for the vehicle population and an average vehicle miles traveled (VMT) for the vehicle population. Equation (7.1) can be defined in the *Emissions* variable.

When determining actual emissions from a population of vehicles, however, we must recognize that emissions from different makes, models, and age of vehicles are not identical. Larger and older vehicles usually have

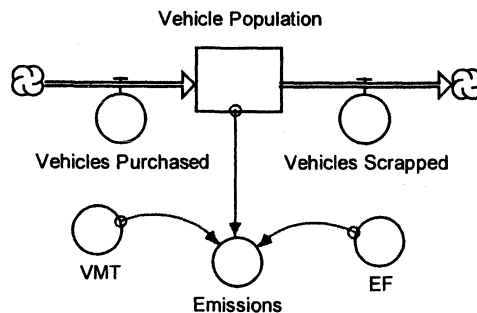


FIGURE 7.1. Simplified system for calculating emissions for a vehicle population.

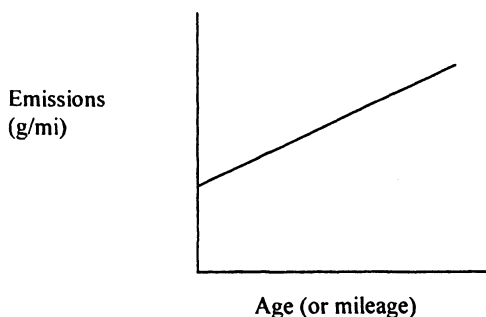


FIGURE 7.2. Emissions versus age relationship.

higher emissions values than do newer and lighter vehicles. Vehicle size is a factor because larger vehicles typically have lower efficiencies (defined in miles per gallon), so they need to consume more fuel per mile driven. Thus, their emissions per mile tend to be higher. Vehicle age is a factor because as a vehicle gets older and accumulates miles, engine wear, improper maintenance, and loss of efficiency in catalytic converters lead to higher emissions. The increase in emissions over the lifetime of a vehicle is called ***emissions deterioration***. This deterioration is reflected in Figure 7.2.

In order to capture the age relationship in modeling, we will now introduce the concept of ***cohort models***. An “age-based cohort model” divides a population (whether cars, people, animals, etc.) into a number of different subpopulations (or ***cohorts***) based upon age. In our problem, a cohort model will allow us to determine more realistically the emissions impacts of a vehicle population of different aged vehicles. A cohort model will also allow us to identify those segments of the vehicle population that contribute most to mobile source emissions.

Cohort models are often used to model populations in which the activities of certain cohorts in those populations need to be tracked, and where the relative sizes of the cohorts can change dynamically over time. For example, recall the population model we have discussed in earlier chapters and shown in Figure 7.3. In this simplified model, we assumed that births and deaths occurred based on the total existing population. In reality, however, births are

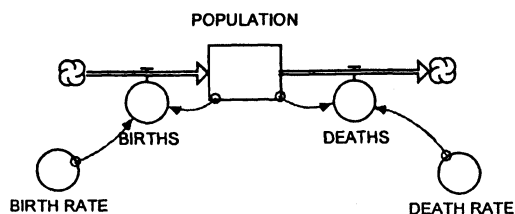


FIGURE 7.3. Population model without cohorts.

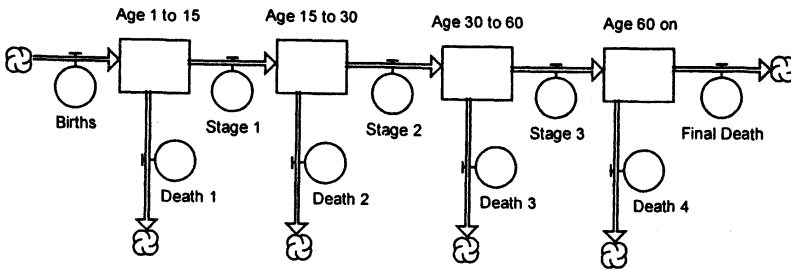


FIGURE 7.4. Population model with cohorts.

more likely a function of the population of younger adults in their “child-bearing” years; similarly, deaths are more likely a function of the population of older adults in their “silver” years. The model shown in Figure 7.3 does not reflect that distinction. In addition, without a cohort model, one has no way of knowing how the population is distributed among age groups and how this distribution may change over time given certain perturbations to the system.

To make this model more realistic and to track particular population segments, one could develop a cohort model that looks something like that in Figure 7.4. Now the population is divided into four cohorts. Each of these cohorts has at least one inflow and at least one outflow. Although not depicted in this model, one could have cohort-specific death rates that are a function of the age of the cohort. In addition, one could have birth rates as a function of cohort age. The birth rates and cohort population sizes would dictate the total number of births in the system.

As an introduction to how cohort models are useful for understanding population growth, open the model titled **CHAP7a.STM** on your CD-ROM. This model is a simple STELLA® model that can be used to demonstrate population growth for a population of humans over time. The model is a cohort model whereby the population is divided into five age cohorts. Birth rates and death rates vary according to the age category, and births are a combined function of each of these values. The model is set up to simulate population growth over a period of 50 years, with $\Delta t = 0.5$ years. Settings for the initial populations of each cohort are shown on the “dials,” which can be adjusted before each run. Birth rates and death rates are shown on the “slider bars” underneath the dials, with default values shown. The graph shows the population for each cohort for the 50-year run period. By clicking on the **Table** icon, the user can also view actual values for each of the cohorts.

Note that each cohort (except the last) looks like a typical stock variable, but has vertical lines running through it. This depicts a stock variable called a **conveyor**. Conveyors were discussed in Chapter 3. Recall that a conveyor works exactly like a conveyor belt—material is loaded onto the conveyor at one end and is unloaded at the other end. The modeler must define the **transit time** during which this material is “conveyed” through the reservoir.

This model uses aging people instead of material, but the idea is the same. Each cohort (except the last) has a transit time equal to 15 years. Thus, people enter the first cohort, “ride” it for 15 years, and then enter the second cohort.

In addition, recall that conveyers allow us to add a “leakage fraction” to each conveyer. This is defined as a percentage of material that “leaks” off the conveyer during the transit time. In this model, the leakage is the number of deaths that occur for a group of people during the transit time. This leakage percentage is defined by the death rates for each of the cohorts. Under default conditions, the amount of material that leaks off a conveyer is distributed evenly over the transit time. The units for the leakage percentage are $(\text{transit time})^{-1}$.

Run model *CHAP7a.STM* by clicking on **Run** in the top toolbar, and then clicking on **Run** on the drop-down menu. The model is set up to run in “flight simulator” mode, which allows the user to modify variables while the model is running. How does each of the cohort populations change over time using the default values? How does the total population change? Change the setting of the birth rate for one or more of the cohorts and run the model again. Now modify various parameters while the model is running and observe your results.

A similar cohort system can be applied to vehicle populations and emissions inventories. A cohort model used for inventories will allow us to account for the higher emission contributions for older vehicles. A cohort model will also allow us to explore technology and policy interventions that impact only certain vehicle populations (e.g., scrappage programs that are aimed at reducing the number of older vehicles on the road, or new alternative fuel vehicles that replace current purchases of gasoline vehicles).

You might guess that you could construct a vehicle cohort model almost identical to the population cohort model of *CHAP7a.STM*. You would first establish a series of cohorts, each representing a different vintage of vehicles. The number of cohorts selected is somewhat arbitrary, but it should reflect the general tendency for vehicles to remain relatively clean initially, and then to experience emissions deterioration over time. This basic design is reflected in Figure 7.5, where we have set up a series of cohorts representing vehicles of various ages. The initial inflow to this series of vehicle

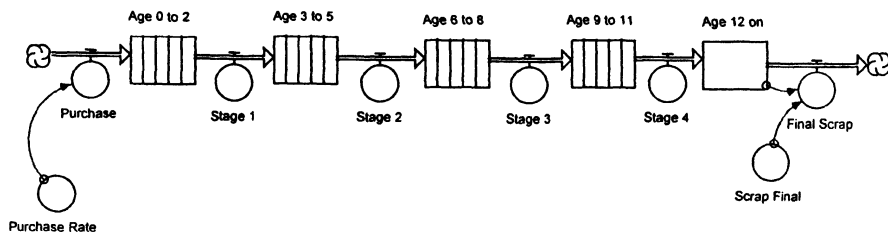


FIGURE 7.5. Auto cohort model without leakage.

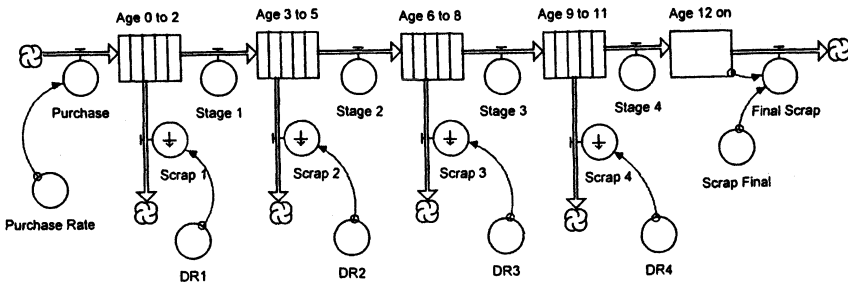


FIGURE 7.6. Auto cohort model with leakage.

cohorts represents vehicle purchases per time, and the final outflow represents vehicles that are finally scrapped after a long lifetime.

We would also need to add leakages for each cohort to this system. These leakages represent untimely vehicle scrappage, either due to accidents or breakdowns. We would certainly expect a higher scrappage rate for older vehicles because these would tend to break down more often than newer vehicles; however, we will see some scrappage of new vehicles due mostly to accidents. Figure 7.6 shows the vehicle cohort model with the leakages added (identified as *Scrap* outflows). Note that each cohort has its own individual scrappage rate (*DR*). Also note that these rates represent the fraction of vehicles scrapped over the transit time for each conveyor.

Finally, because we want to use the vehicle model to conduct an emissions inventory, we need to add an emissions component. This addition is shown as an Emissions Submodel in Figure 7.7. To calculate emissions correctly, we need to specify emissions factors for each vehicle cohort. We may also wish to determine the annual average miles traveled for each cohort (e.g., newer vehicles may drive more miles per year than older vehicles). For this example, however, we have used a constant vehicle miles traveled across all cohorts.

You may note that the model described so far only addresses one *type* of vehicle (e.g., average light-duty passenger vehicles), whereas we may want a model to consider a number of different types of vehicles (e.g., heavy-duty vehicles, light-duty trucks, etc.). We will continue to develop this model under the assumption of only one type of vehicle on the road (let us call it a generic light-duty passenger car); one can make the model more complex by creating separate cohort models for each type of vehicle. This is left as an end-of-chapter exercise.

The mathematical relationships we use to calculate emissions in this model are still based on Equation (7.1), although now we need to apply this equation to each cohort and sum across the total number of cohorts we have. We can show this mathematically, as:

$$E_t = \sum_j \sum_k V_j \cdot VMT_{jk} \cdot EF_{ijk} \quad (7.2)$$

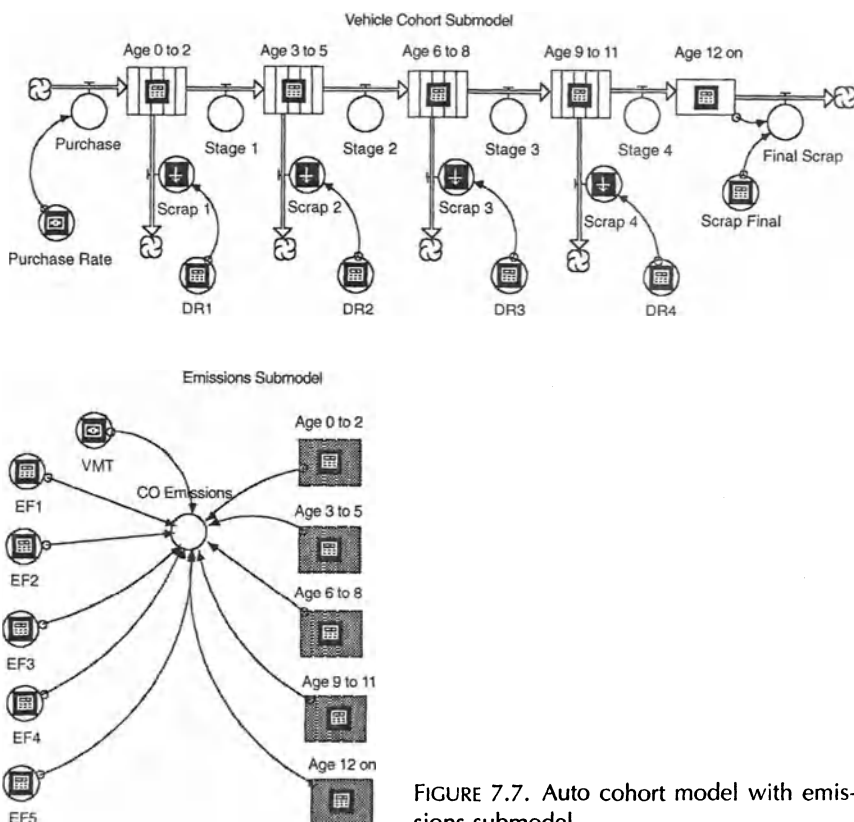


FIGURE 7.7. Auto cohort model with emissions submodel.

This equation has an additional identifier, k , which depicts each cohort (e.g., $k = 1$ may represent the first cohort of vehicles 0–2 years old; $k = 2$ may represent the second cohort of vehicles 3–5 years old, etc.).

7.2.3 Policy and Technology Options for Reducing Mobile Emissions

There are a variety of policy and technology options that may be employed to reduce emissions from mobile sources. This section provides a list of some of these options. In exploring the problem in this chapter, you may attempt to model one of these options as an intervention, or devise a policy or technology option of your own.

Policy Options

Policies that encourage people to drive less or to use public transportation or carpools have been widely instituted throughout many industrialized countries. Some specific policies include:

- High-occupancy vehicle lanes—this option reduces the number of vehicles on the road by providing an incentive (faster transit time) to people who carpool.
- Restricted roadways into urban areas—this option reduces travel on certain urban roads in order to encourage people to use mass transit.
- Bicycle paths—this option provides a safe and often time-saving alternative means of transport.
- Restrictions on inner-city parking (through space limitations or fees)—this option creates economic disincentives to people who drive their cars into urban areas.
- Tolls on certain roadways—this option creates a disincentive for driving a personal vehicle on roadways with tolls.
- Rebates for mass-transit usage—this option offers an economic incentive for people who use mass transit.
- Vehicle scrappage programs and incentives—this option reduces the number of older model year vehicles on the road, because these vehicles are thought to be the largest emitters of urban pollutants.
- Corporate Average Fuel Economy (CAFE) standards—this option forces car manufacturers to improve average energy efficiency in vehicles (measured in miles/gallon) so that less fuel is burned (and thus, fewer pollutants emitted) for a given travel distance.

Technology Options

Technologies that increase vehicle efficiencies, decrease emissions factors for certain pollutants, or reduce the need to drive altogether have been used worldwide to reduce mobile source emissions. Some technologies include:

- High-efficiency vehicles
- Improved catalytic converters
- Reformulated gasoline/oxygenated gasoline
- Alternative transportation fuels (e.g., natural gas, ethanol, methanol, propane, hydrogen, and electric vehicles)
- Telecommuting
- New forms of efficient mass-transit (e.g., Mag-Lev trains)

7.3 Difference Equations and Steady-State Solutions

In this section, we will explore the difference equations for reservoir variables in a cohort model. Difference equations for cohort models are fairly straightforward. To put it simply, the value of a cohort at a point in time is the amount of inflow from the previous cohort, minus the amount of outflow to the next cohort. In addition, if our cohort variables have a leakage component, then we need to consider this as an additional outflow.

It should be obvious that these difference equations will be coupled equations. Every inflow to a cohort is an outflow of a previous cohort (except,

of course, for the first cohort). Every outflow from a cohort is similarly an inflow to a subsequent cohort.

Let us now suppose that we have a generic cohort model with four cohorts (X_1, X_2, X_3 , and X_4). For the time being, let us ignore our leakage component. Each cohort, therefore, will have an inflow defined as I_1, I_2, I_3 , and I_4 , respectively, and an outflow defined as O_1, O_2, O_3 , and O_4 , respectively. Now, recognize that the amount flowing into a cohort is defined as the amount flowing out of the previous cohort. This in turn was the amount flowing *into* the previous cohort at a time equal to $t - T_{i-1}$, where T_{i-1} is the transit time for the previous cohort. Note also that the outflow for any cohort is equal to the inflow to that cohort at a time equal to $t - T_i$. Thus, we would have the following series of equations.

For X_1 :

$$X_1(t + \Delta t) = X_1(t) + I_1\Delta t - O_1\Delta t = X_1(t) + I_1(t)\Delta t - I_1(t - T_1)\Delta t$$

$$X_1(t + \Delta t) - X_1(t) = \Delta X_1 = [I_1(t) - I_1(t - T_1)]\Delta t$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta X_1}{\Delta t} = \frac{dX_1}{dt} = I_1(t) - I_1(t - T_1)$$

For X_2 :

$$X_2(t + \Delta t) = X_2(t) + I_2\Delta t - O_2\Delta t = X_2(t) + I_1(t - T_1)\Delta t - I_2(t - T_2)\Delta t$$

$$\text{Note } I_2(t - T_2) = I_1(t - T_2 - T_1)$$

$$X_2(t + \Delta t) - X_2(t) = I_1(t - T_1)\Delta t - I_1(t - T_2 - T_1)\Delta t$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta X_2}{\Delta t} = \frac{dX_2}{dt} = I_1(t - T_1) - I_1(t - T_2 - T_1)$$

For X_3 :

$$X_3(t + \Delta t) = X_3(t) + I_3\Delta t - O_3\Delta t = X_3(t) + I_2(t - T_2)\Delta t - I_3(t - T_3)\Delta t$$

$$\text{Note } I_2(t - T_2) = I_1(t - T_2 - T_1) \text{ and } I_3(t - T_3) = I_1(t - T_3 - T_2 - T_1)$$

$$X_3(t + \Delta t) - X_3(t) = I_1(t - T_2 - T_1)\Delta t - I_1(t - T_3 - T_2 - T_1)\Delta t$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta X_3}{\Delta t} = \frac{dX_3}{dt} = I_1(t - T_2 - T_1) - I_1(t - T_3 - T_2 - T_1)$$

For X_4 :

$$X_4(t + \Delta t) = X_4(t) + I_4\Delta t - O_4\Delta t = X_4(t) + I_3(t - T_3)\Delta t - I_4(t - T_4)\Delta t$$

$$\text{Note } I_3(t - T_3) = I_1(t - T_3 - T_2 - T_1) \text{ and } I_4(t - T_4) = I_1(t - T_4 - T_3 - T_2 - T_1)$$

$$X_4(t + \Delta t) - X_4(t) = I_1(t - T_3 - T_2 - T_1)\Delta t - I_1(t - T_4 - T_3 - T_2 - T_1)\Delta t$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta X_4}{\Delta t} = \frac{dX_4}{dt} = I_1(t - T_3 - T_2 - T_1) - I_1(t - T_4 - T_3 - T_2 - T_1)$$

One observation that can be made for these rate equations is that they can all be simplified down to functions of I_1 , the inflow into the first cohort. The difference equations simply track this initial inflow throughout each of the individual cohorts. If we know what the value for I_1 is for a given point in time, then we can calculate the expected values of each of these rate equations. Thus, if we have a spike or lull in our initial inflow, we expect to see that spike or lull transported through each of the cohorts over time.

The steady-state conditions occur at the point where all of the following are true:

$$\begin{aligned}\frac{dX_1}{dt} &= 0 = I_1(t) - I_1(t - T_1) \\ \frac{dX_2}{dt} &= 0 = I_1(t - T_1) - I_1(t - T_2 - T_1) \\ \frac{dX_3}{dt} &= 0 = I_1(t - T_2 - T_1) - I_1(t - T_3 - T_2 - T_1) \\ \frac{dX_4}{dt} &= 0 = I_1(t - T_3 - T_2 - T_1) - I_1(t - T_4 - T_3 - T_2 - T_1)\end{aligned}$$

Each of the cohorts will thus be at steady state when,

$$\begin{aligned}I_1(t) &= I_1(t - T_1) = I_1(t - T_2 - T_1) = I_1(t - T_3 - T_2 - T_1) \\ &= I_1(t - T_4 - T_3 - T_2 - T_1)\end{aligned}$$

This, of course, is a situation where the initial inflow to the first cohort remains constant over a period of time equal to $T_1 + T_2 + T_3 + T_4$.

7.4 Modeling the Dynamic Mobile Source Emissions System

We are now prepared to analyze the problem presented in the beginning of this chapter and to explore how your urban area's mobile source CO emissions inventory may be affected by various technology and policy initiatives. You are asked to determine the annual CO emissions inventory from mobile sources over the next 20 years. Many things will happen in the next 20 years that will affect the results of your model.

Let us suppose you conduct a number of demographic, consumer demand, and engineering studies and you have determined the information found in Table 7.2.

Open the model **CHAP7b.STM** located on your CD-ROM. This model is written in the STELLA® software language and is very similar to Figure 7.7. The model is designed to run over 20 years with a Δt equal to 1 year. The model includes two "submodels." The first submodel (the vehicle cohort submodel) is a standard cohort model that can predict vehicle

TABLE 7.2. Data for mobile source inventory model.

Current vehicle population	
0-2 years old	100,000
3-5 years old	100,000
6-8 years old	75,000
9-11 years old	75,000
>11 years old	50,000
Avg. annual VMT for the next 20 years	12,000 miles/year
Emissions factors	
0-2 years old	3.0 g/mile
3-5 years old	3.4 g/mile
6-8 years old	3.8 g/mile
9-11 years old	4.2 g/mile
>11 years old	5.0 g/mile
Scrap rates (fraction of vehicles scrapped while in the cohort)	
0-2 years old	0.10
3-5 years old	0.15
6-8 years old	0.20
9-11 years old	0.25
>11 years old	0.40
Annual vehicle purchases for the next 20 years	35,000

populations by cohort. There are five cohorts, with the first four representing vehicle vintages in increments of 3 years. The last cohort represents vehicles greater than 12 years old. Scrappage rates are identified for each cohort. In addition, there is an inflow into the first cohort that equals the purchase rate (defined in Table 7.2 as the annual vehicle purchases per year).

The second submodel (the emissions submodel) is designed to calculate total CO emissions from the entire vehicle population for each year. The emissions are calculated using Equation (7.2). Thus, critical inputs include the number of vehicles of each cohort, the emissions factor for each cohort, and the average vehicle miles traveled.

Graphs are included near the bottom of the model. The first graph depicts the vehicle population for each cohort over time. The second graph shows the total annual CO emissions in metric tonnes per year. To the right of the CO emissions graph is a set of three lists of values for initial vehicle populations, emissions factors, and scrappage rates. Initial vehicle populations are shown as the first list. Click on the down arrow next to the list title (i.e., next to *Initial Vehicle Population*) and click on the list you wish to view. You can change the values for each of these parameters by clicking on the appropriate box in your chosen list.

Finally, under the list box are two slider bars. The first is for *Purchase Rate* and defines the number of vehicles purchased each year. These are the vehicles that enter the first cohort in the model annually. The second slider bar

is the average annual vehicle miles traveled (*VMT*). When the model is in “flight simulator” mode (which is its default mode), you can change these parameters while the model is running and explore the results.

Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** on the drop-down menu. Study the output from the model and answer the following questions:

- Each of the cohort stocks begins to level off (i.e., achieve a steady state) at a certain point in time. Why is this happening?
- Total CO emissions tend to decrease and then begin to level off. Why is this the case? Can you tell which vehicle cohort contributes the largest to annual CO emissions? How can you modify the model so that emissions from each vehicle cohort are shown explicitly? (If you are comfortable in the STELLA® working environment, modify the model to do this).
- Modify some of the parameters in the model (e.g., scrappage rates, emissions factors, VMT, or purchase rate) and explain your results. Use the strategies discussed in Chapter 3. Of the variables you studied, which seems to have the greatest impact on annual CO emissions?
- Choose a technology or policy initiative from the list in Section 7.2.3 (or invent your own). Discuss how you would modify the model to incorporate this initiative. If you are comfortable programming in STELLA®, modify the existing model or build a new model to reflect your initiative. Run your model and discuss your results.

7.5 Exercises

1. Define a cohort model and explain why the cohort idea is a useful modeling tool. Identify at least three other environmentally related problems that may cause you to employ a cohort model for analysis.

2. In the system depicted in Figure 7.3, how would you expect the birth rates and death rates to change as we move from younger to older cohorts? How would you modify this model to show how the flows are affected by the sizes of the stocks? Redraw the model as shown, but include appropriate converters and connectors as needed.

3. Assume a population of 250 million vehicles. Assume that each travels an average of 12,000 miles/year, and that the hydrocarbon emissions factor for each vehicle is equal to 0.25 g/mile. How many kilograms of hydrocarbons are emitted from this population of vehicles annually?

4. In Problem 3 we assumed a homogenous population of vehicles. Now assume that 100 million of the 250 millions vehicles are heavy-duty vehicles (HDVs) and the remaining 150 million are light-duty vehicles (LDVs). Each of the LDVs travel an average of 12,000 miles/year and emits 0.25 g

of hydrocarbons per mile. Each of the HDVs travel an average of 15,000 miles/year and emits 0.60g of hydrocarbons per mile. How many kilograms of hydrocarbons are emitted now from the total vehicle population annually?

5. Redesign the model entitled **CHAP7b.STM** on your CD-ROM so that you can predict emissions from each individual cohort. Which cohort is the largest emissions contributor? How might this result impact policy and technology decisions designed to mitigate HC emissions from your total population of vehicles?

6. We may expect scrap rates to decline over time as vehicles become more durable and are better designed for longer life. What impact might this have on CO emissions? Redesign the model entitled **CHAP7b.STM** on your CD-ROM to explore this possibility.

7. We may expect that better engine technologies will reduce the amount of emissions deterioration over time. What impact might this have on CO emissions? Redesign the model entitled **CHAP7b.STM** on your CD-ROM to explore this possibility.

8. The market penetration of new vehicles will not likely follow constant growth as shown in the model **CHAP7b.STM**. Different penetration rates, such as exponential growth, exponential decay, or logistic growth, may be likely. Modify the model entitled **CHAP7b.STM** on your CD-ROM to account for new purchases following one of these alternative patterns.

9. The model discussed in this chapter was simplistic in the fact that it only looked at one type of vehicle. Redesign the model so that you can include two types of vehicles (e.g., light-duty vehicles and heavy-duty vehicles, or gasoline vehicles and alternative fuel vehicles). Conduct some research to identify typical populations and emissions from the two types of vehicles in your locality, and make appropriate changes to your model. Run the model and discuss your results. (*Hint: You will probably have to build another cohort submodel for the second type of vehicle*).

10. In this chapter, we focused on CO emissions for simplicity. In reality, emissions of hydrocarbons (HC), oxides of nitrogen (NO_x), carbon dioxide (CO₂), and various air toxics (e.g., formaldehyde, acetaldehyde, and benzene) are of concern to air pollution officials. Choose one of these non-CO pollutants and conduct some research to identify typical vehicle emissions, then modify the model from this chapter so that emissions of your selected pollutant will be modeled. Run the model and discuss your results.

11. Choose one or more of the policy and technology options for reducing vehicle emissions, as presented in Section 7.2.3. Modify the model from this chapter (or build a new one) that will allow you to analyze those mitigation options. Run the model and discuss your results.

7.6 References Cited and Suggested Readings

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8

Greenhouse Gases and Global Warming

Chapter Objectives—

After you finish this chapter you should be able to:

1. *Describe the greenhouse effect and the processes by which the earth balances energy flows, and how these flows affect global surface temperature.*
2. *Explain the important parameters within an earth energy balance model that lead to increased surface temperatures.*
3. *Explore and build a dynamic model that depicts the relationships among the many variables associated with earth's energy balance system.*
4. *Using a systems model, evaluate how greenhouse gas emission control strategies may affect the variables associated with earth's energy balance and ultimately its surface temperature.*

8.1 The Problem

Since the industrial revolution, concentrations of carbon dioxide (CO₂) and other greenhouse gases (GHGs) in our atmosphere have risen dramatically. These gases can lead to the warming of the earth's biosphere, with accompanying unprecedented climatic changes. The international community is currently discussing ways to mitigate the emissions of GHGs in order to prevent the potentially harmful impacts of global warming.

The concentration of CO₂, the GHG given most attention, is expected to double by the year 2050, inducing an earth surface temperature increase on the order of 2–5°C (Houghton et al. 1996). Such temperature changes are estimated using global climate models that predict the temperature of the earth based on the electromagnetic absorption and radiation characteristics of the earth and its atmosphere. In this chapter you will explore the earth's energy system and how the accumulation of GHGs in our atmosphere can affect earth's surface temperatures. In particular, you will address the following problem.

Problem 8.1 Many countries are concerned about the continued increase in GHG emissions from anthropogenic sources. The primary source of concern is CO₂ from the burning of fossil fuels. This CO₂ (currently at a

concentration of about 355 parts per million in the atmosphere) may increase global surface temperatures due to its ability to absorb and reradiate outgoing infrared radiation back toward the earth. An international organization has asked you to determine the impact of increased CO₂ concentrations on average global surface temperatures. In addition, you are asked to evaluate the potential impact of CO₂ mitigation options on these surface temperature predictions.

8.2 Background Information

8.2.1 Earth's Energy Balance and Black Body Radiation

The earth and all its ecosystems rely on energy from the sun. Solar energy drives matter cycles and provides one of the critical inputs to the photosynthetic process. Without this energy, life as we know it would not exist.

The amount of solar energy received by the earth per unit area per time is referred to as *energy flux* or *solar flux*. Solar flux is often described in units of watts per square meter (W/m²). (Recall that a *watt* is a measure of *power* measured in energy per unit time, where 1 W = 1 J/s). The solar energy that strikes the earth's surface is a function of the earth's distance from the sun, solar activity, and other variables.

Imagine the sun radiating energy out toward space in all directions over time. The planet earth is a sphere that intercepts a portion of that energy. Earth intercepts solar energy over an area equal to the area of a disk with a radius equivalent to the earth's radius (R). To see this experimentally, hold a small ball several inches away from a wall in a dark room. Hold a flashlight at arm's length from the ball, shining the light directly on the ball. The amount of light intercepted by the ball is the amount of light passing through a disk with a radius equal to the radius of the ball. The shadow of the ball on the wall shows the actual area of the disk that intercepts this light.

At a point just above the earth's atmosphere, the amount of solar flux is equal to 1,372 W/m². Although this value sometimes changes by a percentage or two due to solar activity, its relative consistency has allowed its value to be dubbed the *solar constant*.

Thus, the amount of energy per time (i.e., watts) being delivered to the earth from the sun (W_s) is equal to this flux times the area of the aforementioned disk; that is,

$$W_s = 1,372 \cdot \pi R^2$$

where R represents the radius of the earth.

Over the course of a 24-hour day, this energy is distributed over the entire sphere of the earth; therefore, we have $1,372\pi R^2$ watts distributed over a sphere of area $4\pi R^2$ (the surface area of a spherical earth). Thus, the average flux of solar energy reaching the earth is:

$$\frac{1372\pi R^2}{4\pi R^2} = 343 \left(\frac{W}{m^2} \right) \quad (8.1)$$

This 343 W/m^2 is called the **solar energy flux**, and is usually represented by the symbol Ω_s . This energy flux covers a large portion of the electromagnetic spectrum, taking on a wide variety of wavelengths (λ). Most of the incoming radiation, however, is in the wavelength range of 0–2,000 nanometers (nm), which covers the ultraviolet and visible portion of the electromagnetic spectrum. Gaseous oxygen (O_2) and ozone (O_3) in the stratosphere absorb a large portion of the incoming ultraviolet (UV) radiation, although most of the radiation in the visible part of the spectrum (i.e., 400–700 nm) reaches the earth.

Aside from the absorption of incoming solar energy by atmospheric gases, another hindrance to solar energy reaching the earth's surface is created by the reflective capacity of the earth and its atmosphere. Some of the incoming solar radiation is reflected off of the atmosphere and the earth's surface. The ability of the earth and its atmosphere to reflect solar energy is called its **albedo**. The albedo of the earth is usually given a value of 0.31 (i.e., about 31% of the incoming solar energy is reflected from the earth back into space).

The remaining 69% of the solar energy flux is absorbed by the earth-atmosphere system and reradiated, usually in the infrared part of the electromagnetic spectrum. This radiation is in the 5,000–30,000 nm wavelength range. It is this process of absorption and radiation that leads to the surface temperatures we experience on earth. Just as your body absorbs and radiates sunlight on a hot summer day—and becomes uncomfortably hot in the process—the earth's biosphere does the same. In addition, part of the earth's radiation is reabsorbed by atmospheric gases. The mechanism by which these gases absorb and reradiate this energy will be discussed in the next section.

Under steady-state conditions, the earth is in an **energy balance**; the amount of incoming solar flux is equal to the amount of energy being reflected and radiated from the earth. This energy balance is shown as:

$$\begin{aligned} \Omega_s &= a\Omega_s + (1 - a)\Omega_s \\ \Omega_E &= \Omega_s - a\Omega_s = (1 - a)\Omega_s \end{aligned} \quad (8.2)$$

where $\Omega_E \text{ (W/m}^2\text{)}$ = the energy flux radiated from the earth

Ω_s = the incoming solar flux [determined in Equation (8.1)
to be 343 W/m^2]

a = the albedo of the earth.

Under these steady-state conditions we can calculate the expected temperature of the earth's surface by applying the Stefan-Boltzman Law of Black Body Radiation. This law holds that the energy flux of a radiating "black body" is a function of the surface temperature of that body raised to the fourth

power. The term *black body* refers to the idea that the body in question is a perfect absorber and radiator of energy in the part of the electromagnetic spectrum under scrutiny. It turns out that earth has characteristics that make it a pretty good black body, with about a 95% radiation-absorption efficiency. The relationship between energy flux and surface temperature is given as:

$$\Omega = \sigma T^4$$

where Ω = energy flux (W/m^2)

σ = the Stefan-Boltzmann constant equal to $5.67 \times 10^{-8} \text{ (W}/\text{m}^2/\text{K}^4)$

T = temperature in degrees Kelvin.

For the solar-earth system described in Equation (8.2), the amount of energy flux absorbed and radiated by the earth (Ω_E) is equal to $(1 - a)\Omega_s$. Thus, the earth's average surface temperature is calculated as:

$$T = \left(\frac{\Omega_E}{\sigma} \right)^{\frac{1}{4}} = \left[\frac{(1 - a)\Omega_s}{\sigma} \right]^{\frac{1}{4}} \quad (8.3)$$

Note that this temperature represents an average surface temperature—one that we would expect if the solar energy flux were distributed equally over all parts of the globe. This, of course, is not the case, but it gives us a point of departure for studying the global warming phenomenon. By substituting values of $a = 0.31$ and $\Omega_s = 343 \text{ W}/\text{m}^2$, we can calculate the value of the earth's surface temperature to be approximately 255 K (or -18°C)! The average surface temperature of the earth, thankfully, is not so low (in fact, it is about 33°C higher, or $+15^\circ\text{C}$), so we must revisit our assumptions to determine why our estimate is inaccurate.

8.2.2 The Mechanics of the Greenhouse Effect

In 1800, Jean-Baptiste Fourier hypothesized that gases in the atmosphere act to “trap” heat within the earth's biosphere. It was unknown how this happened until scientists started to investigate how gas molecules respond to interactions with energy. When molecules are struck by photons of energy, those molecules absorb this energy and become “excited.” If the energy is in the ultraviolet (UV) part of the electromagnetic spectrum (i.e., low wavelength, high frequency, high energy), then the energy may break the bonds of the molecule. If the energy is in the infrared (IR) part of the spectrum (i.e., high wavelength, low frequency, low energy), then it may set the molecule spinning or may cause the atoms in the molecule to vibrate. Figure 8.1 depicts these possibilities.

Certain molecules, due to their composition and structure, will start vibrating when struck with IR radiation. These molecules absorb the incoming radiation, become “excited,” and then reradiate this energy (in all directions) as they move from this excited state back into a stable, or “ground,” state. Carbon

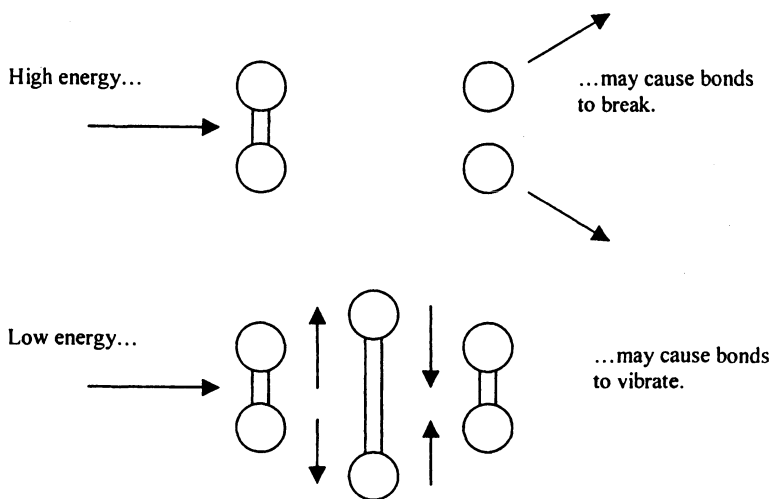


FIGURE 8.1. Results due to bombardment of molecules with electromagnetic radiation.

dioxide (CO_2), water vapor (H_2O), methane gas (CH_4), nitrous oxide (N_2O), and chlorofluorocarbons (CFCs) are all molecules that behave in such a manner.

Thus, the gases mentioned earlier have the ability to absorb *outgoing* infrared radiation from the earth and *incoming* radiation from the sun, and to reradiate this energy in all directions. On average, half of the reradiated energy will be emitted up (i.e., toward space) and the other half will be emitted down (i.e., toward earth), where it will be reabsorbed by the earth and ultimately reradiated. Through this process, the amount of energy flux directed at the earth is increased. Earth now has two inputs: direct energy from the sun and reradiated energy from atmospheric gases. Because earth is in an energy balance, this also means that the amount of energy flux *output* from the earth is increased. This increase in radiated energy increases the temperature of the earth based on the black body radiation model discussed earlier.

8.2.3 Mathematical Relationships and Our Systems Diagram

To build a valid model of earth's energy system, we must determine all the inflows and outflows of energy within the system. This is a complicated task. Among other things, we must consider solar inputs, albedo, and the absorption-reradiation of energy from atmospheric gases. Figure 8.2 shows a diagram depicting the energy flows of this system. These flows will be discussed shortly.

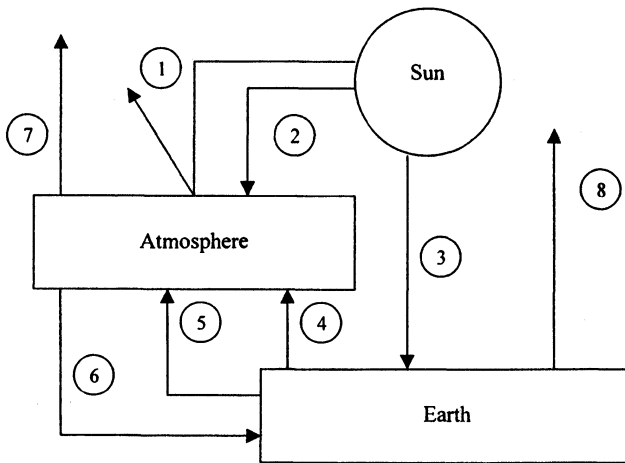


FIGURE 8.2. Sun-atmosphere-earth energy flow diagram.

Arrows represent each energy flow in Figure 8.2. These arrows are numbered from (1) through (8). Table 8.1 identifies each flow according to its corresponding number, describes that flow, presents a mathematical relationship for that flow, and offers a current estimate of flow values (note that all “flows” here are in units of flux, or Watts per meter squared).

The system depicted in Figure 8.2 and the relationships shown in Table 8.1 identify a system of equations that model the energy flows of the sun-earth-atmosphere system. We will now build a systems diagram similar to Figure 8.2, but which explicitly identifies each of the mathematical relationships within it.

Before we start constructing the systems diagram, an important decision must be made. Carefully studying the problem tells us that we are interested in exploring the surface temperature on earth. This temperature is a function of Ω_E , which we have described as a flow variable representing the outflow of energy from the earth. In fact, all the variables discussed thus far (and presented in Table 8.1) are flow variables. One could model this system and explore the problem without any use of reservoir variables.

The use of reservoirs, however, allows us to analyze situations where we wish to study the lag times between energy inflows and energy outflows. The reservoirs would have units of energy per area (or Joules per meter squared) and can be considered *energy densities*. Thus, we would be able to explore the case where energy “builds up” in one of the reservoirs over time, and is then released through flow processes. We will start our diagram, therefore, by identifying two reservoir variables—one representing our energy density for the earth (E), the other representing our energy density for the atmosphere (A). We will construct our model assuming no significant lag period between the time energy is absorbed in a reservoir and is emitted

TABLE 8.1. Description of energy flows for Figure 8.2.

Flow #	Description	Mathematical equation	Flux (W/m ²)
(1)	Solar flux that is reflected off the atmosphere/earth and back into space	$a\Omega_s$	107
(2)	Solar flux that is absorbed in the atmosphere by atmospheric gases	$f_b(1 - a)\Omega_s$	67
(3)	Solar flux that is absorbed directly by the earth's surface	$(1 - f_b)(1 - a)\Omega_s$	168
(4)	Latent and thermal heat transfer from the earth's surface to the atmosphere (nonradiative energy)	$t_e\Omega_E$	102
(5)	Earth's radiated energy flux that is absorbed by atmospheric gases	$f_a(1 - t_e)\Omega_E$	350
(6)	Atmospheric radiated energy flux that is radiated toward and absorbed by the earth	$R_a\Omega_A$	324
(7)	Atmospheric radiated energy flux that is radiated toward space	$(1 - R_a)\Omega_A$	195
(8)	Earth's radiated energy flux that is not absorbed by the atmosphere and goes directly to space	$(1 - f_a)(1 - t_e)\Omega_E$	40

a = the albedo of the atmosphere-earth system (≈ 0.31);

f_b = the fraction of incoming solar energy absorbed by the atmosphere;

t_e = the fraction of earth's flux that is transferred to the atmosphere through latent and thermal processes;

f_a = the fraction of outgoing radiation from the earth that is absorbed by the atmosphere;

R_a = the fraction of atmospheric radiation that is radiated toward and is absorbed by the earth;

Ω_s = the total solar flux entering the earth-atmosphere system;

Ω_E = the earth energy flux radiated from the earth;

Ω_A = the atmospheric energy flux radiated from the atmosphere.

from that reservoir. Thus, for our model E and A are cognitive instruments that allow us a better mental picture of what is occurring, but their use is not absolutely required in order for the system to be valid. Figure 8.3 shows these two reservoirs as the basis for our systems diagram.

We will next add our solar energy flows [energy flows (1), (2), and (3) listed in Table 8.1]. These flows are shown in Figure 8.4. First, we have the solar energy that is reflected off the earth-atmosphere system, represented by the solar flux times the albedo of the earth system, or $a\Omega_s$. Next, we have the solar energy that is not reflected, and is also absorbed by the atmosphere. This is equal to $f_b(1 - a)\Omega_s$. Note that this accounts for the solar energy not reflected (i.e., $1 - a$), as well as some fractional parameter, f_b , that is less than 1 and represents the ability of atmospheric gases to absorb solar energy. Because most atmospheric gases are good absorbers in the IR part of the spectrum and most of the incoming solar energy is in the visible/UV part of

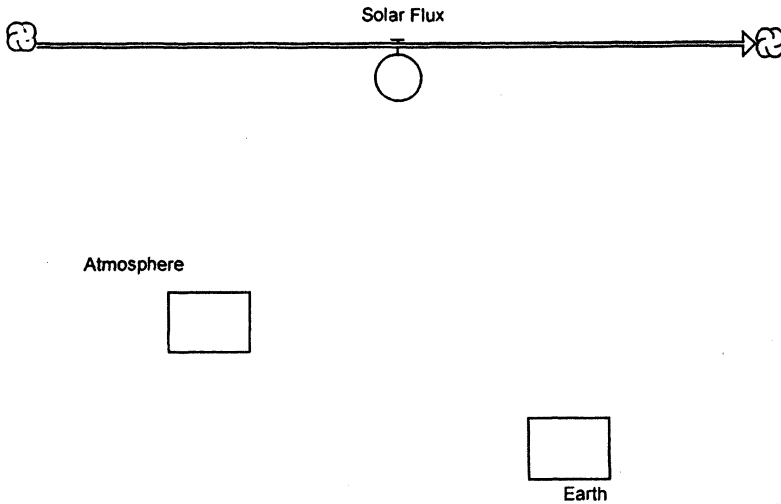


FIGURE 8.3. Identified energy reservoirs for our earth-atmospheric system.

the spectrum, we expect f_b to be a small number. Finally, we have the remainder of the solar energy that is absorbed by the earth itself. This is represented as $(1 - f_b)(1 - a)\Omega_s$. Defining these relationships in this way assures that all the solar energy is accounted for because $1 - f_b$ will account for all the solar energy not absorbed by the atmosphere.

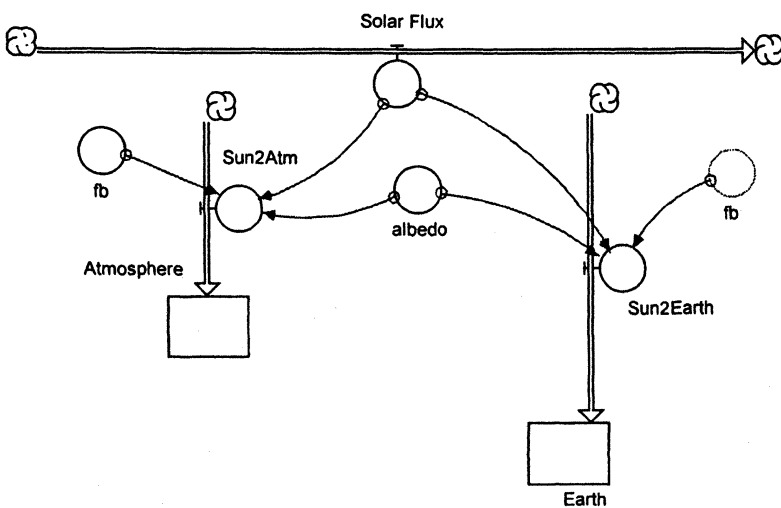


FIGURE 8.4. Earth system with solar flows added.

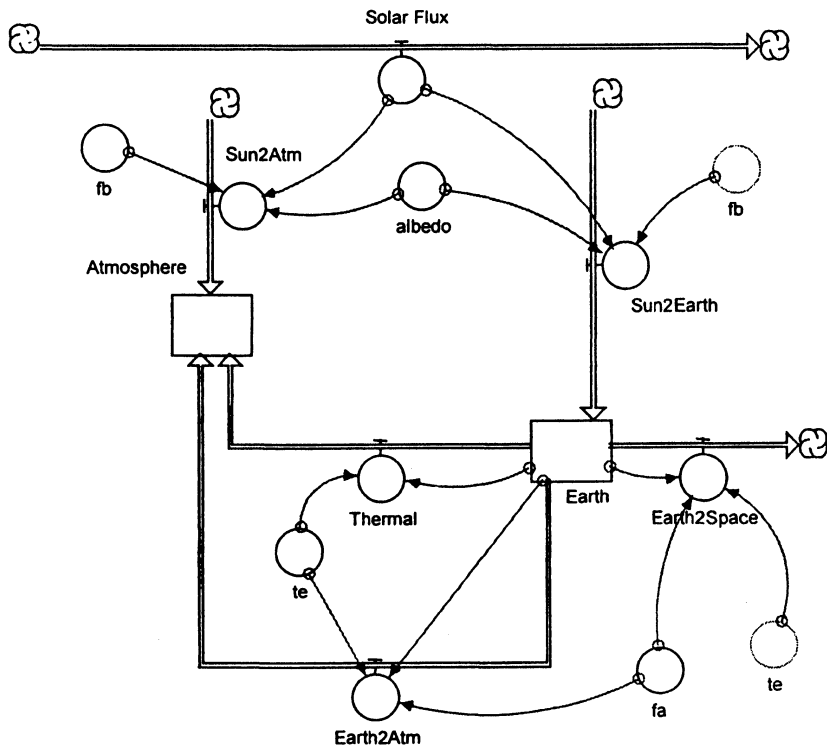


FIGURE 8.5. Earth system with earth energy flows added.

We will consider next all of the energy that is reradiated from the earth's surface. This is described in flows (4), (5), and (8) in Table 8.1. These new flows are shown in Figure 8.5. First, we have a flow of thermal energy that is transported from the earth's surface to the atmosphere via convection and evaporation-transpiration processes. If we define E as the total energy reservoir of the earth, then this thermal flow is represented as $t_e E$ where t_e represents the fraction of the earth's thermal energy flux. Second, we have a flow of *radiated* energy from the earth that is captured by atmospheric gases. This flow is identified as $f_a(1 - t_e)E$, where f_a represents the fraction of the radiated energy that is absorbed. The value of f_a tends to be high because the atmosphere is a very good absorber of IR radiation. Finally, we have a flow of radiated energy that is not captured by the atmosphere, escaping instead into space through a radiation "window." This flow is represented by $(1 - f_a)(1 - t_e)E$. By denoting our flows using fractional values, we have accounted for all the energy in the earth's reservoir at each time unit.

Last, we need to consider energy radiated from the atmosphere. The flows from the atmosphere are identified as flows (5) and (6) in Table 8.1 and are added in Figure 8.6. The atmosphere will radiate some of its energy back

toward the earth (this, in fact, is the cause of the greenhouse effect), and will radiate the remainder toward space. We will define the amount of energy radiated back toward earth as $R_a A$, where R_a represents the fraction of atmospheric radiation that is directed toward and is absorbed by the earth. The value $(1 - R_a)A$, therefore, will represent the atmospheric radiation directed into space.

From this systems diagram, we have accounted for and have balanced all energy inputs and outputs. All that is needed is a component that will calculate earth surface temperatures. This is shown in the bottom part of Figure 8.6. Notice that in calculating surface temperatures, we are only concerned

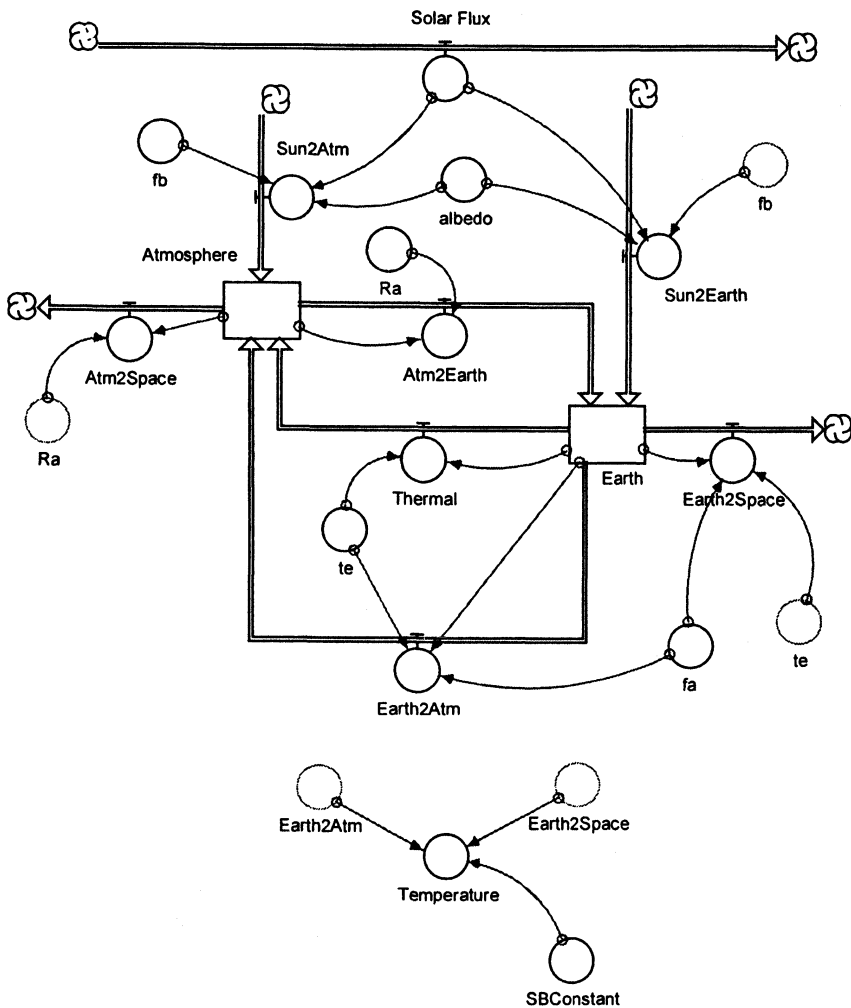


FIGURE 8.6. Earth system with atmospheric flows added.

with the energy radiated from the earth (and not energy transported by thermal processes). The converter *Temperature* in this figure is merely a calculation based on Equation (8.3).

Greenhouse gases enter this system by directly affecting the way in which the atmosphere absorbs and radiates energy. These effects are mostly observed through the variables: f_a and R_a . First, it is quite obvious that an increase in GHGs in the atmosphere will have a direct effect in the ability of the atmosphere to absorb outgoing radiation from the earth (i.e., an increase in f_a). This is what GHGs do, as shown earlier in Figure 8.1; therefore, an increase in their concentrations will mean more molecules to intercept, absorb, and reradiate energy photons.

Second, one might expect that an increase in GHGs might also increase R_a , the fraction of radiated atmospheric energy that is directed toward the earth. The reasons for this are less intuitive. To consider this effect, think of a one-molecule-thick layer of GHGs in the atmosphere. This is shown on left-hand side of Figure 8.7. For this case, assume an amount of energy equal to one unit is radiated toward this layer. Assume that this energy is absorbed and reradiated in all directions. For this case, we would expect that 0.50 of the reradiated energy would be directed in the upward direction, and 0.50 would be directed downward.

We will now add another layer of GHGs in the atmosphere, again equal to one molecule thick and shown on the right-hand side of Figure 8.3. Assume a unit of energy is emitted toward the first layer. This energy is absorbed and reradiated—0.50 upward and 0.50 downward. Now, however, the upward radiated energy is reabsorbed by GHGs in the second atmospheric layer. Of this energy (one half of the original total), 0.50 is radiated upward (or $0.5 \times$

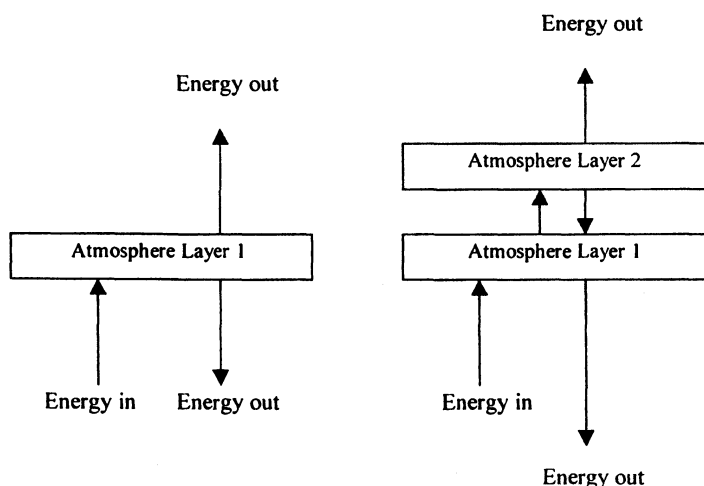


FIGURE 8.7. Illustration of the R_a effect.

0.5 = 0.25 of the original), and the remainder is radiated back down toward the first atmospheric layer (again, 0.25 of the original). Assume this energy is again absorbed and reradiated by the first atmospheric layer—with 0.50 of it radiated upward, and 0.50 radiated downward. Although this downward radiated energy now only comprises $0.5 \times 0.5 \times 0.5 = 0.125$ of the original, it should be clear that it adds to the amount of energy being radiated back toward the earth (i.e., increases R_a). Thus, we expect an increase in GHGs to lead to a direct increase in R_a .

We may also expect that f_b would be affected by GHG levels in the atmosphere; however, because most of the incoming solar flux is in the UV and visible part of the electromagnetic spectrum, and because GHGs such as CO_2 absorb mostly in the IR part of the spectrum, this effect would likely be minor.

8.3 Difference Equations

Because we used two reservoir variables (one for the earth and one for the atmosphere), we can identify difference equations for each of these variables. Depending on the values and formula given for the flow parameters we may or may not have energy accumulation within these reservoirs. The key question is whether these values are such that the inflows to the reservoirs equal the outflows.

In Table 8.1 and in the preceding discussion, we have defined a situation where the sum of all inflows and outflows are equal. The difference equations that follow are based on those constructs, as you will see. Thus, we expect our steady-state solutions to depict a balancing of energy flows into and out of the energy reservoirs.

For the earth reservoir, E , we have:

$$E(t + \Delta t) = E(t) + (1 - f_b)(1 - a)\Omega_s \Delta t + R_a A(t) \Delta t \\ - t_e E(t) \Delta t - f_a(1 - t_e)E(t) \Delta t - (1 - f_a)(1 - t_e)E(t) \Delta t$$

$$\frac{\Delta E}{\Delta t} = (1 - f_b)(1 - a)\Omega_s + R_a A - [t_e + f_a(1 - t_e) + (1 - f_a)(1 - t_e)]E$$

$$\frac{\Delta E}{\Delta t} = (1 - f_b)(1 - a)\Omega_s + R_a A - E$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta E}{\Delta t} = \frac{dE}{dt} = (1 - f_b)(1 - a)\Omega_s + R_a A - E$$

In these equations, E represents the amount of energy per area in the earth reservoir (Joules per meter squared), Ω_s represents the solar flux (Watts per meter squared), and A represents the amount of energy per area in the atmosphere reservoir (Joules per meter squared). These equations require us to consider the parameters R_a and t_e in units of *inverse time* (t^{-1}). This is so

because we need to achieve units of Watts per meter squared instead of Joules per meter squared for our inflows and outflows. R_a , therefore, is the fraction of the atmosphere's energy density (Joules per meter squared) that is radiated toward the earth per time, and t_e is the fraction of earth's energy density (Joules per meter squared) that is transferred to the atmosphere through thermal processes per time.

Note that the *outflow* for this rate equation simplifies to be the reservoir value itself (E). This tells us that for each period, the total amount of energy in the reservoir flows out of the reservoir (i.e., there is no accumulation). This occurs because of how we defined the flows mathematically. By summing fractional parameters (e.g., t_e) with one minus these parameters (e.g., $1 - t_e$) we are sure to cancel out the parameters themselves. One way to change this system to a more general form in which accumulation in the reservoir is possible is to modify the flows so that the fractional parameters do not cancel each other out. We leave this as an exercise for the reader.

For the atmosphere reservoir, we have:

$$\begin{aligned}
 A(t + \Delta t) &= A(t) + (f_b)(1 - a)\Omega_s \Delta t + t_e E(t) \Delta t \\
 &\quad + f_a(1 - t_e)E(t) \Delta t - R_a A(t) \Delta t - (1 - R_a)A(t) \Delta t \\
 \frac{\Delta A}{\Delta t} &= (f_b)(1 - a)\Omega_s + t_e E + f_a(1 - t_e)E - R_a A - (1 - R_a)A \\
 \frac{\Delta A}{\Delta t} &= (f_b)(1 - a)\Omega_s + [t_e + f_a(1 - t_e)]E - A \\
 \lim_{\Delta t \rightarrow 0} \frac{\Delta A}{\Delta t} &= \frac{dA}{dt} = (f_b)(1 - a)\Omega_s + [t_e + f_a(1 - t_e)]E - A
 \end{aligned}$$

Again, the reader should note that the outflow in the final rate equation is equal to the reservoir value itself (and thus there is no accumulation). This, again, is true based on the same arguments presented earlier for the earth reservoir.

8.4 Modeling the Dynamic Greenhouse Gas System

We are now ready to develop and explore a model for the scenario described in this chapter and to explore how the system responds to the perturbations discussed in that scenario.

We are asked to determine the impact an increased level of carbon dioxide would have on average surface temperatures of the earth. This increase in carbon dioxide is expected to affect two primary parameters in our system directly. First, it will affect f_a , the fraction of outgoing IR radiation from the earth that is absorbed by the atmosphere.

Second, it will affect R_a , the fraction of outgoing IR radiation from the atmosphere that is directed back toward the earth (see the earlier discussion on this).

Before we begin to see the impacts of increased CO₂ on our system specifically, however, let us be sure we thoroughly understand the system. Open the model entitled *CHAP8a.STM* on your CD-ROM. This model is written in the STELLA® software language and is identical to Figure 8.6. The model also includes the values for each of the parameters shown later; however, the parameters are modified in the model so that our time units are in years.

- Solar Flux = 342 W/m²
- Atmosphere = 518.15 J/m²
- Earth = 491.31 J/m²
- $f_a = 0.897$
- $f_b = 0.285$
- $R_a = 0.624$
- albedo = 0.313
- $t_e = 0.207$

The model uses the equations identified in Section 8.2 (so that accumulation in the atmosphere and earth reservoirs will not occur). The model is in “flight simulator” mode and the user can run the model while changing a number of parameters shown in the slider bars. The graph under the model shows the average surface temperature of the earth. The table to the right of the graph depicts a value for these temperatures for each year of the model run. The reader should be able to check mathematically to see whether the preceding values represent steady-state conditions using the rate equations from Section 8.3.

Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** in the drop-down menu. Study the output from the model and answer the questions that follow.

- Is the model in steady-state given its default values?
- Run the model several times, adjusting each of the parameters individually as the model runs. Then adjust the parameter values in different combinations (e.g., increase albedo while also increasing R_a). Use the analytical strategy outlined in Chapter 3. Explain your results. Which variables do you think drive this system?
- Describe, verbally, a scenario under which GHGs begin to accumulate, and how this accumulation affects the surface temperature of the earth.

Once you are familiar with the operation of model *CHAP8a.STM*, we can add the CO₂ component to this model to address the problem at the beginning of this chapter. To do this, we might envision a stock of CO₂ (or other GHGs) in the atmosphere that changes over time. This reservoir would impact values of R_a and f_a . The relationship between R_a and f_a with CO₂ in the atmosphere is not straightforward; however, we would expect that as CO₂ concentrations increase, our values for R_a and f_a would increase as well. We also note that there is some saturation effect, whereby the marginal increase in R_a or f_a decreases as CO₂ concentrations are increased.

Open the model **CHAP8b.STM**. This model is a variation of model **CHAP8a.STM**. This second model includes a stock of CO_2 with a net inflow/outflow (representing the net build-up or depletion of CO_2 in the atmosphere). This stock is in units of parts per million (ppm) and is initially set at 350 ppm. The inflow is set as positive 5 ppm/year (i.e., an increase in CO_2 concentrations of 5 ppm annually). Note in this model as well that R_a and f_a are graphical functions of CO_2 . The graphical relationship can be seen (and modified) by clicking on the appropriate graph icon.

- Based on the verbal description you provided in the third preceding bullet, analyze your scenario using model **CHAP8b.STM**. If you are comfortable working in the STELLA® environment, you can modify the model by adding population or economic submodels that link to the CO_2 accumulation variables. Once you have made the appropriate modifications, run your model and explain your results.

8.5 Exercises

1. Use Equation (8.1) to show that under the conditions given so far, the temperature of the earth would be approximately 255 K. If the albedo of the earth was given as $a = 0.50$, what would you expect the earth's temperature to be using Equation (8.3)? Does your result make sense?

2. Suppose the amount of energy reradiated from atmospheric gases back toward the earth was equal to 100 W/m^2 . What would the total energy flux absorbed by the earth be? Modify Equation (8.1) to determine the surface temperature you would expect the earth to have under these conditions.

3. Using the flux values given in Table 8.1, check to see whether the energy flows into and out of each of the reservoirs in Figure 8.2 are equal.

4. Using the mathematical relationships in Table 8.1 as a guide, qualitatively explore the impacts due to modifications in the variables identified below the table. For example, what do you expect will happen to the earth's energy flux if the variable f_b is increased? Decreased?

5. For the set of coupled rate equations given in Section 8.3, determine the relationship between A and E under steady-state conditions.

6. There is some debate in the scientific community on how atmospheric temperatures will change under global warming conditions. As global surface temperatures increase, will temperatures at the "top" of the atmosphere increase or decrease? Modify the model **CHAP8b.STM** to explore how temperatures at the "top" of the atmosphere will change. Note that these temperatures will be driven by IR radiation emitted from the top of the atmosphere toward space. Explain your results.

7. From your knowledge of the global warming phenomenon, identify a positive or negative feedback effect associated with global climate change. For example, many scientists believe that increased average surface temperatures will melt the earth's ice caps, thereby reducing the albedo of the earth. This loss in reflectivity will lead to even higher temperatures as the earth absorbs more solar flux, thus leading to more ice cap melting, lower albedo, higher temperatures, and so on. Once you have identified a particular feedback effect, modify the model **CHAP8b.STM** (or any new model you have created from this baseline model) to explore the impacts of such feedback. Explain your results.

8. In global climate models, we often wish to determine temperatures and other climate effects at a much smaller scale than over the entire earth. For example, we may wish to model various "layers" in the atmosphere in order to explore temperature differences as we move vertically from the earth's surface. Modify model **CHAP8a.STM** (or any new model you have created from this baseline model) to include at least two atmospheric layers. Be sure to think about how such layers might affect your parameters (e.g., your R_s value for each layer of the earth). Run your model and create output that shows temperature at the earth's surface and each layer of the atmosphere. Explain your results.

9. There are some scientists that argue that the increases in our global surface temperature over the past 100 years have been driven by changes in the solar flux. Analyze this hypothesis using the models provided, or a model of your own.

8.6 References Cited and Suggested Readings

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9

Atmospheric Chemistry and Pollution Transport

■ Chapter Objectives—

After you finish this chapter you should be able to:

1. *Describe the sources and impacts of acid deposition.*
2. *Describe the basics of chemical kinetics and why kinetics are important in understanding atmospheric pollution problems, such as acid deposition.*
3. *Describe the important variables that affect the rate at which acid deposition occurs at a given distance from a source.*
4. *Explore and build a dynamic model that depicts the relationships among the variables associated with pollution transformation and transport.*

9.1 The Problem

Air pollution found in a geographic area is often not generated in that area; rather, it has been transported from other places. The location that generates the pollution is called the pollution *source*; the location that receives the pollution is called the pollution *receptor*. The reason air pollution often displays this **source–receptor** characteristic is because many of the physical and chemical processes that form air pollution take place in a medium (the atmosphere) that is spatially dynamic. In other words, the atmospheric “chemical reactor” in which pollution forms is transported with the wind and the weather. Dynamic meteorological conditions, such as temperature, humidity, wind direction, and wind speed, affect the way pollutants are mixed, dispersed, and transported across geographic boundaries.

An excellent example of this phenomenon is the problem of acid deposition. In many parts of the world, a leading cause of acid deposition in one geographic region is due to sulfur dioxide (SO_2) emissions generated at coal burning power plants elsewhere. Sulfur dioxide is emitted by power plants and is transported via prevailing winds to other locations. Through its journey, this SO_2 gas undergoes a series of chemical reactions that ultimately transform it into sulfuric acid (H_2SO_4) or sulfurous acid (H_2SO_3), which is deposited as acid rain.

In this chapter we will explore the dynamic nature of atmospheric chemistry, focusing on the acid deposition problem as our illustrative example. The scenario you will examine is as follows:

Problem 9.1 Acid deposition, particularly sulfuric acid formed from SO_2 , is a grave concern in many parts of the world. Your local air quality authority is concerned about a new air pollution source being constructed 60 km away. This new source will emit significant amounts of SO_2 . You are asked to explore what impact those emissions will have on acid deposition locally. You are also asked to consider the role SO_2 pollution control technologies may have on mitigating these impacts.

9.2 Background Information

9.2.1 Acid Deposition

Acids are substances that release hydrogen ions (H^+), usually in aqueous solutions. Common examples of acids include sulfuric acid (H_2SO_4), nitric acid (HNO_3), and hydrochloric acid (HCl). Acid deposition is the process by which acids formed in the atmosphere are deposited on land. This deposition can be in a dry form (in which acid particles fall to rest on land), or in a wet form (in which acid in its aqueous phase is deposited through precipitation mechanisms, such as rain, snow, or fog).

Because hydrogen ions are extremely reactive and corrosive, a number of direct and indirect problems are associated with acid deposition. Acids may destroy property directly, especially limestone statues and buildings. Acids may also destroy human and animal lung tissue when breathed in. Acids may indirectly weaken the immune systems of plants, preventing them from effectively fighting off other diseases. Acids also increase the solubility of some metals in soil, so acid precipitation that infiltrates soil may dissolve metal (e.g., aluminum) and carry them into bodies of water where they can harm aquatic life (Mohnen 1988).

The acidity of a solution is denoted by its *pH* value. pH is a measure of the number of hydrogen ions in a given sample of water. pH is mathematically the negative log of the hydrogen ion concentration, or

$$\text{pH} = -\log[\text{H}^+]$$

where $[\text{H}^+]$ is the concentration of hydrogen ions in units of *moles per liter*. From this equation, we see that as hydrogen ion concentrations increase the pH level decreases. Thus, lower pH levels represent more acidic water.

Pure water has a pH of 7.0; however, the natural pH of rainwater is approximately 5.6. This is primarily due to water's ability to absorb carbon dioxide (CO_2) gas from the atmosphere. The absorption of CO_2 in rain droplets creates carbonic acid (H_2CO_3), which lowers the pH of the rain (i.e., makes it more acidic). pH values much less than 5.6, however, have been observed in the eastern part of the United States. In fact, some areas of the eastern United States regularly see rainfall with a pH of 4.0 or less. Because pH is

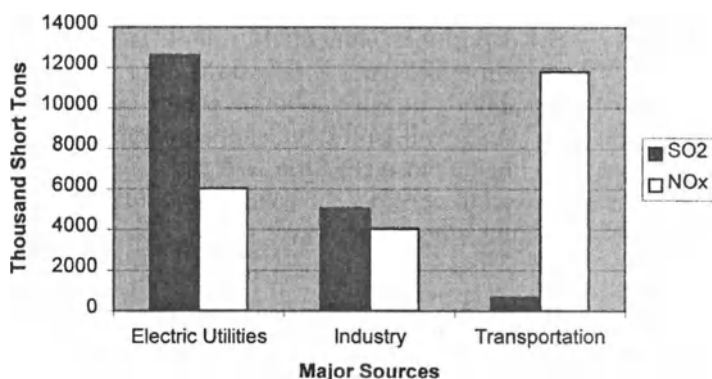


FIGURE 9.1. Major anthropogenic sources of SO₂ and NO_x in the United States. (Source: Data from U.S. Environmental Protection Agency; 1998.)

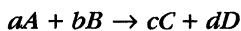
measured on a logarithmic scale, rain with a pH value of 4 is more than *10 times* more acidic than water with a pH value of 5.6.

The largest contributor of acid deposition in many parts of the world is sulfur dioxide (SO₂). Sulfur dioxide is formed from many sources, but its predominant anthropogenic source is coal burning power plants. Figure 9.1 shows the major sources of SO₂ in the United States. Once SO₂ is emitted into the atmosphere, it undergoes a series of chemical reactions that transform it to either sulfurous acid (H₂SO₃) or sulfuric acid (H₂SO₄).

9.2.2 Some Basic Chemical Kinetics

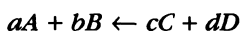
Chemical kinetics is the study of how reactants become products in a chemical reaction. The rate at which chemical reactants become products is a function of many variables, including the quantities of the reactants, the temperature, and the pressure (especially for gaseous reactions).

Readers should be familiar with the standard, one-directional chemical equation demonstrated by the following equation:



This reaction is a *forward* reaction. In a forward reaction, we assume that all the reactants become products. The rate at which this reaction takes place may be fast or slow.

For most reactions, however, not all of the reactants necessarily become products. There may also be a *reverse* reaction, of the form:



If both of these reactions are occurring at the same time, we can rewrite the chemical equation as follows:



Under these conditions, there comes a point at which the amount of reactants becoming products in the forward reaction exactly equals the amount of reactants becoming products in the reverse reaction. Thus, an equilibrium is achieved, where the concentrations of all the reactants and products remain constant. The ratio of these concentrations is defined as the *equilibrium constant* and is given by

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

where the square brackets represent concentration units. K_c is a useful tool because it can tell us what to expect in terms of final concentrations of reactants and products given a chemical reaction. We can also use K_c as a benchmark to determine which direction a reaction will proceed. For example, if we are given the concentrations of reactants and products, we can determine the *reaction quotient*, Q , as:

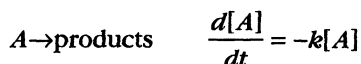
$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

If Q is larger than K_c (i.e., $Q > K_c$), then we know that the product concentrations are too high, and that the reaction will proceed in a reverse (or backward) direction. If Q is less than K_c (i.e., $Q < K_c$), then we know that the reactant concentrations are too high and that the reaction will proceed in a forward direction.

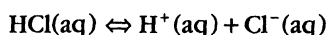
Aside from the equilibrium values for reactants and products, and whether the reaction will occur in the forward or backward direction, we may be interested in how fast a chemical reaction takes place (i.e., its *reaction rate*). These rates are normally not constant (i.e., the reaction rate decreases as the concentrations of the reactants decrease). For simple, single-step reactions, the reaction rate is defined as:

$$\text{Reaction rate} = k[A]^a[B]^b$$

Notice that this reaction rate increases as concentrations of A and B increase. When our concentrations are set to "1.0," then our reaction rate equals k , which is called the *rate coefficient*. The units for k will depend on the reaction under study, but for the simple case of a single reactant (e.g., A), the units will be in inverse time (t^{-1}). For example,



As mentioned earlier, an acid can be considered a substance that dissociates "to" release a hydrogen ion in water. For example, HCl dissociates as follows:



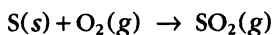
Notice that this reaction is reversible and the concentrations of the reactants and products will achieve equilibrium. The equilibrium constant for this reaction is known as an **acid-dissociation constant** (K_a). The formulation for K_a for HCl is:

$$K_a = \frac{[H^+][Cl^-]}{[HCl]}$$

The strength of the acid can be determined by K_a . A strong acid will have a high K_a ; a weak acid will have a low K_a . It should be noted that some strong acids (HCl, H_2SO_4 , and HNO_3) are in fact almost 100% dissociated in water, thus, they have extremely high K_a values.

9.2.3 Reactions Involving SO_2

Anthropogenic SO_2 emissions are released during combustion of substances that have high sulfur concentrations (e.g., coal and oil). Sulfur (S) that is trapped in these fossil fuels undergoes oxidation during the combustion process, leading to the following reaction:



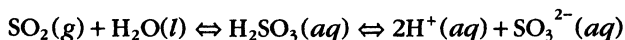
where the (s) represents a solid phase and (g) represents a gaseous phase. This gaseous SO_2 is normally released into the atmosphere.

There are several reactions that can ultimately transform gaseous SO_2 to either sulfuric acid (H_2SO_4) or sulfurous acid (H_2SO_3). Each of the reactions involves the oxidation of SO_2 to some other form. This oxidation can occur in a homogeneous gaseous phase, a homogeneous aqueous phase (i.e., within raindrops), or a heterogeneous gaseous–aqueous phase. The predominant source of oxidation is in the homogeneous aqueous phase, where SO_2 is absorbed in a water molecule. Once in the form of sulfurous or sulfuric acid, acid deposition can occur. Although much of the chemistry involved is complicated and even uncertain, the following general reactions summarize the oxidation and deposition processes that occur (Bunce 1994).

1. Some of the emitted SO_2 is oxidized to SO_3 . This oxidation is usually due to a catalyst in the atmosphere that spurs a reaction between SO_2 and atomic oxygen (O). Another possibility, however, is the reaction of SO_2 with other compounds (e.g., NO_2) to form SO_3 . We will simply write the overall reaction as shown later, with a pseudo–first-order rate constant given by k_1 . This rate constant varies according to meteorological conditions, with high humidity and sunlight both leading toward higher reaction rates. Bunce (1994) has identified a good first-order approximation for k_1 as 0.1 hr^{-1} . This translates into approximately a 10% conversion of SO_2 to SO_3 per hour.



2. Some of the emitted SO_2 combines with water (H_2O) to form H_2SO_3 . This H_2SO_3 further dissociates to release hydrogen ions and is deposited in the wet or dry form on land. These reactions are simplified and shown in the following equation.



As shown earlier, H_2SO_3 in the aqueous state, releases H^+ ions and SO_3^{2-} (sulfite) ions. We can approximate a pseudo-first-order reaction rate that captures both the transformation rate and the average deposition rate of this acid. The reaction and deposition processes occur at an average rate of k_2 , approximated as 0.03 hr^{-1} . From Bunce (1994) we can write the summary reaction as follows:



3. Finally, some of the SO_3 generated in the first reaction identified earlier combines with water to form H_2SO_4 , or sulfuric acid. This acid dissociates to release H^+ ions and SO_4^{2-} ions. Again, we use a general reaction to illustrate this chemical transformation and deposition process:



As shown earlier, H_2SO_4 can release H^+ ions and SO_4^{2-} (sulfate) ions. We can approximate a pseudo-first-order reaction rate that captures the reaction and deposition process as follows:



Thus, we have three generalized reactions that depict the transformation of SO_2 gas in the atmosphere and the ultimate deposition of this sulfur in the form of sulfites (SO_3^{2-}) and sulfates (SO_4^{2-}). These reactions have relatively slow reaction rates, which allow for the transport of this pollutant over long distances. One should recognize, however, that under precipitation events, the rate constants (k_1 , k_2 , and k_3) are quite high (i.e., close to 1.0), and so under such situations a “washout” may occur with all pollutants being deposited over the area of the precipitation event. The concern for the downwind receptor site, of course, is how much deposition of sulfites and sulfates will occur on their site. This will be a function of the concentrations of sulfite/sulfate, as well as the deposition rate. These issues are discussed in more detail in the sections that follow.

9.2.4 Mathematical Relationships and Our System Diagram

Calculating acid deposition at a receptor site due to SO_2 emissions from a source many kilometers away is a complex and challenging task. We must account for the emissions quantity, the dispersion of the emissions, the transformation of the emissions into acid, the travel time of the emissions to the

receptor site, and the deposition rate of the acid. With several simplifying assumptions, however, we can model the expected deposition of SO_3^{2-} and SO_4^{2-} at various points between the source and receptor site.

For the problem introduced at the beginning of this chapter, we are interested in determining the expected levels of deposition of sulfate and sulfite at our receptor site and the role that various mitigation policies or technologies may have on those levels. To begin our understanding of this problem, let us make the following assumptions:

1. Assume the receptor site is directly downwind of the source site.
2. Assume a constant wind direction and wind velocity during the modeling period.
3. Assume a constant emissions rate of SO_2 from the source.
4. Assume a constant rate of chemical transformation and deposition (i.e., k_1 , k_2 , and k_3 are constant).
5. Assume a constant natural inflow of SO_2 into the SO_2 reservoir due to various natural processes.
6. Assume a one-time inflow of SO_2 from the facility—this does not mean that the facility only pollutes for one time unit; rather, it means that the facility contributes pollution into the parcel of air over one time unit and then this parcel moves downwind. The facility will continue to pollute subsequent parcels of air that pass overhead during later time periods. The pollution from the facility is used to calculate the conditions in the parcel of air during the first time period.

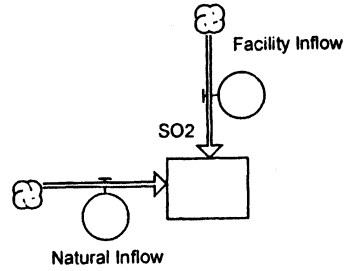
We can begin to address the transport problem by first identifying a stock of SO_2 in the atmosphere. The primary contributor to this stock is a one-time influx of SO_2 (modeled in the first time period) from a smokestack somewhere upwind of a receptor site. For now, we can picture this stock as a parcel of air of a given volume with a certain mass of SO_2 (i.e., a parcel of air with a given concentration of SO_2) whose concentration is given by the emissions rate of the source and other natural inflows.

The parcel of air will begin to travel downwind from the source toward the receptor site. As it moves downwind, three dynamic effects will occur. The first effect is simply the addition of a natural inflow of SO_2 into the reservoir due to naturally occurring processes (one might call this “background” SO_2). We will use the constant A to represent this natural inflow. This inflow is shown in Figure 9.2, which shows the *Natural Inflow* and the contribution from the facility (*Facility Inflow*). The Natural Inflow is assumed to be constant over time and is identified as

$$\text{Natural Inflow} = A$$

The second effect will be an outflow of SO_2 as dictated by the first chemical reaction in the previous section. Here, SO_2 is being converted into SO_3 . In order to show this, we need to construct a reservoir variable for SO_3 and show a flow out of the SO_2 reservoir and into the SO_3 reservoir. Because the

FIGURE 9.2. System diagram of natural SO_2 inflow.



flow of $\text{SO}_2 \times \text{SO}_3$ is determined by the concentration of SO_2 and some rate constant, k_1 , we identify these relationships in Figure 9.3. This is a *donor-controlled* flow (see Chapter 6). This flow is identified as

$$\text{Transformation} = k_1 \times \text{SO}_2$$

The third effect will be determined by the second chemical equation in the previous section, whereby SO_2 is deposited as H_2SO_3 or SO_3^{2-} . This represents a second outflow of SO_2 from the SO_2 reservoir. Figure 9.4 shows this outflow, which is also a donor-controlled flow, regulated by k_2 . This flow is given mathematically by

$$\text{Sulfite Deposition} = k_2 \times \text{SO}_2$$

Last, we will have an outflow associated with the SO_3 reservoir as it is transformed and deposited as H_2SO_4 or SO_4^{2-} . This is also a donor-controlled flow, as shown in Figure 9.5. The outflow is represented as

$$\text{Sulfate Deposition} = k_3 \times \text{SO}_3$$

Finally, we need to recognize that these processes occur over time in a parcel of air that is being transported due to wind speed. Thus, we can translate time into distance based on the velocity of the wind. In this way, we can track the concentration, transformation, and deposition of SO_2 over time and distance. In this problem, we want to understand relationships among

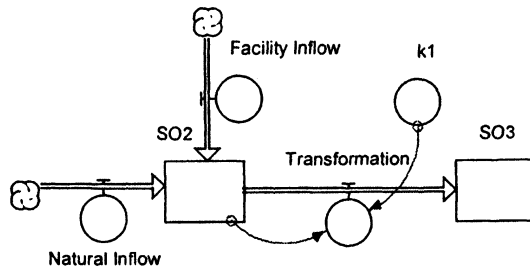
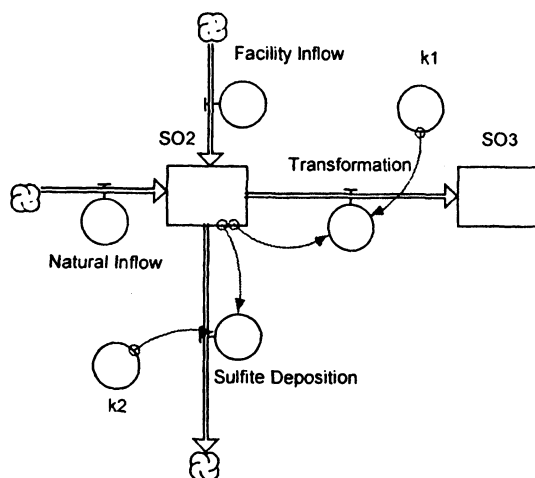


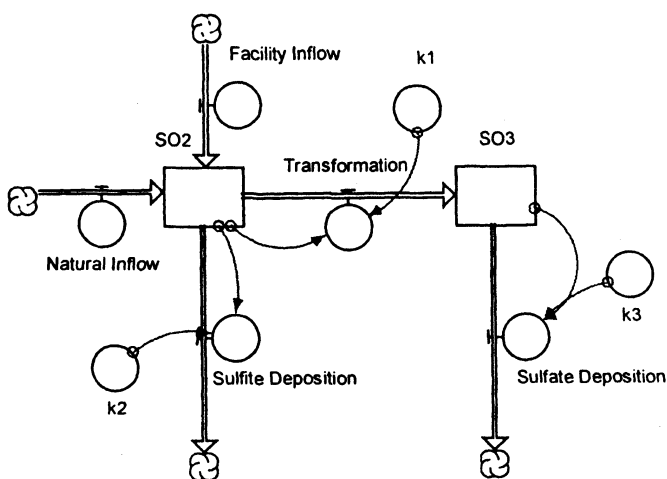
FIGURE 9.3. System diagram with $\text{SO}_2 \rightarrow \text{SO}_3$ transformation.

FIGURE 9.4. System diagram with SO_2 outflow.

these variables at the point where this parcel of air passes over the receptor site.

9.3 Difference Equations and the Steady-State Solution

Because we only have two reservoir variables, we will only have two difference equations to explore. The first is for SO_2 . The difference equation and rate equation can be derived as follows:

FIGURE 9.5. System diagram with SO_3 outflow.

$$SO_2(t + \Delta t) = SO_2(t) + A\Delta t - k_1SO_2(t)\Delta t - k_2SO_2(t)\Delta t$$

$$\frac{SO_2(t + \Delta t) - SO_2(t)}{\Delta t} = A - (k_1 + k_2)SO_2(t)$$

$$\frac{\Delta SO_2}{\Delta t} = A - (k_1 + k_2)SO_2$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta SO_2}{\Delta t} = \frac{dSO_2}{dt} = A - (k_1 + k_2)SO_2$$

Note that we do not include the Facility Inflow in these equations because this has been defined as a one-time input during the first time period. The rate equation accounts for changes in the reservoir variable after this one-time influx occurs. This rate equation shows a constant increase in SO_2 due to background effects, and a decreasing term that is a function of the concentration of SO_2 itself (i.e., exponential decay would be expected). The rapidity of the exponential decline is dependent on the values for k_1 and k_2 , which are both dependent on prevailing meteorological conditions.

For our SO_3 reservoir, the difference equation and rate equation can be derived as follows:

$$SO_3(t + \Delta t) = SO_3(t) + k_1SO_2(t)\Delta t - k_3SO_3(t)\Delta t$$

$$\frac{SO_3(t + \Delta t) - SO_3(t)}{\Delta t} = k_1SO_2(t) - k_3SO_3(t)$$

$$\frac{\Delta SO_3}{\Delta t} = k_1SO_2 - k_3SO_3$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta SO_3}{\Delta t} = \frac{dSO_3}{dt} = k_1SO_2 - k_3SO_3$$

This rate equation shows that the rate at which SO_3 increases is a function of the concentration of SO_2 and the rate constant of the $SO_2 \rightarrow SO_3$ transformation (k_1). The rate at which SO_3 decreases is a function of its own concentration and the rate constant k_3 . Thus we expect two competing terms: a decreasing term that is clearly exponential, and an increasing term that will start high, but will decrease exponentially.

The steady-state solutions to these two rate equations are determined by setting each equation to zero and solving.

For SO_2 :

$$\frac{dSO_2}{dt} = A - (k_1 + k_2)SO_2 = 0$$

$$SO_2 = \frac{A}{k_1 + k_2}$$

For SO_3 :

$$\frac{d\text{SO}_3}{dt} = k_1\text{SO}_2 - k_3\text{SO}_3 = 0$$

$$\overline{\text{SO}_3} = \frac{k_1 A}{k_3(k_1 + k_2)} \text{ when } \overline{\text{SO}_2} = \frac{A}{k_1 + k_2}$$

9.4 Modeling the Dynamic Acid Deposition System

We now have the tools needed to solve the problem introduced at the beginning of this chapter; however, first we need to specify the problem in more detail.

We are asked to determine the impact of a new SO_2 source on a receptor site downwind from that source. In particular, we wish to determine how increased concentrations of SO_2 due to the new source will potentially affect SO_3^- and SO_4^{2-} at the receptor site. From the preceding equations we know, depending on meteorological conditions, that if SO_2 and SO_3 concentrations are high, then we can expect high deposition of sulfites and sulfates (especially in the case of a precipitation event over the receptor site). Our model should also allow us to explore the impact of various technologies or policies that might reduce the deposition at the receptor site.

Let us suppose we acquire the following information about the facility and ambient conditions within a parcel of air traveling over the facility and toward the receptor site:

- $A = 0.25 \text{ ppb}$
- $k_1 = 0.075 \text{ hr}^{-1}$
- $k_2 = 0.030 \text{ hr}^{-1}$
- $k_3 = 0.030 \text{ hr}^{-1}$
- $\text{SO}_2(0) = 2.38 \text{ ppb}$
- $\text{SO}_3(0) = 5.95 \text{ ppb}$
- $V = 4 \text{ km/hr}$
- $D = 60 \text{ km}$
- $F = 30 \text{ ppb}$ (one time inflow, and then parcel moves on)

Here, A represents the natural inflow of SO_2 from the surroundings, V represents wind speed (in kilometers per hour), D represents the distance between source and receptor site, and F represents the one-time emission inflow of SO_2 to the parcel of air as it passes over the facility stack. Again, note that F is a one-time inflow—the air parcel passes over the facility, receives an influx of F , and then moves on. Using these values, we should be able to model the system and explore options that the facility may undertake to reduce the overall impact on the receptor site.

Open the model **CHAP9a.STM** located on your CD-ROM. This model is written in the STELLA® software language and is similar to Figure 9.5. The

model uses the equations identified in Section 9.2. The model is in “flight simulator” mode, and the user can run the model while changing a number of parameters shown in the slider bars. Four graphs are under the model. The first two show the concentrations of SO_2 and SO_3 over distance. The second two show deposition of sulfates and sulfites over distance. The user can also double click on the graph icons to the right of the slider bars in order to view stocks and flows versus distance.

- Run the model by clicking on **Run** in the top toolbar, and then clicking on **Run** in the drop-down menu. Run the model several times, adjusting each of the parameters individually as the model runs. Use the strategies for analyzing a model as discussed in Chapter 3. Explain your results. Which variables do you think drive this system?
- Based on the problem given at the beginning of the chapter, should your locale be concerned about the new facility?
- Does the model achieve steady state? If yes, is this steady state at a value that you feel is correct?

9.5 Exercises

1. Calculate the pH of the following liquids, using the hydrogen ion concentrations given:
 - a. Blood with an $M_{\text{H}^+} = 4.5 \times 10^{-8}$
 - b. Orange juice with an $M_{\text{H}^+} = 3.2 \times 10^{-4}$
 - c. Wine with an $M_{\text{H}^+} = 1.6 \times 10^{-3}$
2. Assume a one-way reaction depicted simply by $a[A] = b[B]$. Assume that the reaction rate of this reaction is $k = 0.2 \text{ hr}^{-1}$. If the initial concentration of $[A]$ is 100 ppb and the initial concentration of $[B]$ is zero, what will be the concentration of $[B]$ after one-half hour.
3. Describe at least one technology and one policy that you could employ to reduce the level of SO_2 and SO_3 at the receptor site, based on the problem discussed in this chapter.
4. Modify model **CHAP9a.STM** to incorporate an SO_2 control technology, run your new model, and discuss the results.
5. Modify model **CHAP9a.STM** to incorporate one policy that would reduce SO_2 emissions, run your new model, and discuss the results.
6. Modify model **CHAP9a.STM** so that you can mimic a precipitation event. One way to do this is by changing k_2 and k_3 to be stochastic variables that will peak during a precipitation event. Run your model and discuss the results.
7. Modify model **CHAP9a.STM** so that the value of k_1 changes with solar energy intensity. For example, we may expect k_1 to increase during

midday, when solar intensity is the greatest, and to decrease during evening hours. Run your model and discuss the results.

8. Research another environmental problem that involves the transformation of one pollutant into another via atmospheric chemistry. Create a dynamic model that simulates the changes of pollutant concentrations in a parcel of air over time.

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Epilogue

This book is intended to provide a better appreciation for the dynamic nature of environmental systems. In working with the practical applications in the text, readers develop modeling skills that will hopefully become a part of the mental paradigm within which they view the world. Once dynamic systems are fully appreciated, people begin to see the world differently. No longer is the environment viewed as a collection of static, unchanging phenomena, but rather as a web of system components linked by active, dynamic processes.

The book also stresses that dynamic modeling is a useful tool for understanding systems. Models enable researchers to simulate these systems, identify potential problems within the systems, and explore solutions to these problems. Although these results-oriented uses of models are important, modeling is not an end in itself, nor is a model a crystal ball that can predict the future in absolute detail and accuracy. Surprising though it may seem, many people unfamiliar with modeling believe that if a model exists, and if it seems complicated enough, then its predictions must be correct.

Of course, this is not true. A bad model that does not accurately reflect reality may be worse than not having a model at all. Think what would happen if all decisions were based on bad models. A testament that models usually never capture reality precisely is reflected in the fact that most of the models widely used in the environmental field are continuously updated and refined. Sometimes, people are surprised to learn that the improvements are significant departures from the original models. How does that make us feel about all the decisions made on earlier versions of the model? Does such a situation mean that all our previous decisions, based on output from earlier versions of the model, were wrong? Well, not exactly, as long as those decisions were made with an understanding that model output must be considered in a special way.

Some final thoughts on modeling are presented below so that you do not develop unrealistic expectations from systems models that you might develop, use, or interpret.

1. *The output from a model is only as reliable as its least reliable input.* This is summarized in the modeling community with the familiar phrase: “Garbage in . . . garbage out.” The computational correlation to this is the concept of significant figures. When working on arithmetic problems, you can never calculate an answer that is more precise than your least precise piece of data. The lesson is that either you must be extremely careful in collecting data, or if your data are imprecise, you must recognize that the value of the model may not lie in its actual output, but instead may be in the relationships that the model helps define and illustrate.

2. *The most important outcome of a modeling effort may be a better understanding of the system, and not necessarily the actual model output.* Sometimes the most important results from a modeling exercise are imbedded in the sensitivity analyses. Understanding how the system changes with respect to important variables is key to understanding the system dynamics. This idea has been highlighted in this text by way of example and discussion, but it is such an important point that it is worth repeating. The greatest value of a model may be that it enables us to observe changes and identify the driving variables that cause these changes.

3. *Keep it simple.* Like a good piece of literature, one way to judge a model is by looking at how much information it imparts with as much economy as possible. Some believe that models must be extensive, exhaustive, and incorporate complex lines of mathematical code. In many cases, however, such a model actually loses its value as an analytical tool. People using the model or relying on its output consider it a “black box” in which data go in and results pop out. Simple models, on the other hand, enable the user and the model builder to effectively communicate about the model and the system that it represents. In this way, the user can have more involvement in the modeling enterprise and can even suggest modifications that might improve the model’s usefulness.

4. *Dynamic modeling is only one area of modeling.* The applications and discussions in this text focused on dynamic models because many environmental systems are dynamic—that is, the system elements are constantly changing in sometimes complicated or surprising ways. Dynamic models are also useful in capturing and understanding the feedback effects often seen in environmental systems. But a dynamic model may not always be the right tool for the job. Many systems can be modeled effectively using static models. For example, simple spreadsheet models are extremely effective at building relationships among variables (or “cells”). Depending on the research question, the dynamic nature of a particular system may not be important. Also, other modeling techniques, such as optimization, are available to answer other kinds of questions that dynamic simulation models may not be able to answer. In truth, we hope that this book has provided enough of a “teaser” in the field of modeling that you are now more willing to investigate other modeling approaches.

5. *Maintain your modeling skills in systems thinking.* Systems thinking is both a tool and a way of viewing the world. As a tool, it must be practiced. We encourage you to pick up this book every once in awhile, or to develop you own models and practice with the tool. Practicing with the tool will hone your skills and will make you a more effective systems thinker.

As you move toward a better appreciation of models and their applications, resources are available to help you. The modeling package throughout the text (STELLA®) is produced by High Performance Systems, which maintains a website at <http://www.bps-inc.com>. That site has much information related to systems thinking and modeling.

For specific applications, M.I.T. hosts the “Creative Learning Exchange” that acts as an exchange service for STELLA® models. The site, located at <http://sysdyn.mit.edu/cle/lom.html>, can direct the user to many models on a number of different subjects. Most models can be freely downloaded.

Finally, we are continuously looking for constructive feedback on our work. Comments on this text can be sent via E-mail to either of the authors, Mike Deaton (deatonml@jmu.edu) or James Winebrake (winebrjf@jmu.edu). Errors found or suggestions for future editions are especially welcome.

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